

EECS240 – Spring 2009

Advanced Analog Integrated Circuits Lecture 1: Introduction



Elad Alon
Dept. of EECS

Course Focus

- Focus is on analog design
 - Typically: Specs → circuit topology → layout
- Will learn spec-driven approach
 - But will also look at where specs come from
- Key point:
 - Especially in analog, some things are much “easier” to do than others
 - Sometimes (often) the right thing to do is change the specs

Administrative

- **Course web page:**
http://bwrc.eecs.berkeley.edu/classes/icdesign/ee240_s09
- **Webcast link:**
<http://webcast.berkeley.edu>
- **Office hours**
 - 565 Cory Hall
 - Tues. and Thurs. 11am-12pm (right after class)
- **All announcements made through web page**
 - Check back often

Lecture Notes

- **Based on material from me, Prof. Bernhard Boser, and Prof. Ali Niknejad**
- **Primary source of material for the class**
 - No required text – reference texts on next slide
- **Notes posted on the web at least 1 hour before lecture**
 - Will hand out limited # of hard copies in class

Reference Texts

- **Analysis and Design of Integrated Circuits**,
Paul R. Gray, Paul J. Hurst, Stephen H. Lewis, Robert G. Meyer,
4th Ed., Wiley, 2001.
- **Design of Analog CMOS Integrated Circuits**,
Behzad Razavi, McGraw-Hill, 2000.
- **The Design of CMOS Radio-Frequency Integrated Circuits**,
Thomas H. Lee, 2nd Ed., Cambridge University Press, 2003.
- **Analog Integrated Circuit Design**,
D. Johns and K. Martin, Wiley, 1997.
- **The Designers Guide to SPICE & SPECTRE**,
K. S. Kundert, Kluwer Academic Press, 1995.
- **Operation and Modeling of the MOS Transistor**,
Y. Tsividis, McGraw-Hill, 2nd Edition, 1999.

Grading

- **Grading:**
 - **HW: 20%**
 - One HW roughly every two weeks
 - Essential for learning the class material
 - Need to setup HSPICE or equivalent simulator (SpectreRF, Eldo, or other favorite tool)
 - **Project: 25%**
 - Groups of 2 – find a partner ahead of time
 - **Midterm: 20%**
 - **Final Exam: 35%**

Homework

- **Homework:**
 - Can discuss/work together
 - But write-up must be individual
 - Drop in box outside Elad's office (565 Cory)
 - Generally due 5pm on Thursdays
- **No late submissions**
 - Start early!

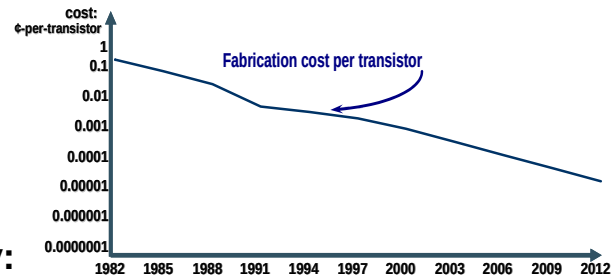
Schedule Notes

- **ISSCC Week: 2/9 - 2/13 (no lectures)**
- **Midterm: March 12 (tentative)**
- **Spring break: 3/23 – 3/27**
- **Project:**
 - Part 1 due Apr. 16
 - Part 2 due Apr. 28
 - Part 3 due May 7 (tentative)
- **Final: Mon., May 18, 8am-11am**

“Analog ICs in a Digital World?”

Digital circuitry:

- Cost/function decreases by 29% each year
- 30X in 10 years



Analog circuitry:

- Cost/function may not scale very well
- Common complaints about scaling analog:
 - Supply voltage is too low, device gain is low, horrible matching...

“Analog will die – everything will be digital!”

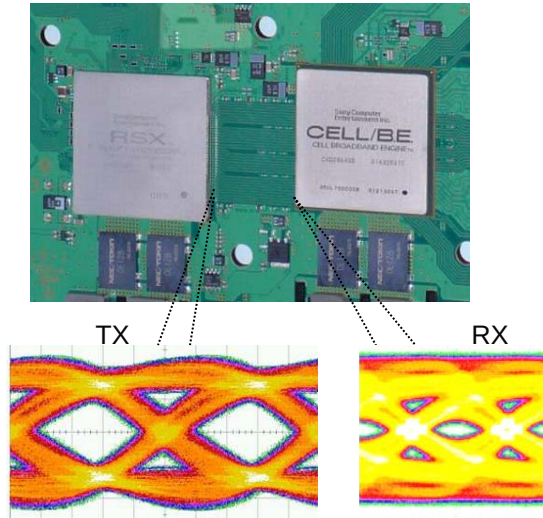
- Who agrees?

(Good) Digital Design Needs Analog Insights

- Can synthesize large blocks at “medium” frequencies in ASIC flow, but
 - Need to know transistors to design the cells
 - Really need to know transistors to design memories
- Lots of analog issues to deal with when push digital performance, power, etc.
 - Charge sharing, interconnect parasitics, etc.
- Matching growing concern in advanced CMOS technologies
 - Especially in memories

Another Example

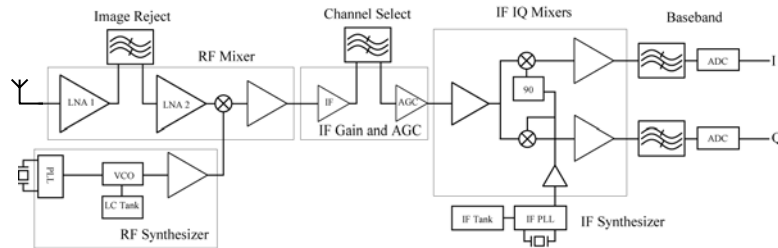
- Look at interface between two digital chips
 - Is received bit a “1” or a “0”?
- Analog circuits critical for receiving bits correctly



The More Fundamental Reason

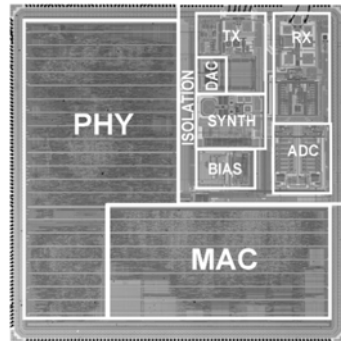
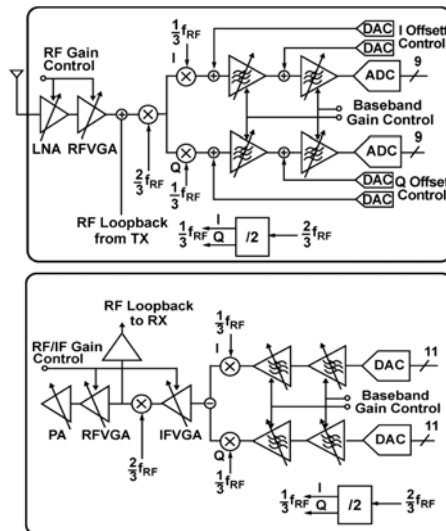
- The “real” or “physical” world is analog
 - Analog is required to interface to just about anything
 - Digital signals have analog characteristics too...
 - In many applications, analog is in the critical path
- Examples:
 - Wireline, optical communications
 - RF transceivers (receiver + transmitter)
 - Sensors and actuators (e.g., MEMS)

RF Receiver



- Why so many RF and analog building blocks?
- Why not just put the ADC right after the antenna?

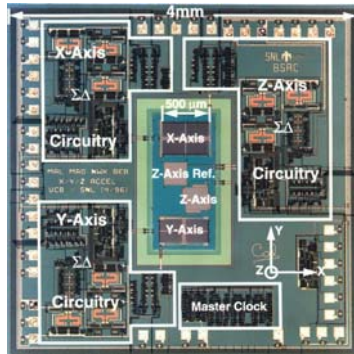
RF Transceiver Layout



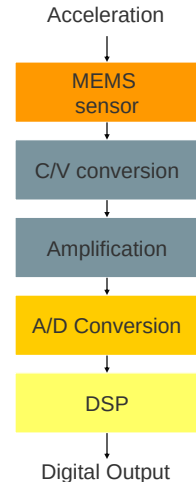
Source: Mehta et al, "An 802.11g WLAN SoC", JSSC Dec. 2005

- Analog building blocks take up significant die area
 - Even in 0.18um...

MEMS Accelerometer



M. Lemkin and B. E. Boser, "A Three-Axis Micromachined Accelerometer with a CMOS Position-Sense Interface and Digital Offset-Trim Electronics," *IEEE J. Solid-State Circuits*, vol. SC-34, pp. 456-468, April 1999



Digital Versus Analog Design

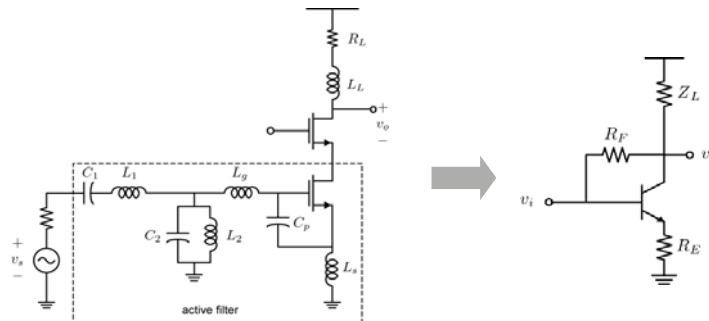
- **Abstraction in digital is Boolean logic (1's, 0's)**
 - Works because of noise margins
- **At a higher level, it's gates and registers (RTL)**
- **Digital layout is often automated**

- **Abstraction in analog is the device model**
 - (BSIM is a few thousand lines long)
- **At a higher level, it's the (opamps) (filters) (comparators)**
 - Abstraction depends on the problem you're solving
- **Analog layout is usually hand crafted**

“Analog” versus “RF”

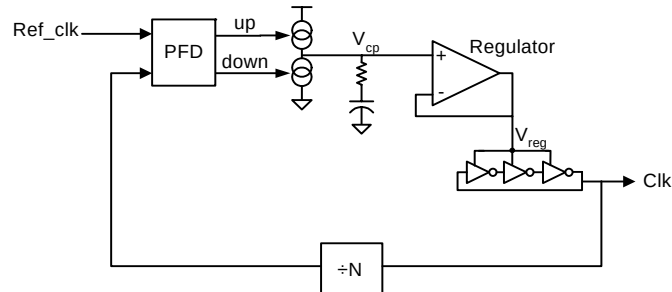
- RF = “Analog with inductors”
- RF signal is usually narrowband (i.e., sinusoidal)
 - Tuned circuit techniques used for signal processing.
- RF impedance levels are relatively low
 - Can’t make antenna impedance too high
- Analog impedances are high (low) for voltage (current) gain.
 - Voltage/current gain versus power gain.
- “Mixed-signal” analog is often discrete time (sampled).

RF Shifting Toward Analog



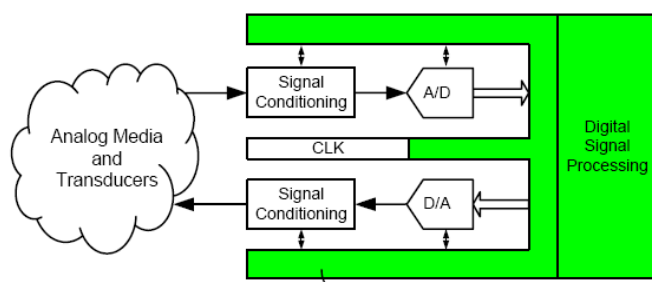
- **Classic RF uses inductors to tune the circuits**
 - Inductors are big – would be nice to get rid of them
- **With increasing f_T , moving towards wideband analog & feedback**
 - What’s the penalty?

Mixed-Signal Design



- Many building blocks involve analog and digital circuit co-design
 - PLLs, ADCs, etc.
- Sometimes hard to even distinguish between analog and digital
 - Is VCO analog, or digital?

Digitally-Assisted Analog



Source: B. Murmann, "Digitally-Assisted Analog Circuits – A Motivational Overview," ISSCC 2007.

- In 90nm, one RF inductor ($200\mu \times 200\mu$) takes same area as a microprocessor!
 - Leverage digital processing to improve analog circuits
- Good analog design doesn't go away though
 - Need to find right partitioning to maximize the benefit

Syllabus

- **Devices (both passive and active):**
 - Models, simulation, layout, and matching
- **Electronic noise**
- **Basic support functions:**
 - Current sources, references, biasing
- **Basic analog “gate”: amplifier**
 - Opamps, OTAs, feedback, settling time, common-mode feedback
- **Application driver: high-speed links**
 - Motivates additional building blocks
 - As well as why you care about certain specs
 - Data converters, comparators, offset cancellation, filters, sample & hold, oscillators, PLLs

EECS 240 versus 247

- **EECS 240**
 - Transistor level building blocks
 - Device and circuit fundamentals
 - A lot of the class at a low level of abstraction
 - SPICE
- **EECS 247**
 - Macro-models, behavioral simulation, large systems
 - Signal processing fundamentals
 - High level of abstraction
 - Matlab

240 versus 242/142

- **142/242 mostly concerned with narrowband circuits operating at a “high” carrier frequency**
 - Signals mostly look like sinusoids
 - Inductors ubiquitous
 - Use of feedback is rare
- **240 focuses on more “wideband”, general-purpose analog and mixed-signal**
 - Signals are “arbitrary”
 - Spend a lot of time worrying about capacitance
 - Feedback common

240 versus 231

- **231 concentrates on device physics**
- **240: device physics abstracted to the extent possible**
 - Device models from a “circuit designer’s perspective”
 - Treat transistor as black box described by complex equations
 - Equations relevant for biasing, nonlinear effects (output swing), and some charge storage effects
 - Mostly outside design loop
 - → “small signal analysis”

EECS240 – Spring 2009

Lecture 2: CMOS Technology and Passive Devices



**Elad Alon
Dept. of EECS**

Today's Lecture

- **EE240 CMOS Technology**
- **Passive devices**
 - Motivation
 - Resistors
 - Capacitors
 - (Inductors)
- **Next time: MOS transistor modeling**

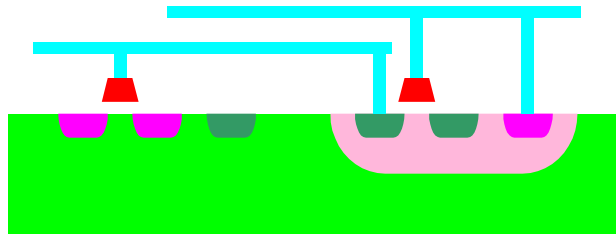
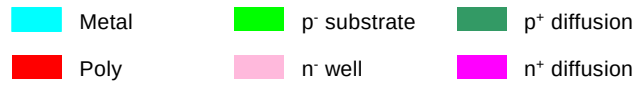
EE240 Process

- **90nm 1P7M CMOS**
 - Minimum channel length: 90nm
 - 1 level of polysilicon
 - 7 levels of metal (Cu)
 - 1.2V supply
 - Models for this process not “real”
- **Other processes you might see**
 - Shorter channel length (45nm / 1V)
 - Bipolar, SiGe HBT
 - SOI

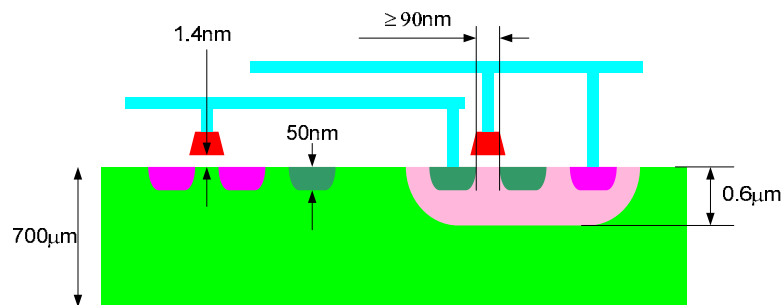
Process Options

- Available for many processes
- Add features to “baseline process”
- E.g.
 - Silicide block option
 - “High voltage” devices (2.5V & 3.3V, >10V)
 - Low V_{TH} devices
 - Capacitor option (2 level poly, MIM)
 - ...

CMOS Cross Section



Dimensions



Why Talk About Passives?

Resistors

- No provisions in standard CMOS
- Resistors are bad for digital circuits →
 - Minimized in standard CMOS
 - But, often want big, well-controlled R for analog...
- Sheet resistance of available layers:

Layer	Sheet resistance
Aluminum	60 mΩ/□
Polysilicon	5 Ω/□
N+/P+ diffusion	5 Ω/□
N-well	1 kΩ/□

How about an N-Well Resistor?

Silicide Block Option

Layer	$R/\square$ [$\Omega/\square$]	T_c [ppm/ $^{\circ}\text{C}$] @ $T = 25^{\circ}\text{C}$	V_c [ppm/V]	B_c [ppm/V]
N+ poly	100	-800	50	50
P+ poly	180	200	50	50
N+ diffusion	50	1500	500	-500
P+ diffusion	100	1600	500	-500
N-well	1000	-1500	20,000	30,000

- Non-silicided layers have significantly larger sheet resistance
- Resistor nonidealities:
 - Temperature coefficient: $R = f(T)$
 - Voltage coefficient: $R = f(V)$
 - Manufacturing Variations

Resistor Example

Goal: $R = 100 \text{ k}\Omega$, $T_C = 1/R \times dR/dT = 0$

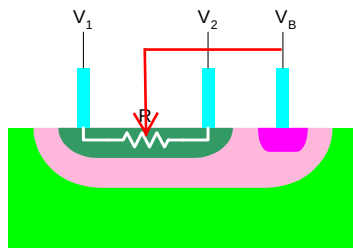
Example Solution: N+ and P+ poly resistors in series

$$\begin{aligned} R &= R_N (1 + T_{CN} \Delta T) + R_P (1 + T_{CP} \Delta T) \\ &= \underbrace{R_N + R_P}_R + \underbrace{(R_N T_{CN} + R_P T_{CP})}_0 \Delta T \quad \Rightarrow \\ R_N &= R \frac{1}{1 - \frac{T_{CN}}{T_{CP}}} = 20 \text{ k}\Omega = 200 \text{ squares} \\ R_P &= R \frac{1}{1 - \frac{T_{CP}}{T_{CN}}} = 80 \text{ k}\Omega = 444.4 \text{ squares} \end{aligned}$$

Voltage Dependence

Voltage Coefficient

p⁻ substrate
 p⁺ diffusion
 n⁺ diffusion
 n⁻ well



Example:

Diffusion resistor

→ Applied voltage modulates depletion width (cross-section of conductive channel)

→ Well acts as a shield

$$R = \frac{V_1 - V_2}{I}$$

$$\approx R_0 \left[1 + T_c (T - 25^\circ) + V_c (V_1 - V_2) + B_c \left(\frac{V_1 + V_2}{2} - V_B \right) \right]$$

Resistor Matching

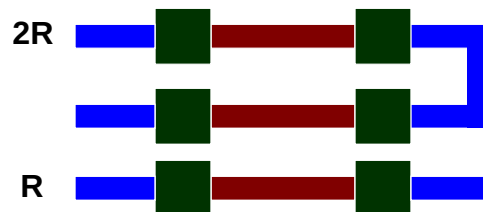
- Types of mismatch:
 - Run-to-run variations
 - Global differences in thickness, doping, etc.
 - Systematic (e.g. contacts)
 - Random variations between devices
- Run-to-run variations in absolute R value: 20+%
 - Can be problematic for termination, bias current, etc.
- Best case: make circuit depend only on ratios
 - E.g., use feedback to control opamp gain
 - With careful layout, can get 0.1 – 1% matching

Systematic Variations from Layout

- Example:



- Use unit element instead:



Common Centroid and Dummies

Example: $R1 : R2 = 1 : 2$

Dummy →



↙ gradient



$0.5 * R2 + \Delta R$



$R1$



$0.5 * R2 - \Delta R$

Dummy →

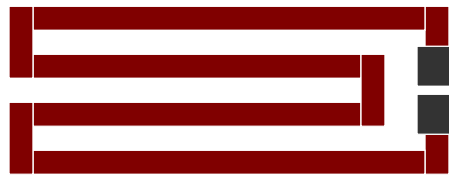


Resistor Layout (cont.)

Serpentine layout for large values:



Better layout (mitigates offset due to thermoelectric effects):



See Hastings, "The art of analog layout," Prentice Hall, 2001.

MOSFETs as Resistors

- Triode region ("square law"):

$$I_D = \mu C_{ox} \frac{W}{L} \left(V_{GS} - V_{TH} - \frac{V_{DS}}{2} \right) V_{DS} \quad \text{for } V_{GS} - V_{TH} > V_{DS}$$

- Small signal resistance:

$$\frac{1}{R} = \frac{\partial I_D}{\partial V_{DS}} = \mu C_{ox} \frac{W}{L} (V_{GS} - V_{TH} - V_{DS})$$

$$R \approx \frac{1}{\mu C_{ox} \frac{W}{L} (V_{GS} - V_{TH})} \quad \text{for } V_{GS} - V_{TH} \gg V_{DS}$$

- Voltage coefficient:

$$V_C = \frac{1}{R} \frac{\partial R}{\partial V_{DS}} = \frac{1}{V_{GS} - V_{TH} - V_{DS}}$$

MOS Resistors

Example: $R = 1 \text{ M}\Omega$

$$R \approx \frac{1}{\mu C_{ox} \frac{W}{L} (V_{GS} - V_{TH})}$$

$$\frac{W}{L} = \frac{1}{\mu C_{ox} R (V_{GS} - V_{TH})}$$

$$= \frac{1}{100 \frac{\mu\text{A}}{\text{V}^2} \times 1 \text{ M}\Omega \times 2 \text{ V}} = \frac{1}{200}$$

$$V_C|_{V_{DS}=0V} = \frac{1}{V_{GS} - V_{TH}}$$

$$= \frac{1}{2 \text{ V}} = 0.5 \text{ V}^{-1}$$

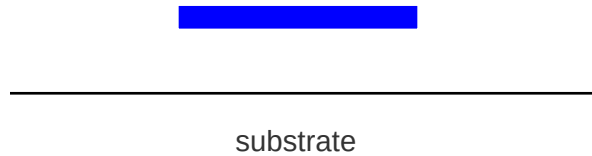
- Large R-values realizable in small area
- Very large voltage coefficient
- Applications:
 - **MOSFET-C filters: (linearization)**
 Ref: Tsividis et al, "Continuous-Time MOSFET-C Filters in VLSI," JSSC, pp. 15-30, Feb. 1986.
 - **Biasing: (>1GΩ)**
 Ref: Geen et al, "Single-Chip Surface-Micromachined Integrated Gyroscope with 50°/hour Root Allen Variance," ISSCC, pp. 426-7, Feb. 2002.

Resistor Summary

- No or limited support in standard CMOS
 - Large area (compared to FETs)
 - Nonidealities:
 - Large run-to-run variations
 - Temperature coefficient
 - Voltage coefficients (nonlinear)
- Avoid them when you can
 - Especially in critical areas, e.g.
 - Amplifier feedback networks
 - Electronic filters
 - A/D converters
 - We will get back to this point

Capacitors

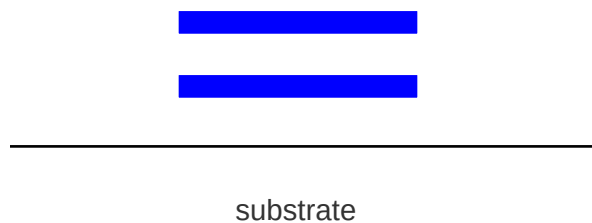
- Simplest capacitor:



- What's the problem with this?

Capacitors

- “Improved” capacitor:

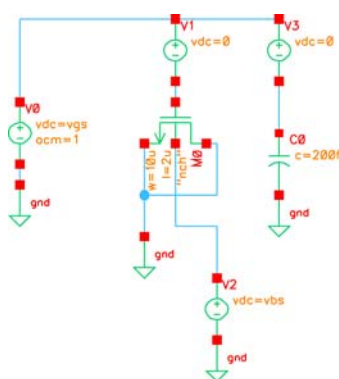


- Is this only 1 capacitor?

Capacitor Options

Type	C [aF/μm ²]	V _C [ppm/V]	T _C [ppm/°C]
Gate	10,000	Huge	Big
Poly-poly (option)	1000	10	25
Metal-metal	50	20	30
Metal-substrate	30		
Metal-poly	50		
Poly-substrate	120		
Junction caps	~ 1000	Big	Big

MOS Capacitor



- High capacitance in inversion

- SPICE:

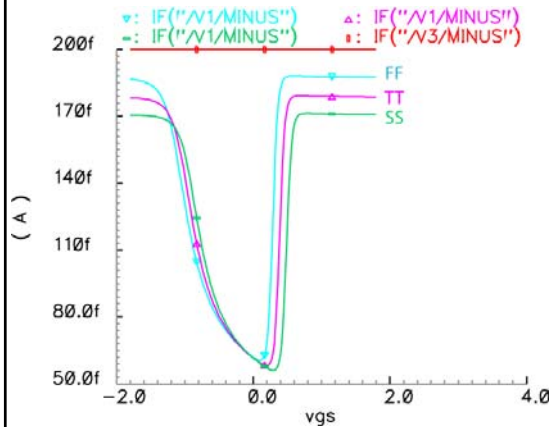
$$C = \frac{I}{\omega V}$$

$$V = 1V$$

$$\omega = 1$$

$$\rightarrow C = I$$

MOS Capacitor



- High non-linearity, temperature coefficient
- But, still useful in many applications, e.g.:
 - (Miller) compensation capacitor
 - Bypass capacitor (supply, bias)

Capacitor Layout

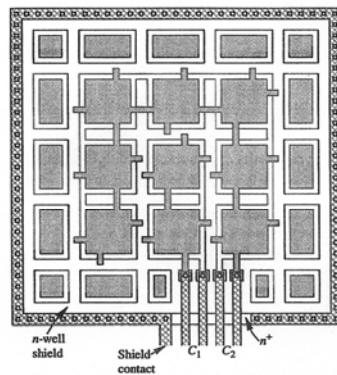


Figure 6.17 A layout of two capacitors with ratio $C_2/C_1 = 1/2$, incorporating several of the techniques discussed in the text (adapted from Ref. 23).

- Unit elements
- Shields:
 - Etching
 - Fringing fields
- "Common-centroid"
- Wiring and interconnect parasitics

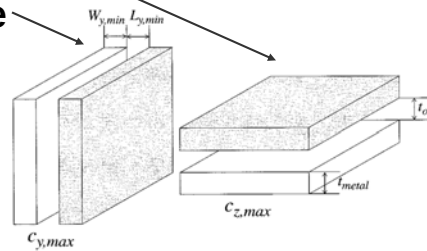
Ref.: Y. Tsividis, "Mixed Analog-Digital VLSI Design and Technology," McGraw-Hill, 1996.

MIM Capacitors

- Some processes have MIM cap as add-on option
 - Separation between metals is much thinner
 - Higher density
- Used to be fairly popular
 - But not as popular now that have many metal layers anyways

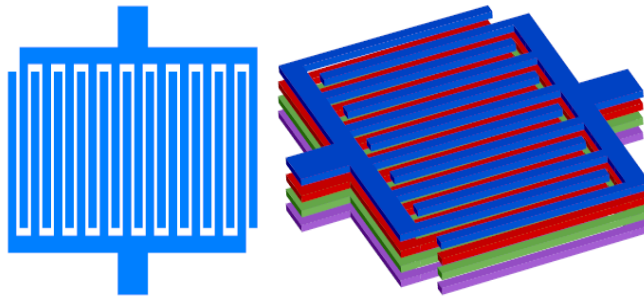
Capacitor Geometries

- Horizontal parallel plate
- Vertical parallel plate
- Combinations



Ref: R. Aparicio and A. Hajimiri, "Capacity Limits and Matching Properties of Integrated Capacitors," JSSC March 2002, pp. 384-393.

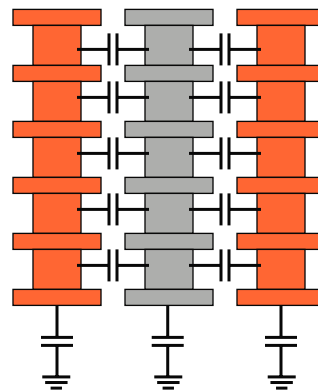
“MOM” Capacitors



- Metal-Oxide-Metal capacitor. Free with modern CMOS.
- Use lateral flux ($\sim L_{\min}$) and multiple metal layers to realize high capacitance values

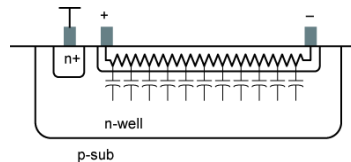
MOM Capacitor Cross Section

- Use a wall of metal and vias to realize high density
- More layers – higher density
 - May want to chop off lower layers to reduce C_{bot}
- Reasonably good matching and accuracy



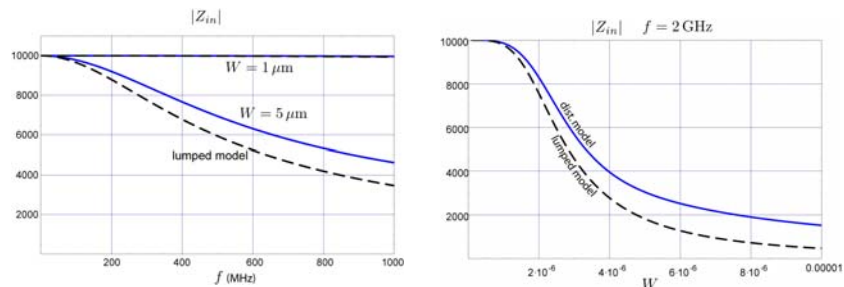
Distributed Effects

- Can model IC resistors as distributed RC circuits.
- Could use transmission line analysis to find equivalent 2-port parameters.
- Inductance negligible for small IC structures up to ~10GHz.



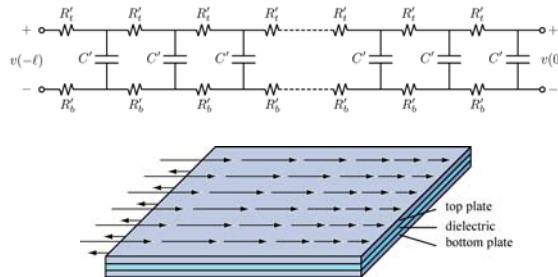
$$R \gg \omega L$$

Effective Resistance



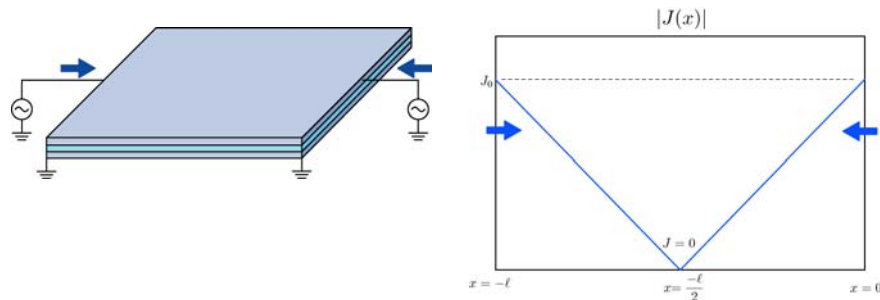
- High frequency resistance depends on W , e.g.:
 - $W=1\mu$ 10k Ω resistor works fine at 1GHz
 - $W=5\mu$ 10k Ω resistor drops to 5k Ω at 1 GHz
- May need distributed model for accurate freq response

Capacitor Q



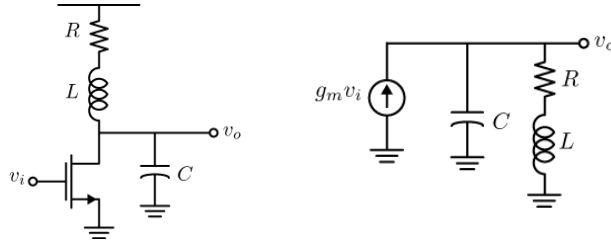
- Current density drops as you go farther from contact edge...

Double Contact Structure



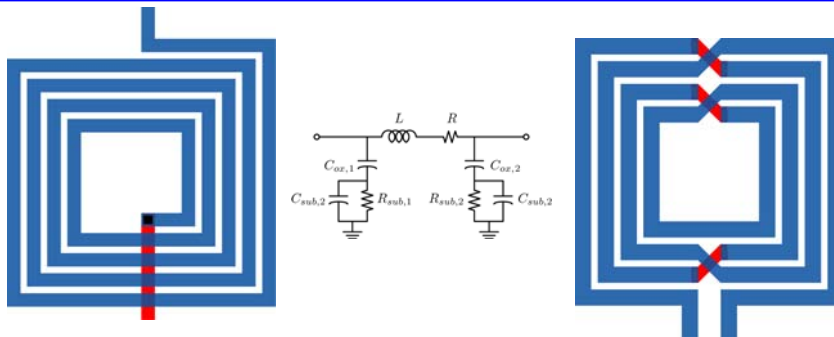
- If contact on both edges,
 - R drops 4X
 - Can be a good idea even if not hitting distributed effects

What About Inductors?



- **Mostly not used in analog/mixed-signal design**
 - Usually too big
 - More of a pain to model than R's and C's
 - But they do occasionally get used
- **Example inductor app.: shunt peaking**
 - Can boost bandwidth by up to 85%!
 - Q not that important (L in series with R)
 - But frequency response may not be flat

Spiral Inductors



- **Used widely in RF circuits for small L (~1-10nH).**
- **Use top metal for Q and high self resonance frequencies.**
 - Very good matching and accuracy – if you model them right

EECS240 – Spring 2009

Lecture 3: MOS Models for Design



Elad Alon
Dept. of EECS

Why Modeling?

- **Analog circuits more sensitive to detailed transistor behavior**
 - Precise currents, voltages, etc. matter
 - Digital circuits have much larger “margin of error”
- **Models allow us to reason about circuits**
 - Provide window into the physical device and process
 - “Experiments” with SPICE much easier to do

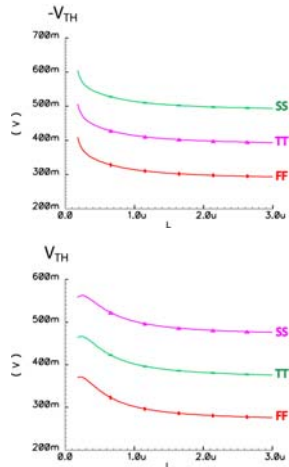
Levels of Abstraction

- **Best abstraction depends on questions you want to answer**
- **Digital functionality:**
 - MOSFET is a switch
- **Digital performance:**
 - MOSFET is a current source and a switch
- **Analog characteristics:**
 - MOSFET described by BSIM with 100's of parameters?
 - MOSFET described by measurement results?

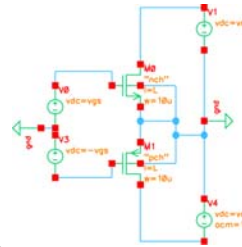
Detour: Device Corners

- **Run-to-run parameter variations:**
 - E.g. implant doses, layer thickness, dimensions
 - Affect V_{TH} , μ , C_{ox} , $R_{\square}$, ...
 - How model in SPICE?
- **Nominal / slow / fast parameters (tt, ss, ff)**
 - E.g. fast: low V_{TH} , high μ , high C_{ox} , low $R_{\square}$
 - Combine with supply & temperature extremes
 - Pessimistic but numerically tractable
→ improves chances for working Silicon

Corner example: V_{TH}



- Corners just shift V_{th}
 - Probably not real
 - (PMOS doesn't look real anyways)
- Variations probably bigger than reality too
 - Fab wants you to buy everything they make



Why not Square Law?

- Square law model most widely known:

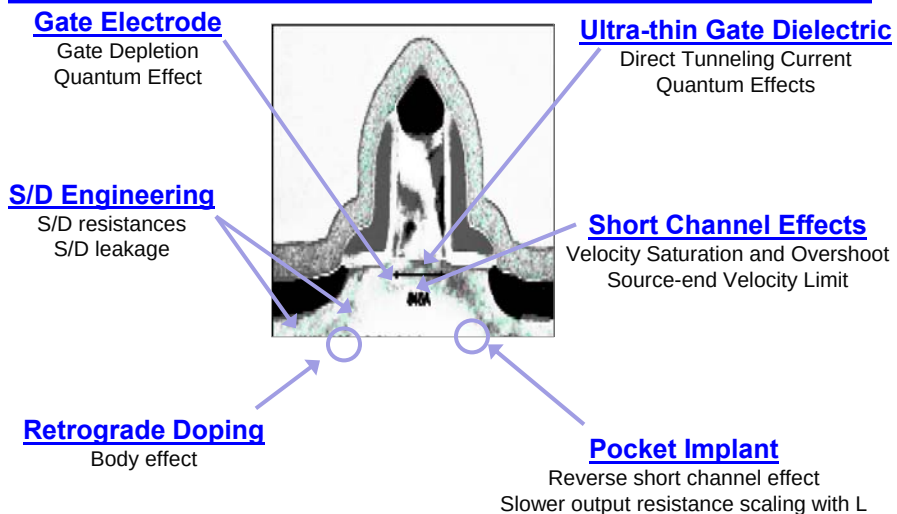
$$I_{D,sat} = \frac{1}{2} \cdot \mu_n \cdot C_{ox} \cdot \frac{W}{L} \cdot (V_{GS} - V_{th})^2$$

- But, totally inadequate for “short-channel” behavior
- Also doesn't capture *moderate* inversion
 - (i.e., in between sub-threshold and strong inversion)

Square Law Model Assumptions

- Charge density determined only by vertical field
- Drift velocity set only by lateral field
- Neglect diffusion currents (“magic” V_{th})
- Constant mobility
- And many more...

A Real Transistor



Now What?

- Rely purely on simulator to tell us how devices behave?
 - Models not always based on real measurements
 - Model extraction is hard
 - Models inherently compromise accuracy for speed
- Need to know about important effects
 - So that know what to look for
 - Model might be wrong, or doesn't automatically include some effects
 - E.g., gate leakage

Output Resistance: CLM

- “Channel Length Modulation”
 - Depletion region varies with V_{DS}
 - Changes effective channel length

- If perturbation is small:

$$I \propto \frac{1}{L - \delta L(V_{ds})} \approx \frac{1}{L} \left(1 + \frac{\delta L(V_{ds})}{L} \right) \longrightarrow \frac{I_{ds}}{I_{ds0}} = (1 + \lambda V_{ds})$$

Output Resistance: DIBL

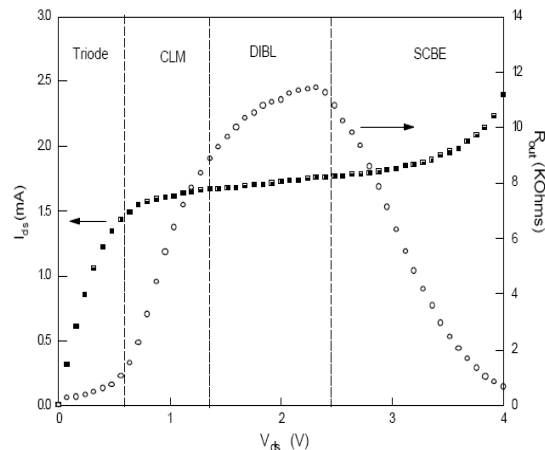
- “Drain Induced Barrier Lowering”
- Drain controls the channel too
 - Charge gets imaged – lowers effective V_{th}
 - Model with $V_{th} = V_{th0} - \eta V_{DS}$

Output Resistance: SCBE

- “Substrate Current Body Effect”
- At high electric fields, get “hot” electrons
 - Have enough energy to knock electrons off Si lattice (impact ionization)
- Extra e^- - h^+ pairs – extra (substrate) current
 - Models usually empirical

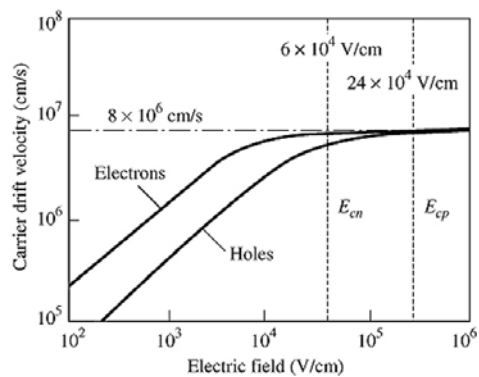
$$I_{sub} = \frac{A_i}{B_i} I_{ds} (V_{ds} - V_{dsat}) \exp\left(-\frac{B_i l}{V_{ds} - V_{dsta}}\right)$$

Output Resistance Mechanisms



- All effects active simultaneously
- CLM at relatively low fields
- DIBL dominates for high fields
- SCBE at very high fields

Velocity Saturation



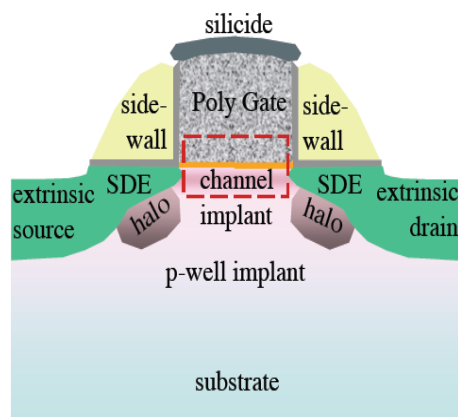
- Drift velocity initially increases linearly with field
- Eventually carriers hit a “speed limit”
- In the limit, $I_D \propto (V_{GS} - V_{th})$

Vertical Field Mobility Reduction

- **Mobility actually depends on gate field**
 - “Hard to run when there is wind blowing you sideways (into a wall)”
- **More technical explanation:**
 - E-field pushes carriers close to the surface
 - Enhanced scattering lowers mobility

$$\mu = \frac{\mu_0}{1 + \theta(V_{GS} - V_T) + \theta_B V_{SB}}$$

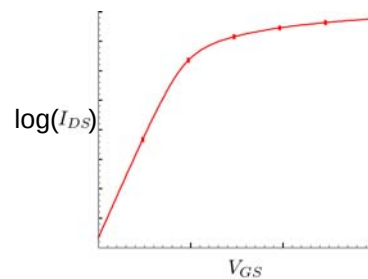
Halo Doping



Reverse Short-Channel Effect

Sub-Threshold Region

- Current doesn't really go to 0 at $V_{GS} = V_{th}$
- Lateral BJT:



Weak Inversion Channel Potential

- “Base” controlled through capacitive divider

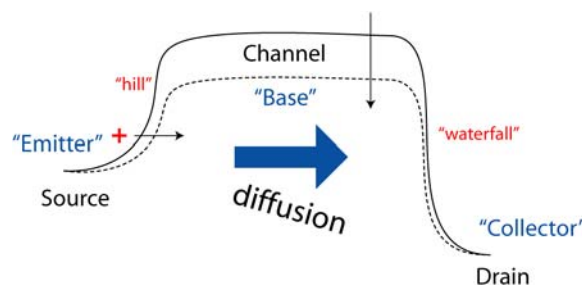
$$\delta V_{ch} \approx \frac{C_{ox}}{C_{dep} + C_{ox}} \delta V_g = \frac{\delta V_g}{n}$$

- Non-ideality factor of channel control $n > 1$:

$$n = 1 + \frac{C_{dep}}{C_{ox}} = 1 + \frac{\epsilon_{dep} t_{ox}}{\epsilon_{ox} t_{dep}}$$

- (n varies somewhat with bias – const. approx. usually OK)

Weak Inversion Current



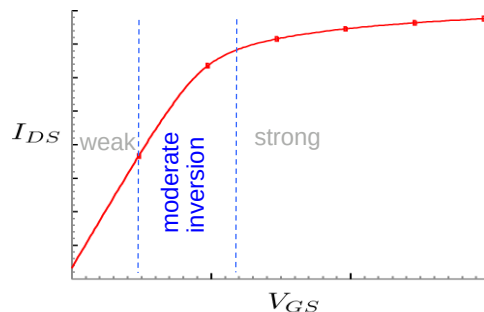
- Current set by diffusion – borrow BJT equation:

$$I_{ds} = \frac{W}{L} I_{ds,0} e^{\frac{q(V_{gs} - V_T)}{nkT}} \left(1 - e^{\frac{-qV_{DS}}{kT}} \right)$$

Operating in Weak Inversion

- Usually considered “slow”:
 - “large” C_{GS} for “little” current drive (see later)
- But, weak (or moderate) inversion becoming more common:
 - Low power
 - Submicron L means “high speed” even in weak inversion
- Not well modeled, matching poor:
 - V_{TH} mismatch amplified exponentially
 - Avoid in mirrors

Moderate Inversion



- Moderate inversion: both *drift* and *diffusion* contribute to the current.
- Closed form equations for this region don't really exist.

Patching Models?

- Have “good” models for weak inversion and strong inversion.
 - Why not just interpolate in between?
- Example (EKV):

$$I_{DS} = \frac{W}{L} \mu C_{ox} (2n) \left(\frac{kT}{q} \right)^2 \left(\left(\ln \left(1 + e^{\frac{q(V_{GB} - V_{T0} - nV_{SB})}{2n kT}} \right) \right)^2 - \left(\ln \left(1 + e^{\frac{q(V_{GB} - V_{T0} - nV_{DB})}{2n kT}} \right) \right)^2 \right)$$

BSIM

- ***Berkeley Short-channel IGFET Model (BSIM)***
 - Industry standard model for modern devices
 - BSIM3v3 is model for this course
- **Typically 40-100+ parameters**
 - Advanced software and expertise needed to perform extraction

BSIM “Hand Calculation” Model

- Requires many, many, many... assumptions
- Vertical mobility degradation:

Define: $u_d = \frac{UA}{t_{ox}}$ mobility degradation coefficient
 $u_d \approx 0.5V^{-1}$ for $t_{ox}=10nm$

- Velocity saturation:

Define: $E_c = \frac{2v_{sat}}{U_0}$ critical E-field for velocity saturation
 $E_c \approx 2 \times 10^4 V/cm$ (typical value)

Strong Inversion Current

$$V_{Dsat} = (V_G - V_T) \left[\frac{1 + u_d(V_G - V_T)}{1 + \left(u_d + \frac{1}{E_c L}\right)(V_G - V_T)} \right]$$

$$I_{Dlin} = \mu_0 C_{ox} \frac{W}{L} \left(V_G - V_T - \frac{V_D}{2} \right) V_D \left[\frac{1}{1 + u_d(V_G - V_T) + \left(\frac{V_D}{E_c L}\right)} \right] = I_{Dlin(long)} \left[\frac{1}{1 + u_d(V_G - V_T) + \left(\frac{V_D}{E_c L}\right)} \right]$$

$$I_{Dsat} = \mu_0 C_{ox} \frac{W}{2L} \left[\frac{(V_G - V_T)^2}{1 + \left(u_d + \frac{1}{E_c L}\right)(V_G - V_T)} \right] = I_{Dsat(long)} \left[\frac{1}{1 + \left(u_d + \frac{1}{E_c L}\right)(V_G - V_T)} \right]$$

Equations of Derivatives

$$g_{msat} = \frac{I_{Dsat}}{(V_G - V_T)} \left[1 + \frac{I_{Dsat}}{I_{Dsat(long)}} \right] = \frac{I_{Dsat}}{(V_G - V_T)} \left[1 + \frac{1}{1 + \left(u_d + \frac{1}{E_C L} \right) (V_G - V_T)} \right]$$

$$r_{out} = \frac{2 \{ (V_D - V_{Dsat}) + [1 + u_d (V_G - V_T)] (V_G - V_T) \} L^2}{\mu_0 C_{ox} W I_{P_{CLM}} [1 + u_d (V_G - V_T)] (V_G - V_T)^2}$$

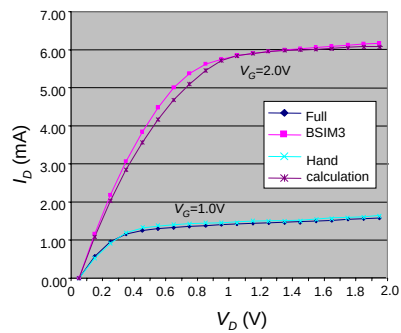
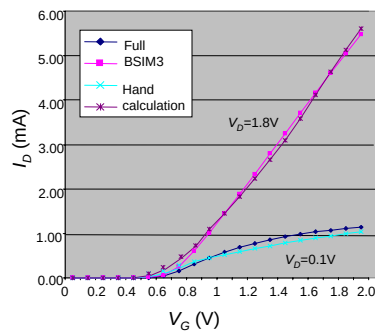
$$= \frac{\{ (V_D - V_{Dsat}) + [1 + u_d (V_G - V_T)] (V_G - V_T) \} L}{I_{Dsat(long)} I_{P_{CLM}} [1 + u_d (V_G - V_T)]}$$

with $l = \sqrt{3t_{ox}x_j}$

- Required parameters **W, L, TOX, U0, UA, VSAT, VTH0, PCLM, XJ**

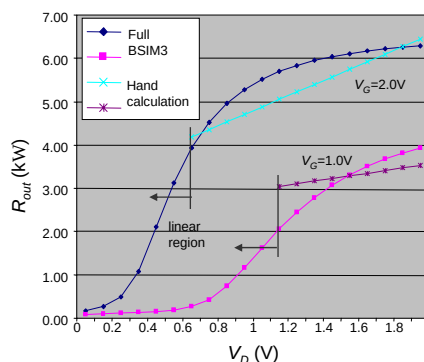
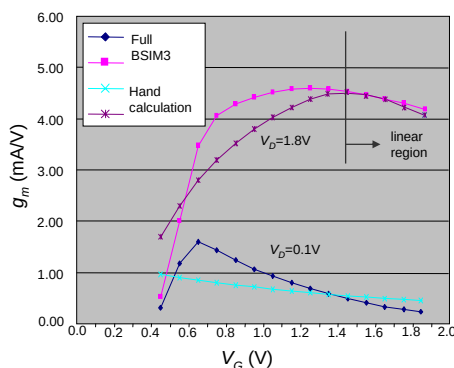
Fitting Results

Comparison between full and simplified model



Parameter detail: **TSMC 0.18um process**
 t_{ox} : 4.1nm, $W=10\mu m$, $V_{T0}=0.39V$

Weakness of Model First Derivatives



“Hand Model” Conclusion

- **Even “simple” model is not convenient**
 - r_o is key for gain, but really hard to model
 - Missing important regions such as moderate inversion
- Hand models really best to build intuition
- But for design (i.e., how to choose W , L , etc.):
 - Will learn how to use the simulator as a “calculator”

EECS240 – Spring 2008

Lecture 4: Design-Driven Small Signal Models



Elad Alon
Dept. of EECS

MOSFET Models for Design

- **SPICE (BSIM)**
 - For verification
 - Device variations
- **Hand analysis**
 - Velocity-sat model (good mostly for intuition)
 - Small-signal model
- **Challenge**
 - How to accurately design when hand analysis models may be way off?

Parameters Designers Care About

- **Layout designer:**
 - Mostly care about just W and L
- **Circuit designer:**
 - Gain $\rightarrow g_m, r_o$
 - Bandwidth $\rightarrow g_m, C_{GS}, C_{GD}, \dots$
 - Power $\rightarrow I_D$
 - Voltage swing $\rightarrow$ minimum V_{DS}
 - Noise
- **Can get many of the circuit parameters without resorting to BSIM**
 - Or rather, by just using BSIM as a look-up table

Low Frequency Model

- **1st order Taylor expansion of I_D :**

$$i_{ds} = \frac{\partial I_{ds}}{\partial V_{gs}} v_{gs} + \frac{\partial I_{ds}}{\partial V_{bs}} v_{bs} + \frac{\partial I_{ds}}{\partial V_{ds}} v_{ds}$$

$$i_{ds} = g_m V_{gs} + g_{mb} V_{bs} + g_{ds} V_{ds}$$

- **Just need to know the coefficients...**

Square Law Model

- In saturation:

$$g_m = \mu C_{ox} \frac{W}{L} \sqrt{\frac{2I_{ds}}{\mu C_{ox} \frac{W}{L}}}$$

$$g_m = \sqrt{2\mu C_{ox} \frac{W}{L} I_{ds}} \propto \sqrt{I_{ds}}$$

$$g_m = \frac{1}{2} \mu C_{ox} \frac{W}{L} (V_{gs} - V_T)^2 \frac{1}{\frac{1}{2}(V_{gs} - V_T)}$$

$$g_m = \frac{2I_{ds}}{V_{gs} - V_T} = \frac{2I_{ds}}{V_{od}}$$

Weak Inversion g_m

- In weak inversion we have bipolar behavior

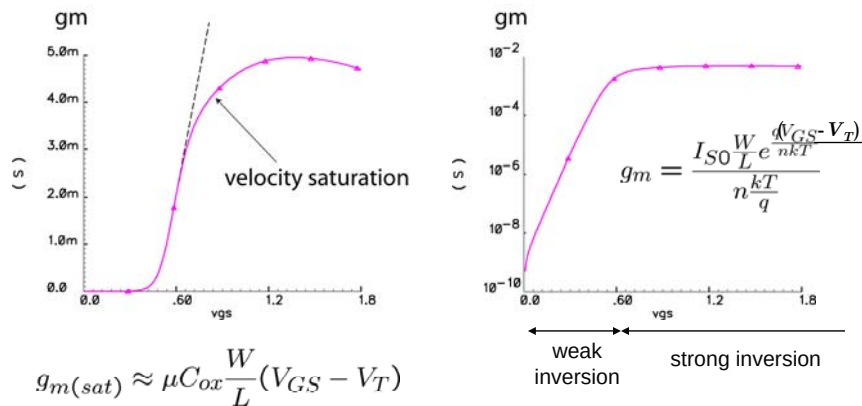
$$I_{ds} \approx \frac{W}{L} I_{ds,0} e^{\frac{q(V_{gs}-V_T)}{nkT}}$$

$$g_m = \frac{\partial I_{DS}}{\partial V_{GS}} = \frac{\frac{W}{L} I_{ds,0} e^{\frac{q(V_{gs}-V_T)}{nkT}}}{n \frac{kT}{q}}$$

- Good model if transistor is actually used in weak inversion

$$g_m = \frac{I_{DS}}{n \frac{kT}{q}} \propto I_{DS}$$

Transconductance



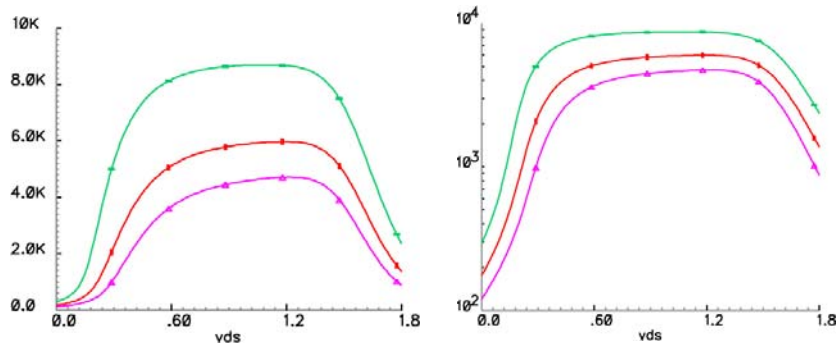
Transconductance (cont)

- Compare g_m of MOSFET and BJT:

$$g_m^{\text{FET}} = \frac{2I_{ds}}{V_{od}} \quad g_m^{\text{BJT}} = \frac{I_C}{V_t}$$

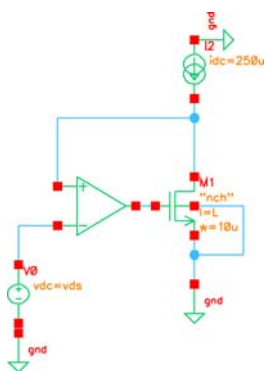
- Since $V_{od} \gg V_t$, BJT has larger g_m for same I_D
- Why can't we make $V_{od} \sim V_t$?
 - You can – if you work in subthreshold
 - Gives great g_m per unit current
 - But pay a penalty in speed (will see shortly)

Output Resistance r_o



Hopeless to model this with a simple equation
(e.g. $g_{ds} = \lambda I_D$)

Open-loop Gain a_{v0}

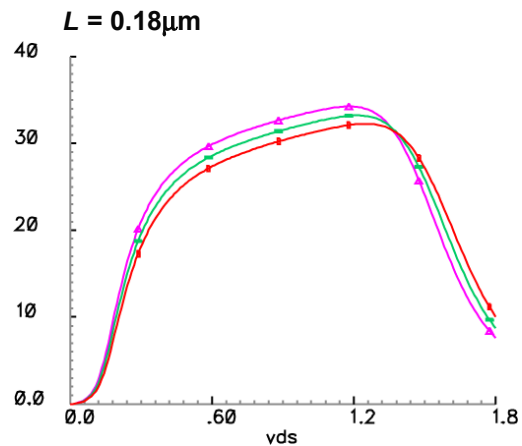


- Represents maximum attainable gain from a transistor
 - May be more useful than r_o

Simulation Notes:

- Bias current i_{dc} sets $V_{GS} - V_T$
- Use feedback to find correct V_{GS} while sweeping V_{DS}
- Use relatively small gain (100) for fast DC convergence

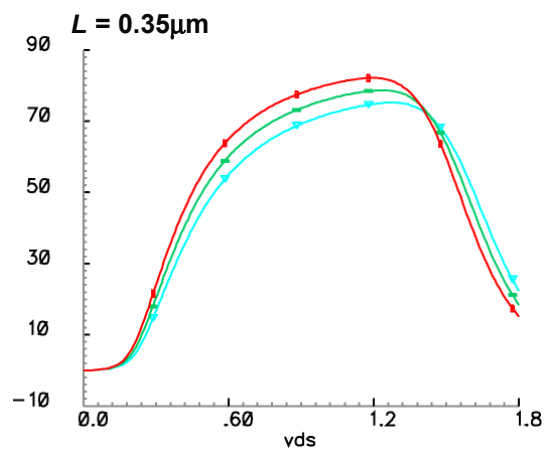
Gain, $a_{v0} = g_m r_o$



- Strong tradeoff: a_{v0} versus V_{DS} range

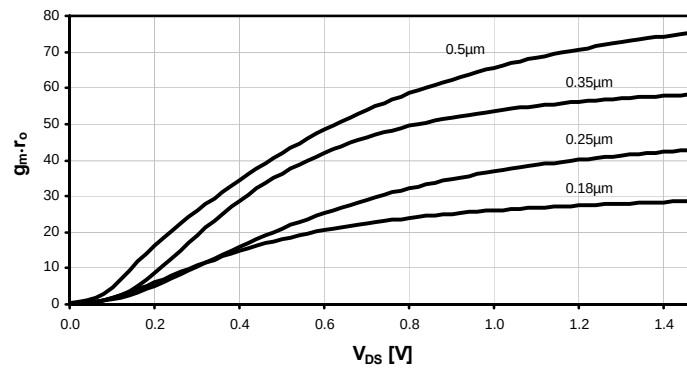
- Create plots for several device lengths

Long Channel Gain



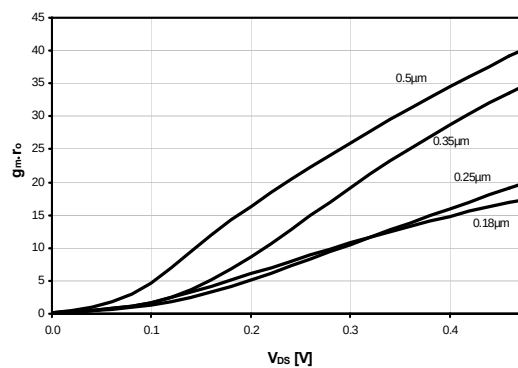
$$L \uparrow \rightarrow a_{v0} \uparrow$$

Technology Trend



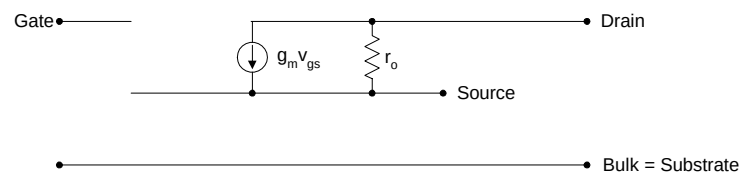
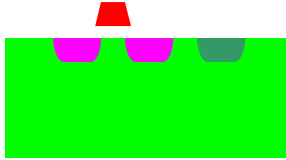
Short channel devices usually have lower peak gain

Transistor Gain Detail

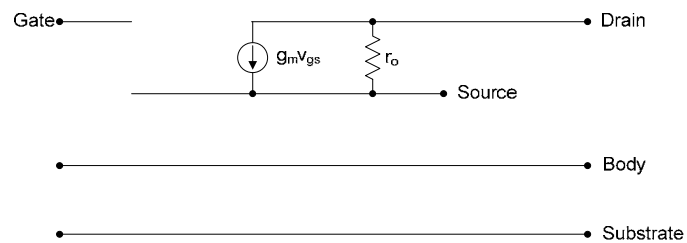
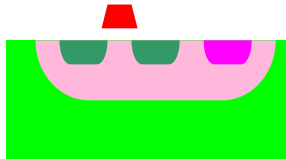


For practical V_{DS} gain penalty is less severe (remember: worst case V_{DS} is what matters!)

Small-Signal AC Model



PMOS AC Model



SPICE Charge Model

- Charge conservation
- MOSFET:
 - 4 terminals: S, G, D, B
 - 4 charges: $Q_S + Q_G + Q_D + Q_B = 0$ (3 free variables)
 - 3 independent voltages: V_{GS}, V_{DS}, V_{SB}
 - 9 derivatives: $C_{ij} = dQ_i / dV_j$, e.g. $C_{G,GS} \sim C_{GS}$
 - $C_{ij} \neq C_{ji}$

Ref: HSPICE manual, "Introduction to Transcapacitance", pp. 15:42, Metasoft, 1996.

Small Signal Capacitances

	Weak inversion	Strong inversion linear	Strong inversion saturation
C_{GS}	C_{ol}	$C_{GC}/2 + C_{ol}$	$2/3 C_{GC} + C_{ol}$
C_{GD}	C_{ol}	$C_{GC}/2 + C_{ol}$	C_{ol}
C_{GB}	$C_{GC} \parallel C_{CB}$	0	0
C_{SB}	C_{jsB}	$C_{jsB} + C_{CB}/2$	$C_{jsB} + 2/3 C_{CB}$
C_{DB}	C_{jdB}	$C_{jdB} + C_{CB}/2$	C_{jdB}

$$C_{GC} = C_{ox} WL$$

$$C_{CB} = \frac{\epsilon_{Si}}{x_d} WL$$

0.35u Process

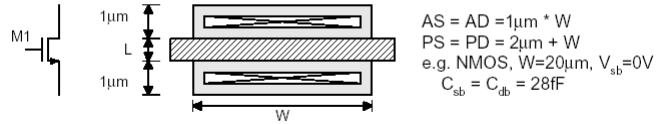
$$C_{ox} = 5.3 \text{ fF}/\mu\text{m}^2$$

$$C_{olN} = 0.24 \text{ fF}/\mu\text{m}$$

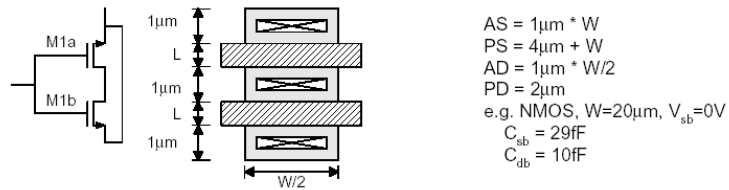
$$C_{olP} = 0.48 \text{ fF}/\mu\text{m}$$

Layout

Individual devices:



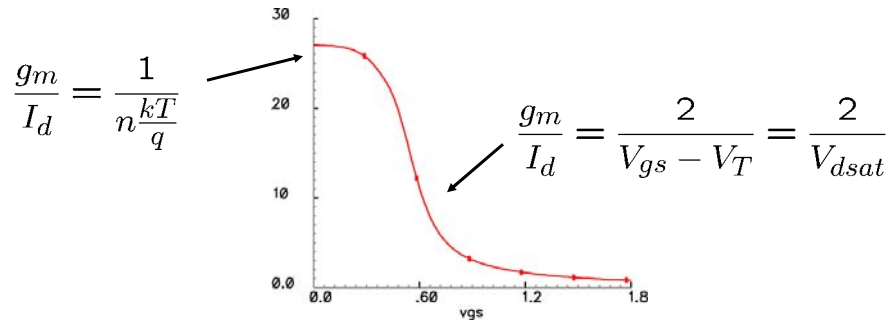
Wide devices consisting of multiple individual ones wired in parallel:



Source/drain Parasitics and HSPICE

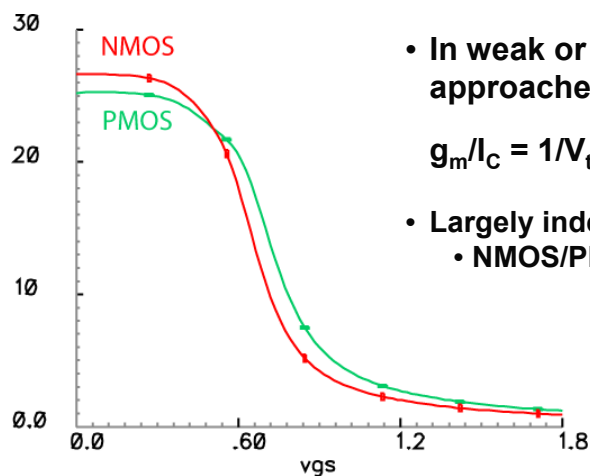
- **ACM=3 model (not in our current library)**
 - **HDIF = half of heavily doped diffusion length**
- **GEO = 0: No sharing**
- **GEO = 1: Drain shared**
- **GEO = 2: Source shared**
- **GEO = 3: Both shared**

Figure of Merit: g_m/I_D



- How much g_m per unit current
- Purely a DC metric
 - Weak and moderate inversion region clearly the most efficient regions to operate in

Efficiency g_m/I_D



- In weak or moderate inversion, approaches BJT

$$g_m/I_C = 1/V_t \sim 40 \text{ V}^{-1}$$
- Largely independent of device type
 - NMOS/PMOS about the same

Efficiency as a Design Parameter

- Why not use g_m/I_D for design?
- Can always determine value (from I_D and g_m)
 - Can do this “independently” of short channel effects (using simulator)
- Units (V^{-1}) and physical interpretation a little strange
 - But we'll just redefine things slightly to fix this

Substitute for g_m/I_D : V^*

- Define:

$$V^* = \frac{2I_D}{g_m} \quad \Leftrightarrow \quad \frac{g_m}{I_D} = \frac{2}{V^*}$$

e.g. $V^* = 200\text{mV} \rightarrow g_m/I_D = 10\text{ V}^{-1}$

- Square-law devices: $V^* = V_{GS} - V_{TH} = V_{od}$

Square law :

$$g_m = \frac{2I_D}{V_{GS} - V_{TH}} = \frac{2I_D}{V^*}$$

- Remember: real devices do **not** obey the square law!

V_{od} vs V^*

- | | |
|---|---|
| <ul style="list-style-type: none"> • Overdrive voltage V_{od} <ul style="list-style-type: none"> • Cannot be measured • Complex equations • “Long channel” devices: <ul style="list-style-type: none"> • $V_{od} = V_{dsat} = V^*$ • $I_D \sim V^{*2}$ • Boundary between triode and saturation • r_o “large” for $V_{DS} > V^*$ • C_{GS}, C_{GD} change | <ul style="list-style-type: none"> • $V^* = 2I_D/g_m$ <ul style="list-style-type: none"> • Measure (simulate) easily • Complex equations • “Short channel” devices: <ul style="list-style-type: none"> • All interpretations of V^* are <i>approximations</i> • Except $V^* = 2 I_D / g_m$ (but $V^* \neq V_{dsat}$) |
|---|---|

Dynamic Figure of Merit

- **Unity current-gain bandwidth**

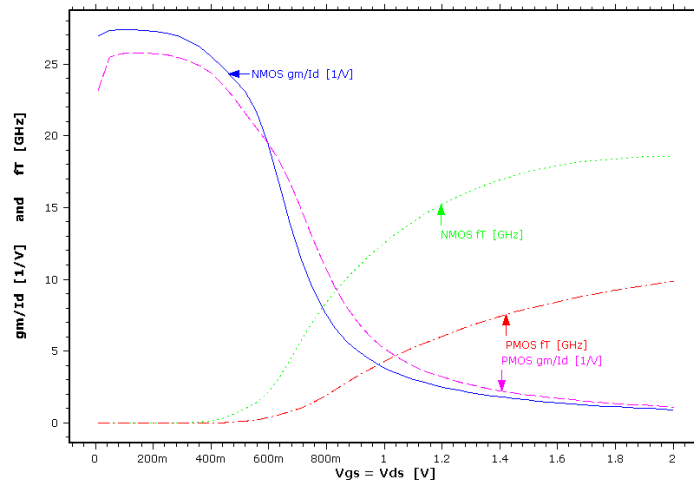
$$\omega_T = \frac{g_m}{C_{gs} + C_{gd}}$$

$$\omega_T = \frac{3\mu V_{od}}{2 L^2} = \frac{3}{2} \omega_0 \quad (\text{Long channel model, } C_{gd}=0)$$

- **For degenerate short channel device**

$$\omega_T = \frac{3\nu_{sat}}{2 L} = \frac{3}{2} \frac{1}{\tau_{sat}}$$

Efficiency g_m/I_D versus f_T

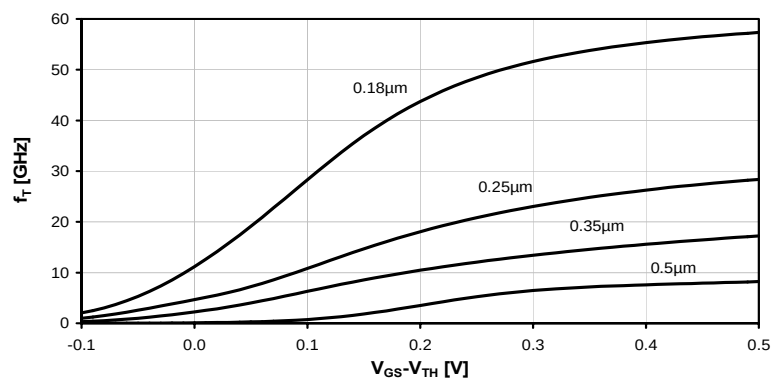


Speed-
Efficiency
Tradeoff

NMOS faster
than PMOS

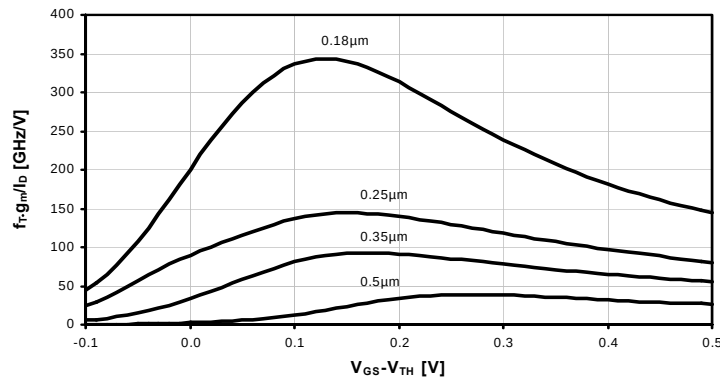
0.35u Process

Device Scaling



Short channel devices significantly faster

Composite Figure-of-Merit: $f_T \cdot g_m / I_D$

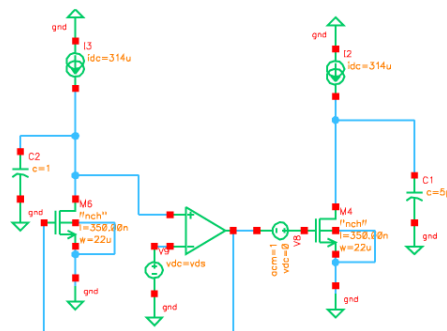


Peak performance for low $V_{GS} - V_{TH}$ (implies low V^*)

Design Example

Example: Common-source amp

$a_{v0} > 70$, $f_u = 100\text{MHz}$ for $C_L = 5\text{pF}$



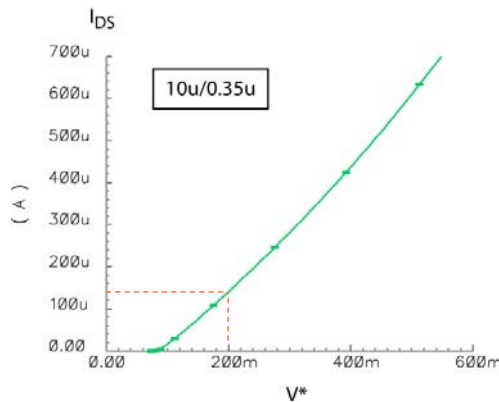
• $a_{v0} > 70 \rightarrow L = 0.35\mu\text{m}$

• $g_m \approx 2\pi f_u C_L = 3.14\text{mS}$

• High f_T (small C_{GS}): $V^* = 200\text{mV}$

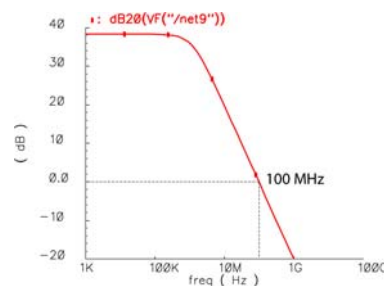
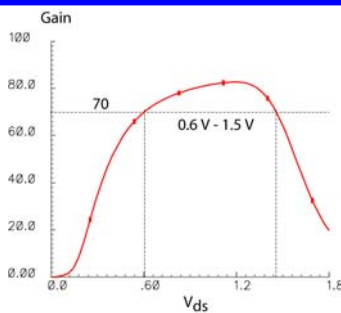
$$I_D = \frac{g_m V^*}{2} = 314\mu\text{A}$$

Device Sizing



- Pick L $0.35\mu\text{m}$
- Pick V^* 200mV
- Determine g_m 3.14mS
- $I_D = 0.5 g_m V^* = 314\mu\text{A}$
- W from graph (generate with SPICE)
 $\rightarrow W = 10\mu\text{m} (314\mu\text{A} / 141\mu\text{A}) = \underline{22\mu\text{m}}$
- Create these graphs for several device lengths

Common Source Verification

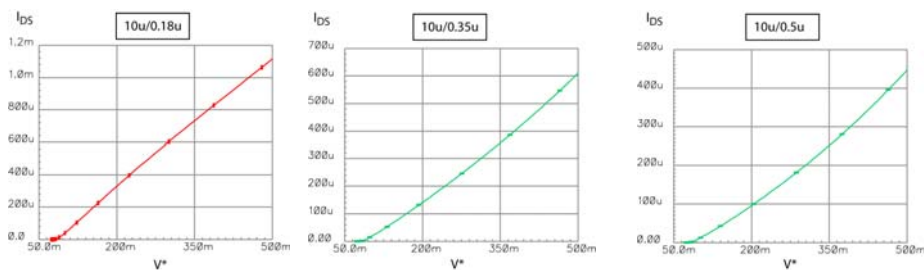


- Amplifier gain > 70
- Amplifier unity gain frequency is “dead on”
- Output range limited to 0.6 V – 1.5 V to maintain gain (about $\pm 0.45\text{V}$ swing)

Small Signal Design Summary

- Determine g_m (from design objectives)
- Pick L
 - Short channel $\rightarrow$ high f_T
 - Long channel $\rightarrow$ high r_o , a_{v0} , better matching
- Pick $V^* = 2I_D/g_m$ based on qualitative interpretation
 - Small $V^* \rightarrow$ large signal swing, high current efficiency
 - High $V^* \rightarrow$ high f_T
 - Also affects noise (see later)
- Determine I_D (from g_m and V^*)
- Determine W (SPICE / plot) $\leftarrow$ takes care of short channel effects, etc.
- Accurate for short channel devices $\rightarrow$ key for design

Device Sizing Chart



Generate these curves for a variety of L 's and device flavors (NMOS, PMOS, thin oxide, thick oxide, different V_{TH})

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Lecture 5: Electronic Noise



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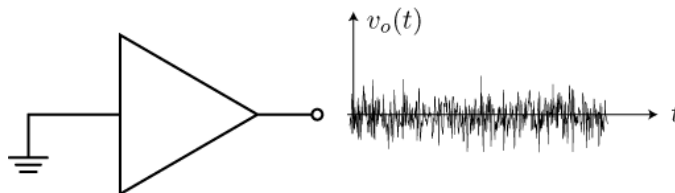
Electronic Noise

- **Why is noise important?**
 - Sets minimum signals we can deal with – often sets lower limit on power
- **Signal-to-noise ratio**
 - Signal Power $P_{\text{sig}} \sim (V_{\text{DD}})^2$
 - Noise Power $P_{\text{noise}} \sim k_B T / C$
 - $\text{SNR} = P_{\text{sig}} / P_{\text{noise}}$
- **Technology Scaling**
 - V_{DD} goes down $\rightarrow$ lower signal
 - Increase C to compensate $\rightarrow$ increases power

Types of “Noise”

- **Interference**
 - Not “fundamental” – deterministic
 - Signal coupling
 - Capacitive, inductive, substrate, etc.
 - Supply noise
- **Device noise**
 - Caused by discreteness of charge
 - “fundamental” – thermal noise
 - “manufacturing process related” – flicker noise

Noise in Amplifiers

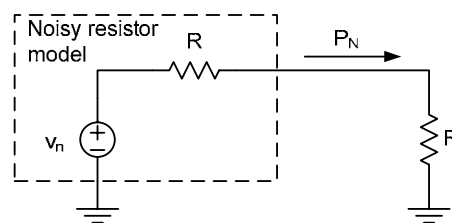


- **All amplifiers generate noise**
 - Comes from carrier random thermal motion and discreteness of charge
- **Noise is random**
 - Has to be treated statistically – can’t predict actual value
 - Deal with mean (average), variance, spectrum

Thermal Noise of a Resistor

- **Origin: Brownian Motion**
 - Thermally agitated particles
 - E.g. ink in water, electrons in a conductor
- **Available noise power:** $P_N = k_B T \Delta f$
 - Noise power in bandwidth Δf delivered to a matched load
 - Example: $\Delta f = 1\text{Hz} \rightarrow P_N = 4 \times 10^{-21}\text{W} = -174\text{ dBm}$
 - Reference: J.B. Johnson, "Thermal Agitation of Electricity in conductors," Phys. Rev., pp. 97-109, July 1928.

Resistor Noise Model



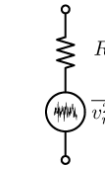
$$P_N = k_B T \Delta f = \frac{\overline{v_n^2}}{4R}$$

Mean square noise voltage:

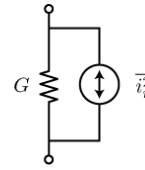
$$\overline{v_n^2} = 4k_B T R \Delta f$$

Thermal Noise

- Present in all dissipative elements
 - I.e., resistors
- Independent of DC current flow
- Random fluctuations of $v(t)$ or $i(t)$
 - Mean is 0
 - Distribution (pdf) is Gaussian
 - Power spectral density is “white”
 - Up to ~THz frequencies
 - $k_B T = 4 \times 10^{-21} \text{ J}$ ($T = 290\text{K} = 16.9^\circ\text{C}$)
- Example:
 - $R = 1\text{k}\Omega \rightarrow 4\text{nV}/\sqrt{\text{Hz}}$
 - 1MHz bandwidth $\rightarrow \sigma = 4\mu\text{V}$



$$\overline{v_n^2} = 4k_B T R B$$



$$\overline{i_n^2} = \frac{4k_B T B}{R}$$

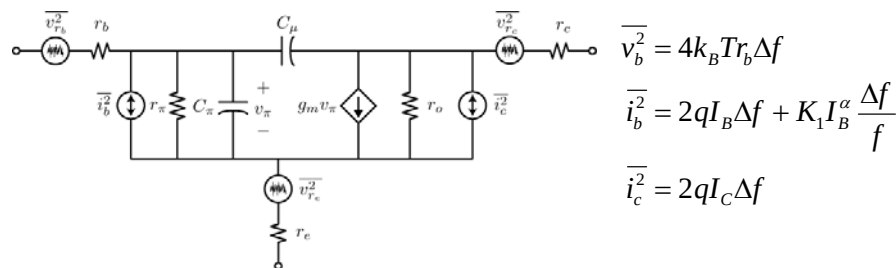
Noise of Passive Networks

- Capacitors and inductors only shape spectrum
- **Noise calculations**
 - **Instantaneous voltages add**
 - **Power spectral densities add**
 - **RMS voltages do NOT add**
- Example: $R_1 + R_2$ in series
- Generalization to arbitrary RLC networks

Noise in Diodes

- **Shot noise**
 - Zero mean, Gaussian pdf, white
 - Proportional to current
 - Independent of temperature
$$\overline{i_n^2} = 2qI_D\Delta f$$
- **Example:**
 $I_D = 1\text{mA} \rightarrow 17.9\text{pA/rt-Hz}$
 $1\text{MHz bandwidth} \rightarrow \sigma = 17.9\text{nA}$
- **Shot noise versus thermal noise**
 - $g_{\text{diode}} = I_D/(k_b T/q)$
 - Thermal noise density: $4k_b T g_{\text{diode}} = 4qI_D$
 - Shot noise half of this (current flow in 1 direction)

BJT Noise



- **Just like diodes: shot noise**
 - Collector and base noise partially correlated
- **Extrinsic resistors contribute noise**
 - Small signal resistors (e.g., r_o) don't
 - These aren't physical resistors

FET Noise

- Channel resistance contributes thermal noise
- Channel conductance:

$$g_{ds0} = \mu C_{ox} \frac{W}{L} (V_{GS} - V_{th}) = g_m$$

- Noise injection is actually distributed across the channel (note γ):

$$\overline{i_d^2} = 4kT\gamma g_m \Delta f$$

More Fundamental Expression

$$\overline{i_d^2} = 4kT \frac{\mu}{L^2} |Q_i| \Delta f$$

- More fundamental equation uses channel charge [Tsividis]
- When $V_{ds} = 0$, device is truly a resistor:

$$\overline{i_d^2} = 4kT \frac{W}{L} \mu C_{ox} (V_{GS} - V_T) \Delta f$$

Strong Inversion Noise

- In saturation, drain current noise is

$$\overline{i_d^2} = 4kT \frac{2W}{3L} \mu C_{ox} (V_{GS} - V_T) \Delta f$$

- For long channel model, can substitute g_m for the above factor.
- In practice, form involving actual inversion charge is more accurate
 - This is what SPICE/BSIM use

Weak Inversion

- Weak inversion: BJT $\rightarrow$ shot noise.
 - Result should be $\sim 2qI_{DS}$
- Get the same result from inversion charge expression:

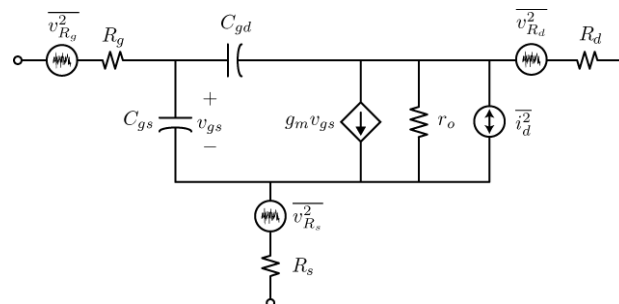
$$Q_i = WL \frac{Q_{i0} + Q_{iL}}{2} = \frac{L^2}{2\mu \frac{kT}{q}} I'_{DS} (1 + e^{-qV_{DS}/kT})$$

$$\overline{i_d^2} = 2qI'_{DS} (1 + e^{-qV_{DS}/kT}) \Delta f$$

Thermal Noise for Short Channels

- **Strong inversion → thermal noise**
 - Drain current: g_{ds0} is what you really care about
- $$\overline{i_d^2} = 4kT\gamma g_{ds0}\Delta f = 4kT\frac{\gamma}{\alpha}g_m\Delta f$$
- **g_m more convenient for input-referred noise**
 - For low field (long L), $\gamma = 2/3$ relates g_m to g_{ds}
 - For high field, use α to capture increase in noise
 - High-field noise can be 2-3 times larger than low field
- **MOS actually has intrinsic gate induced noise (142/242 topic)**
- **Gate leakage → shot noise**

FET Noise Model



- **Model neglects intrinsic gate noise**
- **BSIM3 does not directly include α**

1/f Noise

- **Flicker noise**
 - $K_{f,NMOS} = 2.0 \times 10^{-29} \text{ AF}$
 $K_{f,PMOS} = 3.5 \times 10^{-30} \text{ AF}$
 - Strongly process dependent

$$\overline{i_{1/f}^2} = \frac{K_f I_D \Delta f}{L^2 C_{ox} f}$$

- **Example:** $I_D = 10\mu\text{A}$, $L = 1\mu\text{m}$,
 $C_{ox} = 5.3\text{fF}/\mu\text{m}^2$, $f_{hi} = 1\text{MHz}$

$$\begin{array}{ll} f_{lo} = 1\text{Hz} & \rightarrow \sigma = 722\text{pA} \\ f_{lo} = 1/\text{year} & \rightarrow \sigma = 1083\text{pA} \end{array}$$

$$\overline{i_{1/f,total}^2} = \int_{f_{lo}}^{f_{hi}} \frac{K_f I_D}{L^2 C_{ox} f} df = \frac{K_f I_D}{L^2 C_{ox}} \ln \frac{f_{hi}}{f_{lo}}$$

1/f Noise Corner Frequency

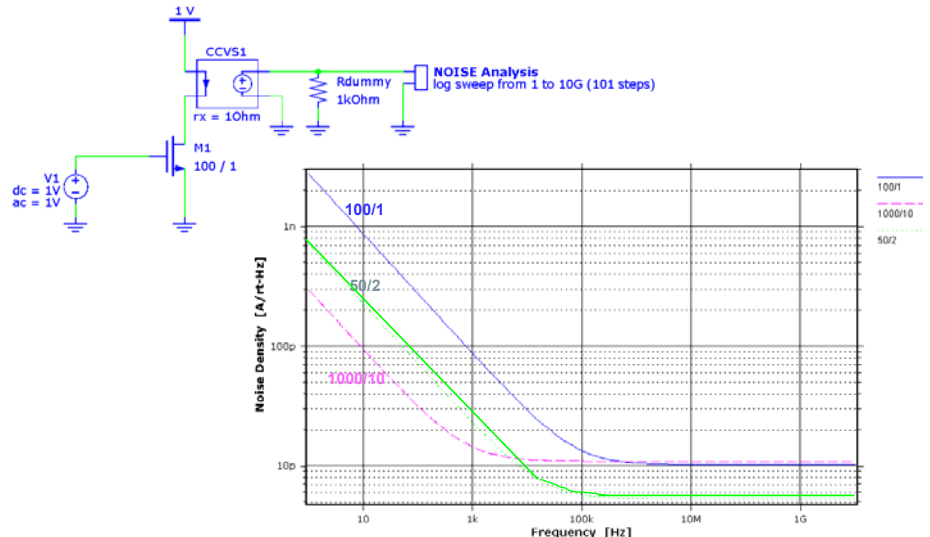
- **Definition (MOS)**

$$\begin{aligned} \frac{K_f I_D}{L^2 C_{ox}} \frac{\Delta f}{f_{co}} &= 4k_B T_r \gamma g_m \Delta f & f_{co} &= \frac{K_f I_D}{L^2 C_{ox}} \frac{1}{4k_B T_r \gamma g_m} \\ & & &= \frac{K_f}{4k_B T_r \gamma C_{ox}} \frac{1}{L^2} \frac{1}{g_m/I_D} \\ & & &= \frac{K_f}{8k_B T_r \gamma C_{ox}} \frac{V^*}{L^2} \end{aligned}$$

- **Example:**
 - $V^* = 200\text{mV}$, $\gamma = 1$

$L = 0.35\mu\text{m}$	$\rightarrow$	NMOS 192kHz	PMOS 34kHz
$L = 1.00\mu\text{m}$	$\rightarrow$	24kHz	4kHz

SPICE Noise Analysis



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Noise Calculations

• Method:

- 1) Create small-signal model
- 2) All inputs = 0 (linear superposition)
- 3) Pick output v_o or i_o
- 4) For each noise source v_x, i_x
Calculate $H_x(s) = v_o(s) / v_x(s)$ (... i_o, i_x)
- 5) Total noise at output is:

$$\overline{v_{on,T}^2(f)} = \sum_x |H_x(s)|_{s=2\pi jf}^2 \overline{v_x^2(f)}$$

simpler notation: $\overline{v_{on,T}^2(f)} = S_n(f)$

Tedious but simple ...

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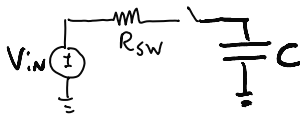
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Lecture 6: Noise Analysis Techniques



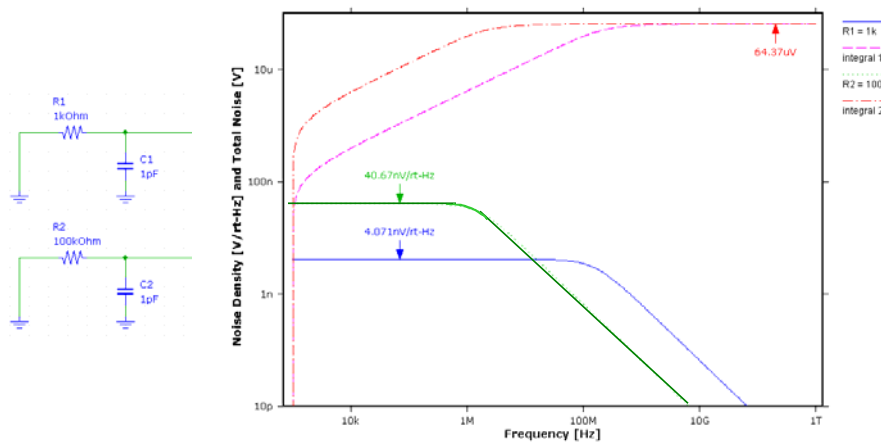
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Noise Variance in a Real Circuit: Sample and Hold



- Noise on the capacitor:
$$\overline{v_{on}^2(f)} = 4k_B T R \left| \frac{1}{1 + sRC} \right|^2$$
$$\rightarrow \overline{v_{oT}^2} = \int_0^{\infty} \overline{v_{on}^2(f)} df = \frac{k_B T}{C}$$
- So effective bandwidth is:
$$4k_B T R \Delta f = \frac{k_B T}{C}$$
$$\rightarrow \Delta f = \frac{1}{4RC} = \frac{\pi}{2} f_o$$

SPICE Verification



Energy-Based Analysis

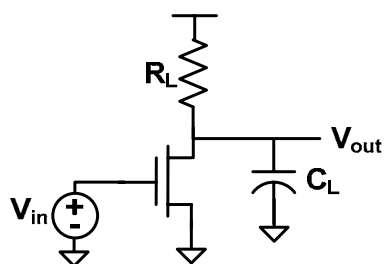
Useful Integrals

$$\int_0^{\infty} \left| \frac{1}{1 + \frac{s}{\omega_o}} \right|^2 df = \frac{\omega_o}{4}$$

$$\int_0^{\infty} \left| \frac{1}{1 + \frac{s}{\omega_o Q} + \frac{s^2}{\omega_o^2}} \right|^2 df = \int_0^{\infty} \left| \frac{\frac{s}{\omega_o}}{1 + \frac{s}{\omega_o Q} + \frac{s^2}{\omega_o^2}} \right|^2 df = \frac{\omega_o Q}{4}$$

$$\int_0^{\infty} \left| \frac{\frac{s}{\omega_z} + 1}{1 + \frac{s}{\omega_o Q} + \frac{s^2}{\omega_o^2}} \right|^2 df = \frac{\omega_o Q}{4} \left(\frac{\omega_o^2}{\omega_z^2} + 1 \right)$$

CS Amplifier



$$\begin{aligned} \overline{v_{on}^2(f)} &= 4k_B T \left(\frac{1}{R_L} + \frac{2}{3} g_m \right) \left| \frac{R_L}{1 + sR_L C_L} \right|^2 \\ \overline{v_{oT}^2} &= 4k_B T \left(\frac{1}{R_L} + \frac{2}{3} g_m \right) R_L^2 \int_0^{\infty} \left| \frac{1}{1 + sR_L C_L} \right|^2 df \\ &= 4k_B T \left(\frac{1}{R_L} + \frac{2}{3} g_m \right) R_L^2 \frac{1}{4R_L C_L} \\ &= \frac{k_B T}{C_L} \left(1 + \frac{2}{3} g_m R_L \right) \\ &= \frac{k_B T}{C_L} \left(1 + \frac{2}{3} |A_{vo}| \right) \\ &= \frac{k_B T}{C_L} n_F \end{aligned}$$

Signal-To-Noise Ratio

- **SNR:** $SNR = \frac{P_{sig}}{P_{noise}}$

- **Signal Power (sinusoidal source):**

$$P_{sig} = \frac{1}{2} V_{zero-peak}^2$$

- **Noise Power (assuming thermal noise dominates):**

$$P_{noise} = \frac{k_B T}{C} n_f$$

- **So:**

$$SNR = \frac{\frac{1}{2} C V_{zero-peak}^2}{n_f k_B T}$$

$$SNR \xrightarrow{\uparrow +6dB} C \xrightarrow{\uparrow \times 4}$$

dB versus Bits

- **Quantization “noise”**

- Quantizer step size: Δ

- Box-car pdf variance: $S_Q = \frac{\Delta^2}{12}$

N	dB
8	50
16	98
24	146

- **SNR of N-Bit sinusoidal signal**

- Signal power $P_{sig} = \frac{1}{2} \left(2^N \frac{\Delta}{2} \right)^2$

- SNR $SNR = \frac{P_{sig}}{S_Q} = 1.5 \times 2^{2N}$

- 6.02 dB per Bit $= [1.76 + 6.02N] \text{ dB}$

SNR versus Power

- 1 Bit $\rightarrow$ 6dB $\rightarrow$ 4x SNR
- 4x SNR $\rightarrow$ 4x C
- Circuit bandwidth $\sim g_m/C \rightarrow$ 4x g_m
- Keeping V^* constant $\rightarrow$ 4x I_D , 4x W
- Thermal noise limited circuit:
 - Each bit QUADRUPLES power!
- Overdesign is expensive
 - Better do the analysis right!

Analog Circuit Dynamic Range

- Biggest signal set by V_{DD} . So, for (single-ended) sinusoid:

$$V_{\max}(rms) = \frac{1}{\sqrt{2}} \frac{V_{DD}}{2}$$

- The noise is

$$V_n(rms) = \sqrt{n_f \frac{k_B T}{C}}$$

- So the dynamic range in dB is:

$$DR = \frac{V_{\max}(rms)}{V_n(rms)} = \frac{V_{DD} \sqrt{C}}{\sqrt{8n_f k_B T}} \quad [V/V]$$

$$= 20 \log_{10} \left(V_{DD} \sqrt{\frac{C}{n_f}} \right) + 75 \quad [\text{dB}] \text{ with } C \text{ in [pF]}$$

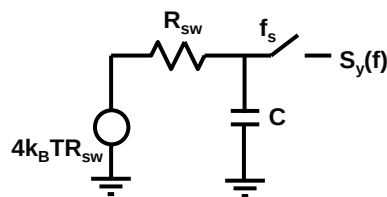
Analog Circuit Dynamic Range

- Biggest swing set by supply voltage V_{DD}

$$SNR = \frac{\frac{1}{2} C V_{DD}^2}{n_f k_B T}$$

- Modern ICs: $V_{DD} = \sim 1V$, $C < \sim 1nF$ ($n_f = 1$)
 - DR < 100dB (~16 Bits)
- PCB circuits with 30V and discrete C of ~100nF:
 - DR < 140dB (23 Bits)
 - A 40dB (~7 Bit) advantage!
- Note: can break this barrier with oversampling

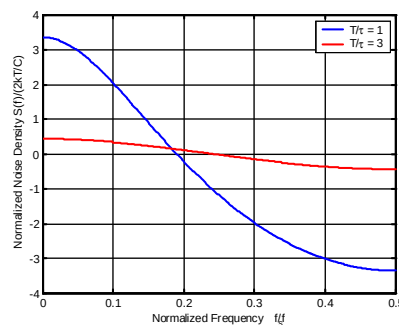
Sampled Noise Spectrum



$$S_y(f) = \frac{k_B T_r}{C} \frac{2}{f_s} \frac{1 - e^{-2a}}{1 + e^{-2a} (1 - \cos 2\pi f T)}$$

$$a = \frac{T}{R_{sw} C} = \frac{T}{\tau} \quad \text{and} \quad T = \frac{1}{f_s}$$

$$\int_0^{f_s/2} S_y(f) df = \frac{k_B T_r}{C}$$



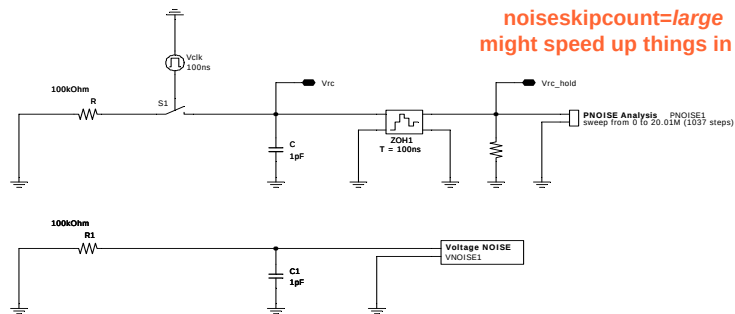
- What if RC doesn't completely settle every cycle?
 - Noise between samples correlated $\rightarrow$ spectrum not white
- If $T/\tau > 3$, correlation small
 - Sampled spectrum white
 - In practice usually the case

Periodic Noise Analysis

Sampling Noise from SC S/H

Netlist
ahd_include "zoh.def"

Netlist
simOptions options reitot=10u vabstol=1n labstol=1p



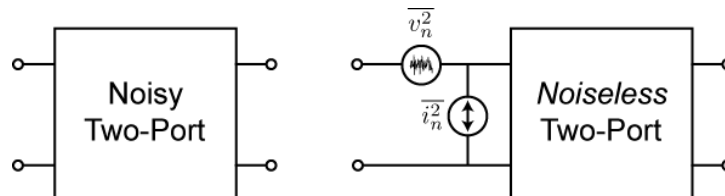
SpectreRF PNOISE: check
noisetype=timedomain
noisetimepoints=[...]
as alternative to ZOH.
noiseskipcount=large
might speed up things in this case.

PSS pss period=100n maxacfreq=1.5G errpreset=conservative
PNOISE (Vrc_hold 0) pnoise start=0 stop=20M lin=500 maxsideband=10

Two-Stage Amplifier

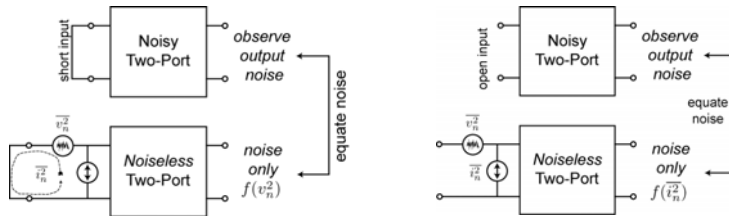
Input Equivalent Noise

Equivalent Noise Generators



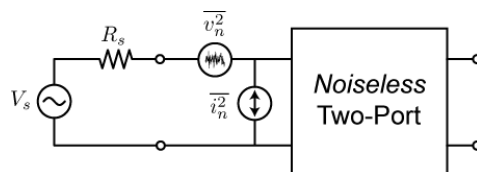
- **Model for noisy two-port:**
 - *Noiseless* two-port
 - Plus equivalent input noise sources
- In general, v_n and i_n are correlated.
 - Ignore that for now

Finding the Equivalent Generators



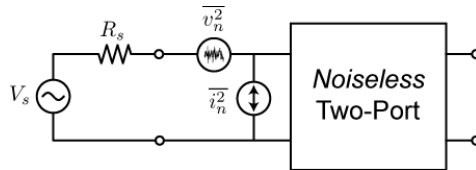
- Find v_n and i_n by opening and shorting the input
- Shorted input:
 - Output noise due only to v_n
- Open input:
 - Output noise due only to i_n

Role of Source Resistance



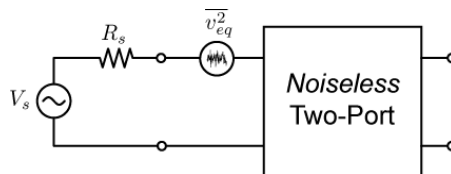
- If R_s is large:
 - Design amplifier with low i_n (MOS)
- If R_s is low:
 - Design amplifier with low v_n (BJT)
- For a given R_s , there is an optimal v_n/i_n ratio
 - Alternatively, for a given amp, there is an optimal R_s

Total Output Noise



$$\begin{aligned}\overline{v_o^2} &= (\overline{v_n^2} A_v^2 + \overline{v_{R_s}^2} A_v^2) \left(\frac{R_{in}}{R_{in} + R_s} \right)^2 + \left(\frac{R_{in}}{R_{in} + R_s} \right)^2 R_s^2 \overline{i_n^2} A_v^2 \\ &= (\overline{v_n^2} + \overline{i_n^2} R_s^2 + \overline{v_{R_s}^2}) \left(\frac{R_{in}}{R_{in} + R_s} \right)^2 A_v^2\end{aligned}$$

New Equivalent Generator



$$\overline{v_{eq}^2} = \overline{v_n^2} + \overline{i_n^2} R_s^2$$

- With known R_s , total noise can be lumped into one v_{eq}

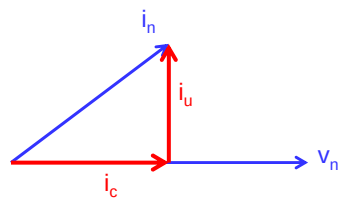
Optimum Source Impedance

- Can use this to optimize source impedance for minimum added noise from two-port (noise figure):

$$R_n \equiv \frac{v_n^2}{4kT\Delta f} \quad G_n \equiv \frac{i_n^2}{4kT\Delta f}$$

$$R_{opt} = \sqrt{\frac{R_n}{G_n}} = \sqrt{\frac{v_n^2}{i_n^2}}$$

Correlated Noise Sources



$$i_n = i_c + i_u$$

$$\langle i_u, v_n \rangle = 0$$

$$i_c = Y_C v_n$$

- Partition i_n into two components:
 - Correlated (“parallel”) to v_n
 - Uncorrelated (“perpendicular”) to v_n

Correlated Noise Sources (cont.)

Finding Y_c :

$$\begin{aligned}\overline{v_{eq}^2} &= \overline{v_n^2} + Z_s^2 \overline{i_n^2} \\ &= \overline{v_n^2} + Z_s^2 \overline{(i_c + i_u)^2} \\ &= \overline{v_n^2} + Z_s^2 \overline{i_c^2} + Z_s^2 \overline{i_u^2} \\ &= \overline{v_n^2} (1 + Z_s^2 Y_c^2) + Z_s^2 \overline{i_u^2}\end{aligned}$$

$$\begin{aligned}\overline{v_n^2} &= \alpha_1^2 \overline{v_1^2} + \alpha_2^2 \overline{i_2^2} \\ \overline{i_n^2} &= \underbrace{\beta_2^2 \overline{i_2^2}}_{\overline{i_c^2} = Y_c^2 \overline{v_n^2}} + \underbrace{\beta_3^2 \overline{v_3^2}}_{\overline{i_u^2}} \\ Y_c^2 &= \frac{\overline{i_c^2}}{\overline{v_n^2}} \\ &= \frac{\beta_2^2 \overline{i_2^2}}{\alpha_1^2 \overline{v_1^2} + \alpha_2^2 \overline{i_2^2}}\end{aligned}$$

Equivalent Noise Voltage (cor)

- Since the above expression is the sum of two uncorrelated noise voltages, we have

$$\overline{v_{eq}^2} = \overline{v_n^2} |1 + Y_C Z_S|^2 + |Z_S|^2 \overline{i_u^2}$$

- Now we can continue as before to find

$$B_{opt} = B_s = -B_c$$

$$G_{opt} = G_s = \sqrt{\frac{G_u}{R_n} + G_c^2}$$

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Lecture 7: Noise and Feedback



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Dept. of EECS

Noise and Feedback

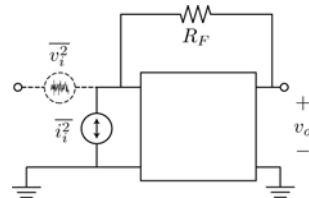
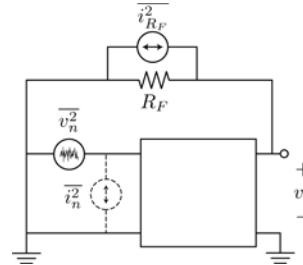
- **Ideal feedback:**
 - No increase of input referred noise
 - No decrease of SNR at output
- **Practical feedback: increased noise**
 - Noise from feedback network
 - Noise gain from elements outside feedback loop

Real Feedback

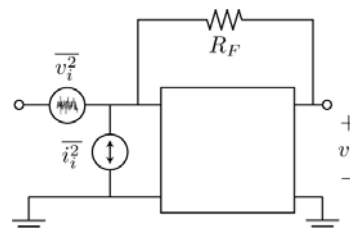
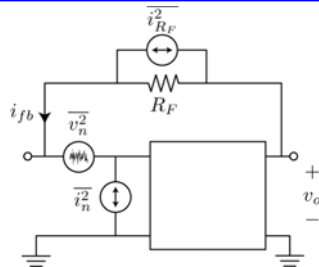
- Conceptually identical to standard two port calculations

- Use $R_s = 0$ to find $v_{i,eq}^2$
 - $R_s = \infty$ to find $i_{i,eq}^2$

- Calculations get tedious...

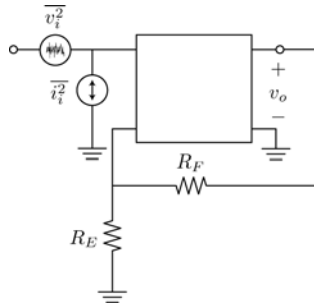


Practical Feedback Analysis



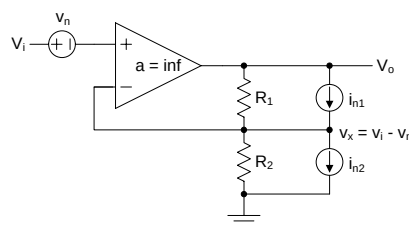
- Quick approximation method:**
 - Consider loading of feedback network on the input
 - Add a noise source associated with this element.
- Example: shunt feedback**
 - Loading at input is $R_F \rightarrow i_i^2 = i_n^2 + 4kT\Delta f/R_F$

Example #2: Series-Shunt Feedback



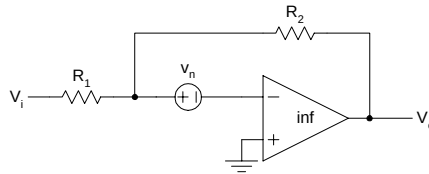
- Loading is $R_F || R_E$
- So, noise voltage becomes:
 - $v_i^2 = v_n^2 + 4kT(R_F || R_E)\Delta f$

Implications: Non-Inverting Amp



- Minimum power from feedback $\rightarrow$ large $R_1 + R_2$
- Example:
 - $A_v = 10$, $R_2 = 100k\Omega$, $R_1 = R_2(A_v - 1) = 900k\Omega$
 - $v_{nfb}^2 = 40nV/\sqrt{Hz}$ (very high)
- Only way to lower noise is increase power...

Example: Inverting Amplifier



- Ignoring noise from R_1 , R_2 :

$$v_o = -v_i \underbrace{\frac{R_2}{R_1}}_{-A_{v0}} + v_n \left(1 + \frac{R_2}{R_1} \right) = -v_i \underbrace{\frac{R_2}{R_1}}_{-A_{v0}} + v_n \frac{R_1 + R_2}{R_1}$$

$$\overline{v_{ieq}^2} = \overline{v_n^2} \left(\frac{R_1 + R_2}{R_1} \frac{R_1}{R_2} \right)^2 = \overline{v_n^2} \left(\frac{R_1 + R_2}{R_2} \right)^2 = \overline{v_n^2} \left(1 + \frac{1}{|A_{v0}|} \right)^2$$

- “Ideal” feedback, why is $v_{ieq}^2 > v_n^2$?

Example

Example

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Lecture 8: Current Sources



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Bias Current Sources

- What makes a current source a current source?
 - High output impedance
- Other important properties:
 - Voltage range ($V_{\min}$)
 - Noise
 - Accuracy
- Techniques: cascoding, gain boosting

Bias Current Source

- Is this a “good” bias current source?

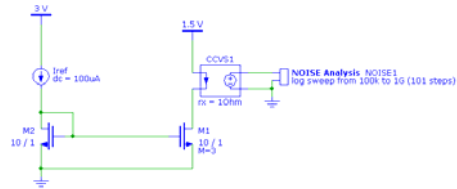
Current Mirror

- Better approach: current mirror

Noise

$$\begin{aligned}\overline{i_{on}^2} &= \overline{i_{d1}^2} + M^2 \overline{i_{d2}^2} \\ &= 4k_B T \gamma (g_{m1} + M^2 g_{m2}) \Delta f \\ &= 4k_B T \gamma g_{m1} (1 + M) \Delta f \\ &= 4k_B T \frac{1}{R_N} \Delta f\end{aligned}$$

$$\begin{aligned}R_N &= \frac{1}{g_{m1}} \frac{\gamma^{-1}}{1 + M} \\ &= \frac{r_o}{a_{v0}} \frac{\gamma^{-1}}{1 + M} \ll R_o = r_o\end{aligned}$$



- **M2 adds noise**
 - Choose small M (power), or
 - Filter at gate of M1
- **Current source FOMs**
 - Output resistance R_o
 - Noise resistance R_N
 - **Active sources boost R_o , not R_N**

Noise cont'd

- I_p^2 from transistor current source much larger than real R with same output impedance
- So why do we use transistors as current sources?

V_{min} versus Noise

- Voltage required for large r_o ("saturation"): $V_{\min} \sim V^*$ $V_{\min} = k \times V^*$ typ. $k = 1 \dots 2$
- Minimum noise (for given I_D):
 → large R_N
 → large V^* (and, hence, $V_{\min}$)

$$R_N = \frac{1}{\gamma g_{m1}} \frac{1}{1+M}$$

$$= \frac{V_{\min}}{2\gamma k I_D} \frac{1}{1+M}$$
- Eats into signal swing...

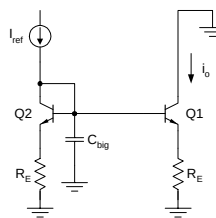
Bipolar's, GaAs, ...

$$\overline{i_{on}^2} = \underbrace{\overline{i_{cn}^2} \left| \frac{1}{1+g_m R_E} \right|^2}_{\text{BJT}} + \underbrace{\overline{i_{Rn}^2} \left| \frac{g_m R_E}{1+g_m R_E} \right|^2}_{R_E} \Delta f \quad (\overline{i_b^2} = 0)$$

a) $g_m R_E = 0$ $\overline{i_{on}^2} = 2k_B T g_m \Delta f$
 $R_N = \frac{2}{g_m} = \frac{2V_T}{I_C}$ set by I_C

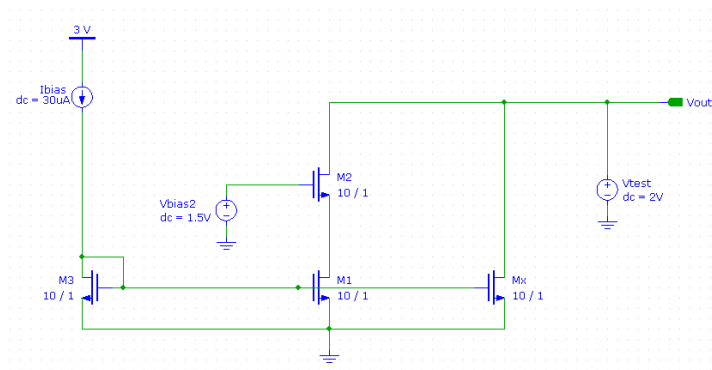
b) $g_m R_E \gg 1$ $\overline{i_{on}^2} = 4k_B T \frac{1}{R_E} \Delta f$
 $R_N = R_E = \frac{V_{\min}}{I_C} \frac{V_{\min} - V_{ce}^{sat}}{V_{\min}}$

compare $R_{N,MOS} = \frac{V_{\min}}{I_D} \frac{1}{2\gamma k}$

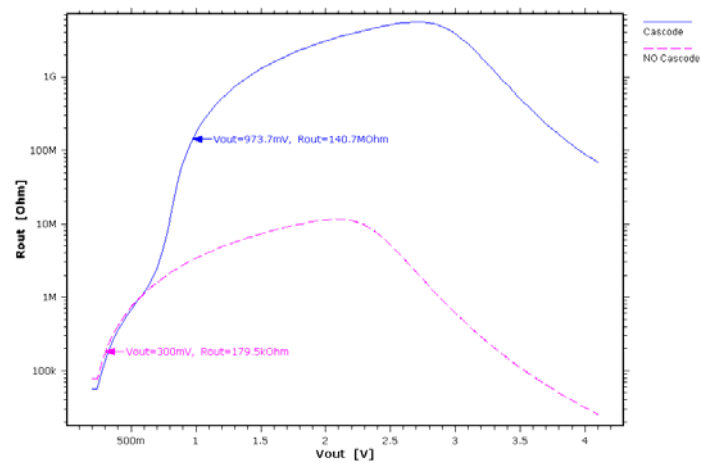


- Increasing R_E lowers noise
 - Same in MOS, BJT, etc.
 - $V_{\min}$ always trades with noise
- Lowest possible noise: resistor (large $V_{\min}$)

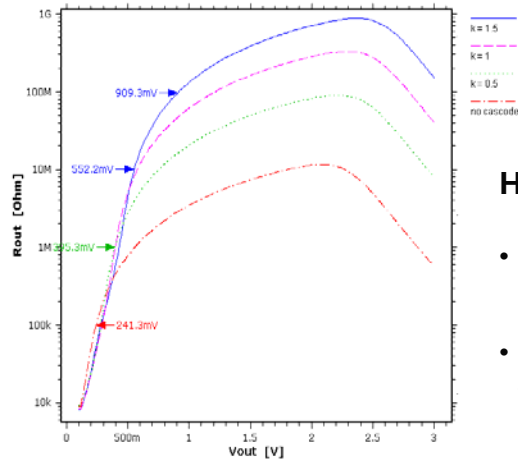
Cascoding



Output Resistance



$$R_{out} = f(k)$$



$$V_{DS1} = kV_1^*$$

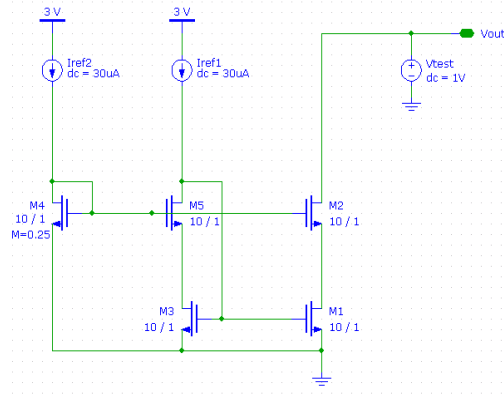
How to choose k?

- Large k useful only for large V_{min}
- But, little penalty for large k and small V_{min}
 - Typically choose $k>1$
 - Get benefit if V_{ds} is big

High-Swing Cascode Biasing

- Need circuit for generating V_{bias2}
- Accuracy important for high V_{ds} /high R_o
 - In practice, not quite as critical (V_{ds} often low)
- Assume you've seen these before
 - Review G & M if not

High-Swing Bias Example



Gain Boosting

- Use feedback to further increase R_{out}
 - No increase of V_{min} (unlike double cascode)

Local Feedback and Stability

Gain-Boosted Z_{out}

Pole-Zero Doublets

If it works, do it again!

2144

IEEE JOURNAL OF SOLID-STATE CIRCUITS, VOL. 39, NO. 12, DECEMBER 2004

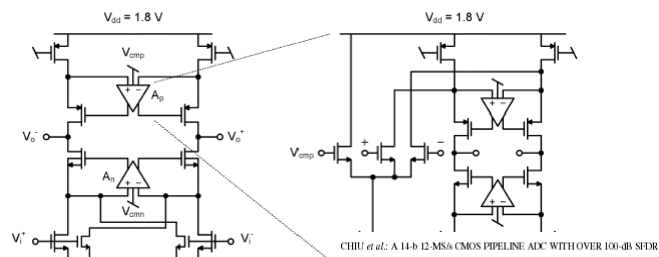
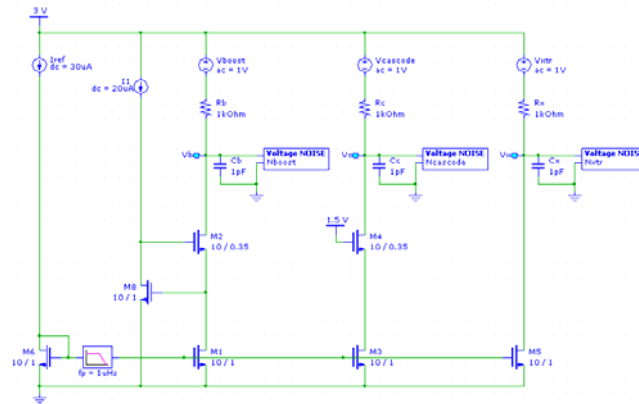


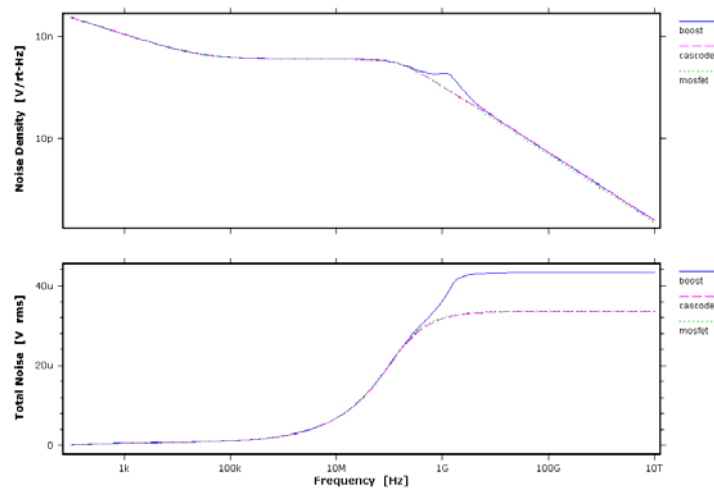
Fig. 8. Nested CMOS gain-boosting technique.

- Since in advanced scaled CMOS $g_m r_o$ is small, we can use nested gain boosting for higher output impedance.
- Watch out for pole-zero doublets!

Noise Analysis



Noise Summary



Cascode Sizing

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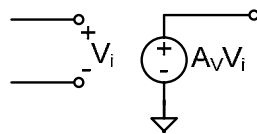
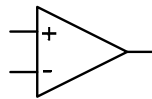
Lecture 9: Amplifiers



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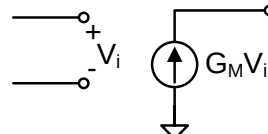
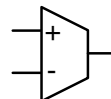
OpAmps and OTAs

OpAmp



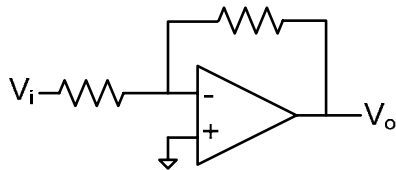
- High voltage gain, high input impedance
- Voltage source output (low impedance)

OTA

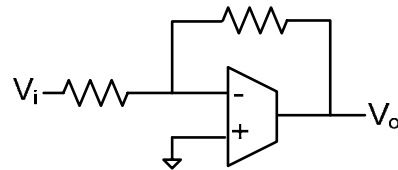


- High “voltage” gain, high input impedance
- Current source output (high impedance)

Resistive Feedback

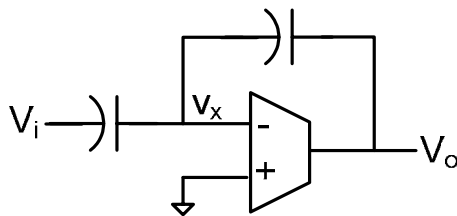


- Open-loop gain: ∞
- (Independent of R_f)



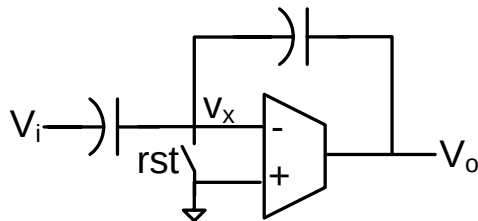
- Open-loop gain: $G_m R_f$
 - Feedback loads the OTA
- How about large R_f ?
 - Lots of area, parasitic poles
- Need a different solution...

How about Capacitive Feedback?



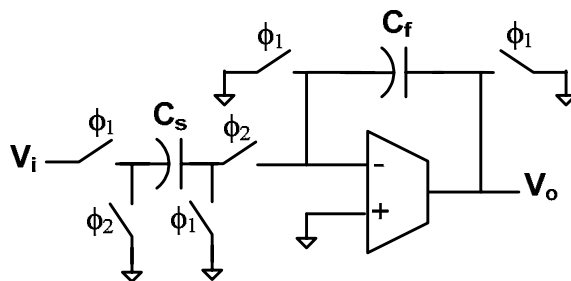
- At low frequency:
 - No loading from feedback network ($|Z_f| = \infty$)
- Gain drops at high frequency
 - But this happens in all amplifiers
- Does this really work?
 - Hint: what happens if you simulate this in SPICE?

Capacitive Feedback cont'd



- Charge on v_x is undefined – needs to be reset to known value
- Can we just do this once at start-up?
 - Depends how long you want to use the amplifier...
- Usually do this “reset” every cycle
 - Why each cycle instead of only once every N cycles?

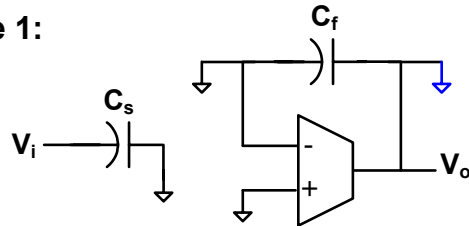
Switched-Capacitor Gain Stage



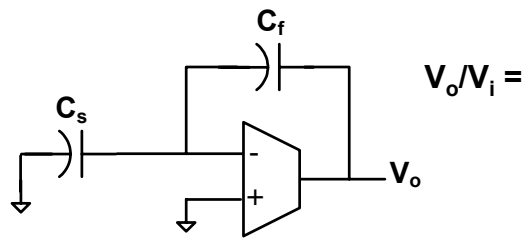
- Many possible topologies – one example shown here
- Clocks generally non-overlapping

SC Gain Stage Phases

- Phase 1:

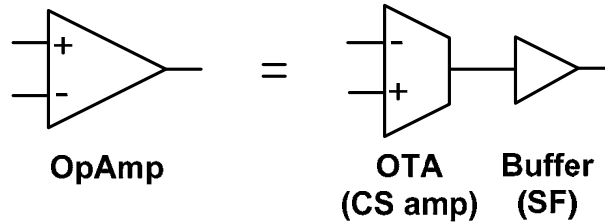


- Phase 2:

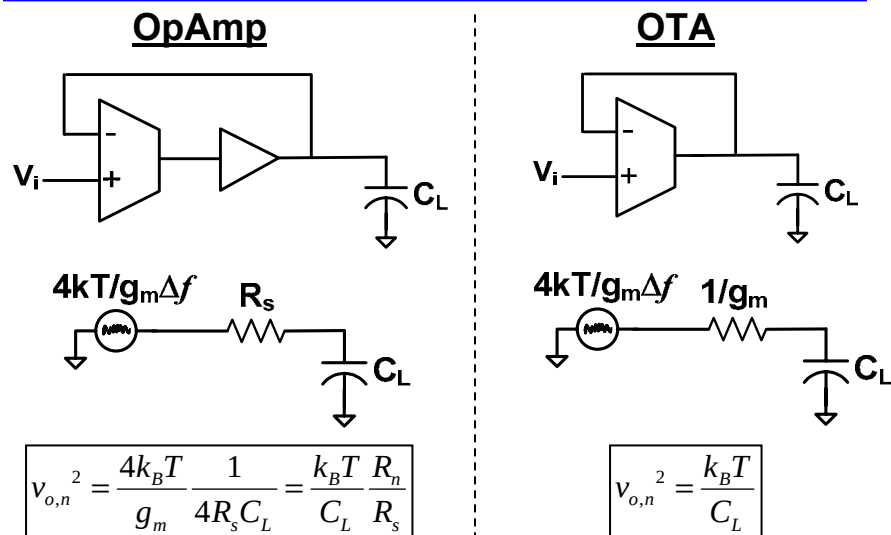


SC Gain Stage Waveforms

Opamp vs. OTA Revisited



Opamp vs. OTA Noise



SC Gain Stage Noise

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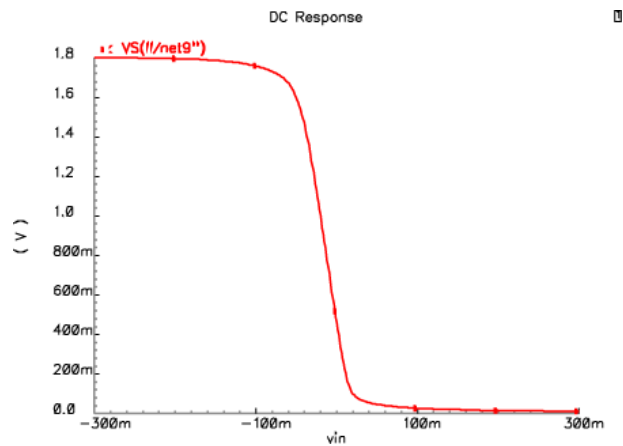
Lecture 10: Single-Ended and Differential OTAs



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Simplest Single-Ended OTA

DC Input/Output, Gain



Gain, Output Range

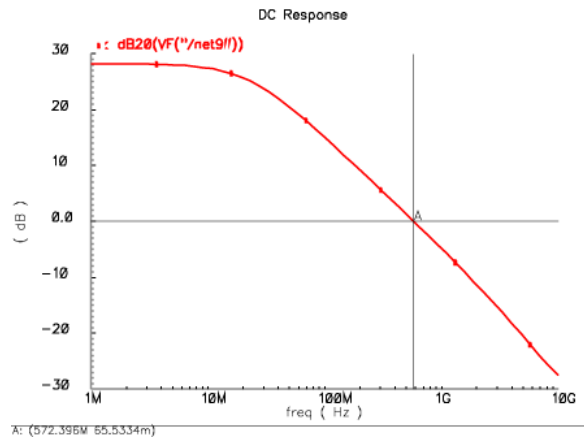
- Small Signal:

$$a_{vo} = \frac{dV_{out}}{dV_{in}}$$

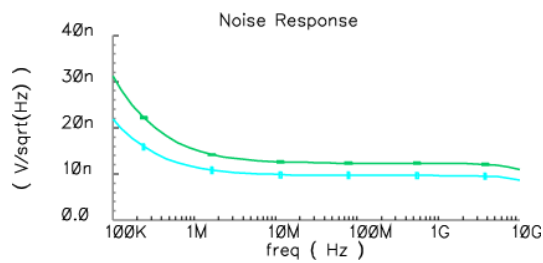
- Large Signal:

$$A_{vo} = \frac{V_{out} - V_{out_o}}{V_{in} - V_{in_o}}$$

Frequency Response

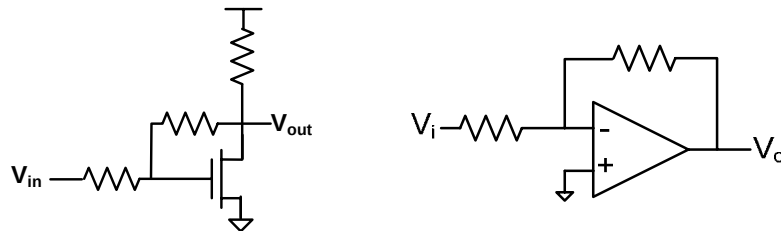


Noise



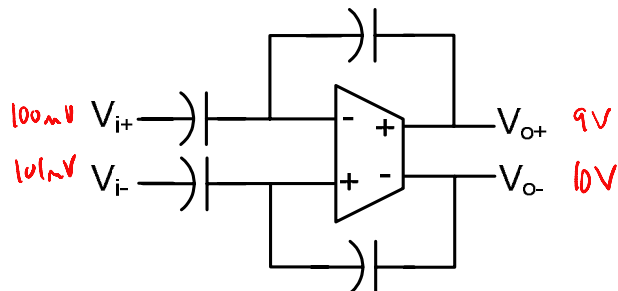
$$\begin{aligned} \frac{\overline{v_{ieq}^2}}{\Delta f} &= 4k_B T \gamma \frac{1}{g_{m1}^2} (g_{m1} + g_{m2}) \\ &= 4k_B T \gamma \frac{1}{g_{m1}} \left(1 + \frac{g_{m2}}{g_{m1}} \right) \\ &= 4k_B T \gamma \frac{1}{g_{m1}} \underbrace{\left(1 + \frac{V_1^*}{V_2^*} \right)}_{n_f} \end{aligned}$$

Differential Input?



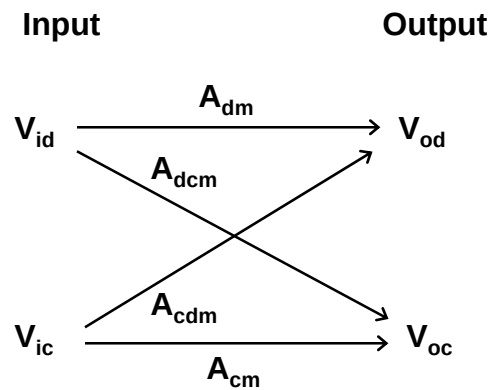
- **Why use a differential input?**
 - Diff. version has extra device(s) – more power, noise, etc.
- **Real reason is systematic offset**
 - All voltages are relative
- **Inherent asymmetry to get single-ended V_{out}**
 - “common-mode” sensitivity

Fully Differential Circuits



- **Fully differential circuits: complete symmetry**
 - $V_{id} = V_{i+} - V_{i-}$ $V_{ic} = (V_{i+} + V_{i-})/2$
 - $V_{od} = V_{o+} - V_{o-}$ $V_{oc} = (V_{o+} + V_{o-})/2$
- **Still need to be careful with common mode**

Fully Differential Amplifier Gains



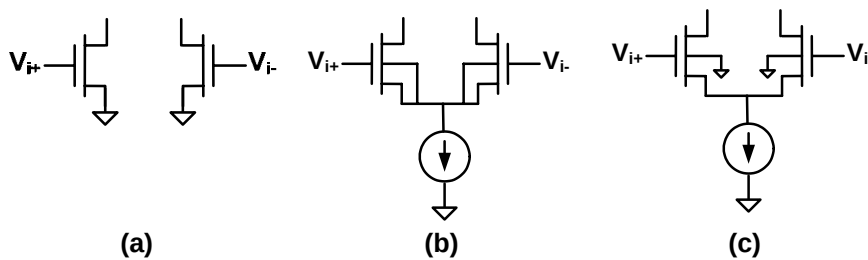
PSRR, CMRR, ...

$$\begin{aligned}
 A_{dm} &= \frac{v_{od}}{v_{id}} \rightarrow \infty & A_{VDD} &= \frac{v_{od}}{v_{DD}} \rightarrow 0 & CMRR &= \left| \frac{A_{dm}}{A_{cdm}} \right| \rightarrow \infty \\
 A_{cm} &= \frac{v_{oc}}{v_{ic}} \rightarrow 0 & A_{VSS} &= \frac{v_{od}}{v_{SS}} \rightarrow 0 & PSRR_{VDD} &= \left| \frac{A_{dm}}{A_{VDD}} \right| \rightarrow \infty \\
 A_{cdm} &= \frac{v_{od}}{v_{ic}} \rightarrow 0 & & & PSRR_{VSS} &= \left| \frac{A_{dm}}{A_{VSS}} \right| \rightarrow \infty
 \end{aligned}$$

- All “terminals” are inputs
 - May not be a node in the circuit – could be e.g. temperature
- Typical metrics: CMRR, PSRR
 - Careful with how you use these

CMRR Example

Differential Input Stage Options



PSRR Example

Baluns

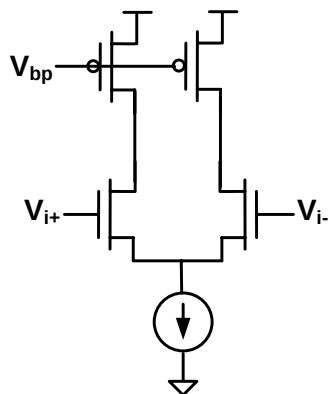
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Lecture 11: Folded Cascode OTA

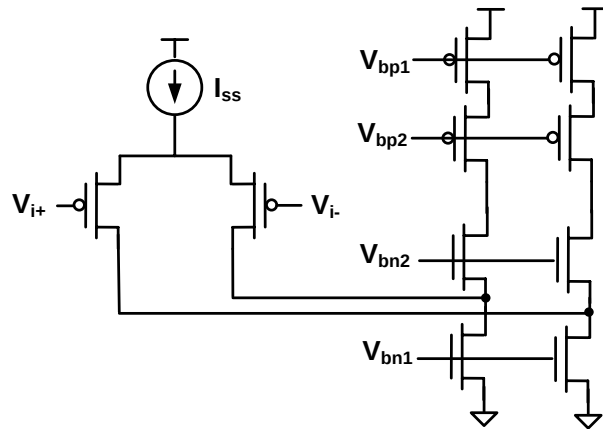


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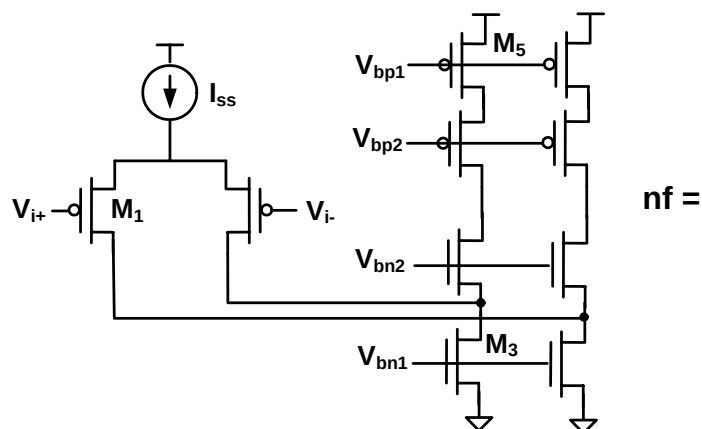
Simple Diff. Input OTA



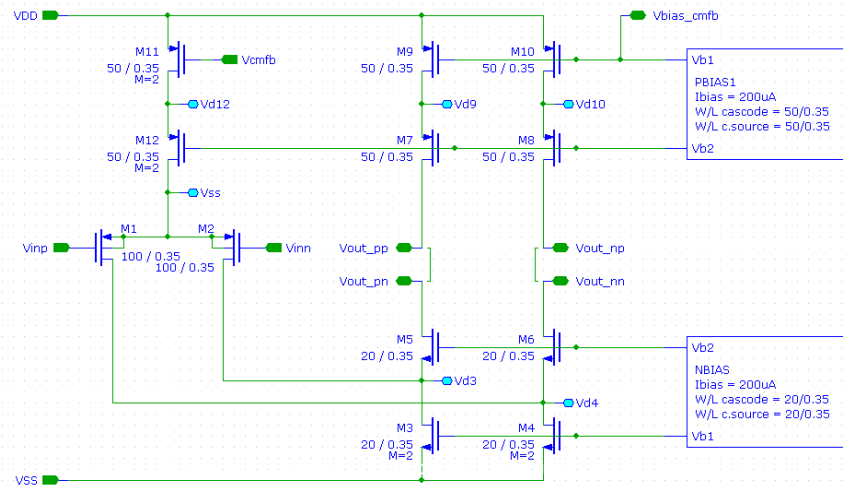
Folded-Cascode Schematic



Folded-Cascode Noise



Simulation Schematic

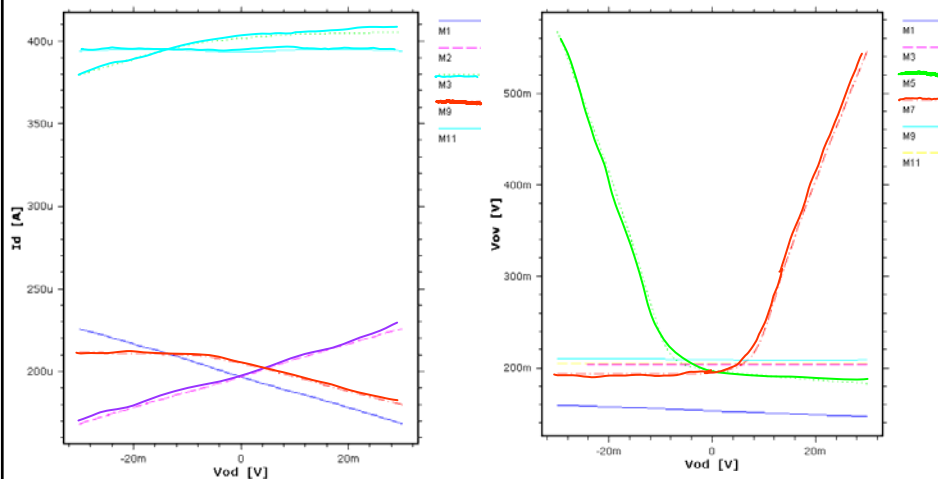


EECS240

Lecture 11

5

Bias, V^*

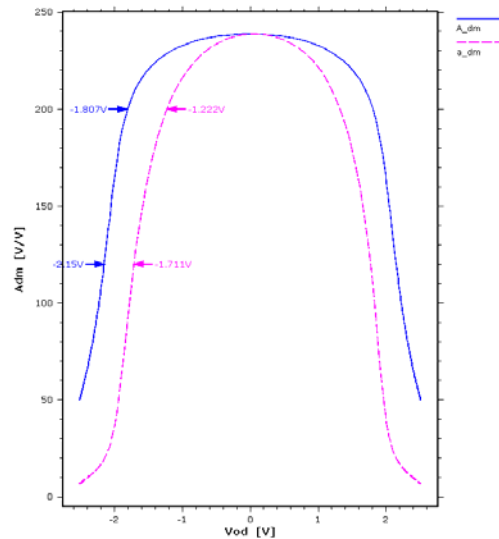


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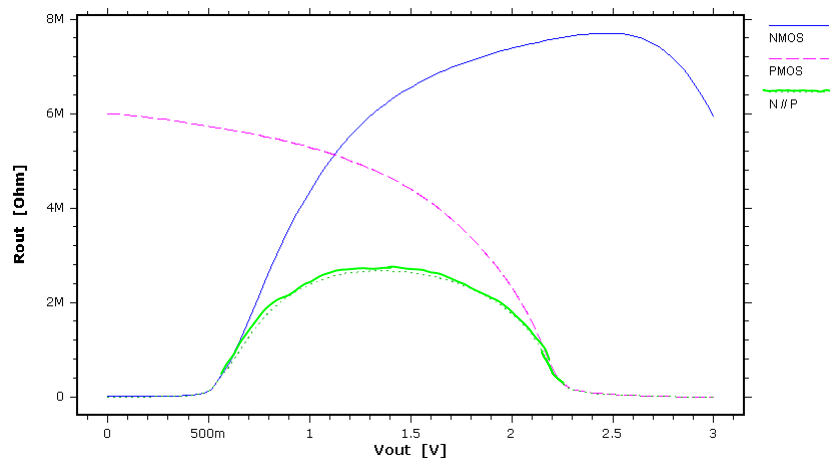
Lecture 11

6

Low Frequency Gain, A_{dm}

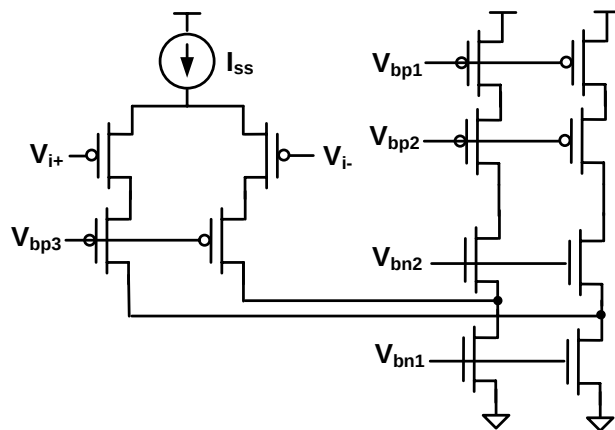


Output Resistance

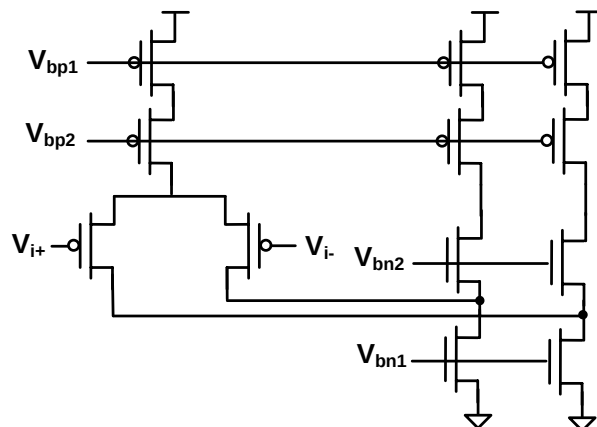


Beware of r_o imbalance between NMOS and PMOS current sources

Input Cascode



Biasing and Parasitic Feedback



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Lecture 12: Feedback

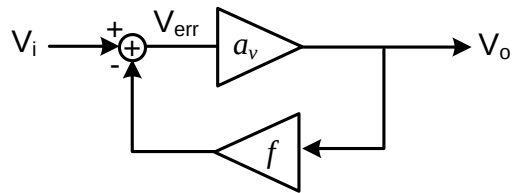


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Feedback

- **Assume you are familiar with feedback benefits, issues**
 - Review: G&M Ch. 8 & 9, Razavi Ch. 8
- **Focus here on:**
 - **Stability**
 - Analysis and simulation
 - **Settling**
 - Often amplifying pulses and not sinusoids
 - More next lecture

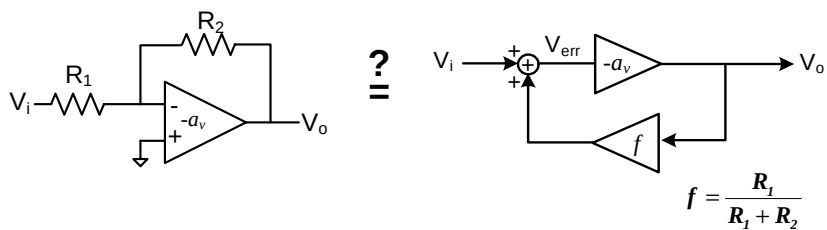
Generic Feedback Circuit



- Open-loop gain: a_v
- Feedback factor: f
- Loop gain: $T = a_v f$
- Closed-loop gain: $A = \frac{V_o}{V_i} = \frac{a_v}{1+T} = \frac{1}{f} \frac{1}{1+\frac{1}{T}} \approx \frac{1}{f}$

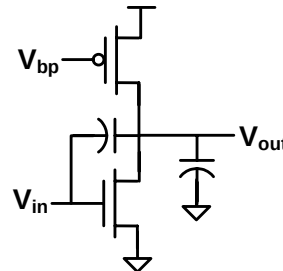
Electronic Feedback Circuit

- Careful with mapping circuit feedback to generic diagram...



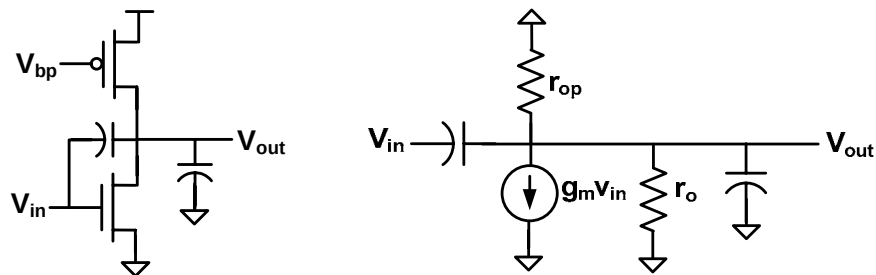
Stability

- Nearly all circuits are actually non-linear and time-varying
 - “Poles” only accurate for given bias, temp., etc.



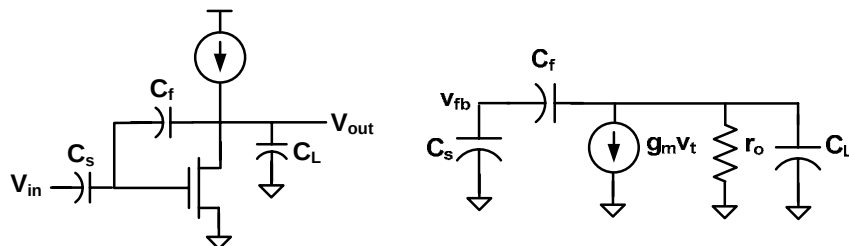
- What we usually mean by stability:
 - Circuit always converges to the “origin” for zero input within finite time
 - (Exponential stability)
 - Another common definition: BIBO stability

Stability In Practice



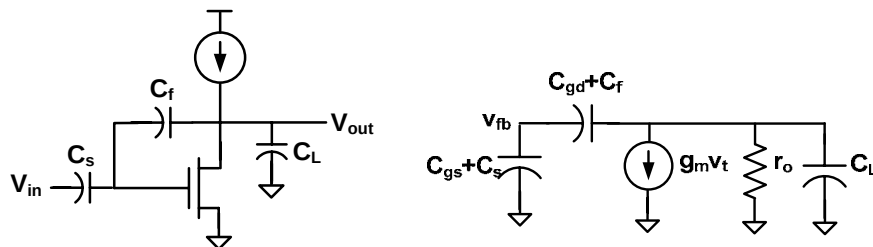
- Linearize the circuit and look at its poles
- Remember: this is only an approximation!
 - Perform linear analysis over several corners, temps, supplies, etc.
 - May want to do a couple of transient sims too

Linear Circuit Stability



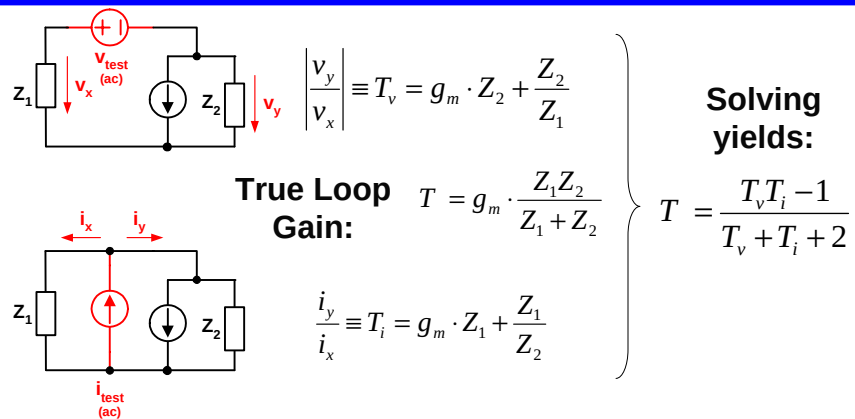
- Stability set by $T(s)$
- $T(s)$ is an open-loop parameter - need to break the loop
 - Easy to do in hand analysis: break at controlled source
 - Not as easy in simulation...

Simulating Stability



Common Approach

Middlebrook Method (1975)



• Measure T_v and T_i , then calculate actual T

Phase Margin

- **Approximate method to evaluate stability: phase margin**
- **Works well for most circuits of interest**
 - Sometimes have to use Nyquist stability test

Multi-Loop Feedback

Multi-Loop Feedback

Multi-Loop Feedback Example

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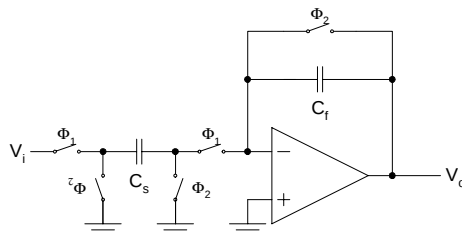
Lecture 13: Settling



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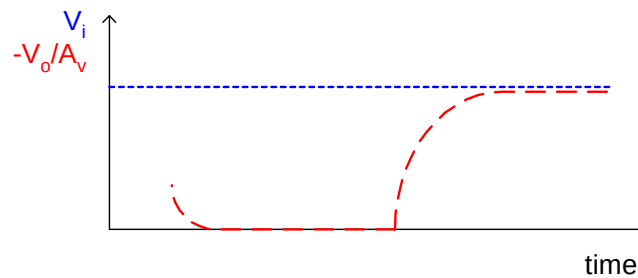
Settling

- Why interested in settling?
 - Oscilloscope: track input waveform without ringing
 - ADC (switched-cap amplifier): gain a signal up by a precise amount within T_{sample}

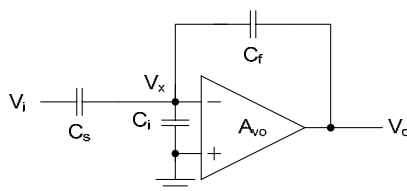


Step Response

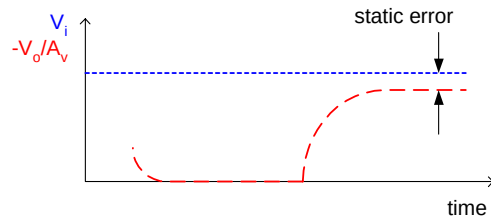
- Two types of settling “errors”:
 - Static
 - Finite gain, capacitor mismatch
 - Dynamic
 - Takes time to reach final value



Static Error



$$\frac{V_o}{V_i} = -\frac{c}{1 + \underbrace{\frac{1}{FA_{vo}}}_{T_o}}$$



$$F = \frac{C_f}{C_f + C_s + C_i}$$

$$c = \frac{C_s}{C_f}$$

Static Error (cont.)

$$\frac{V_o}{V_i} = -\frac{c}{1 + \frac{1}{FA_{vo}}} \approx -c \left(1 - \underbrace{\frac{1}{FA_{vo}}}_{\text{relative error}} \right)$$

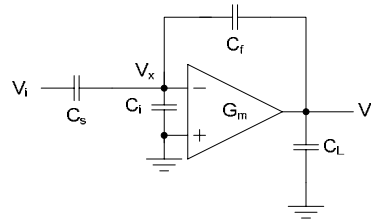
- **Example:**
 - Closed loop gain: $c = -4$, $C_f = 1\text{pF}$, $C_s = 4\text{pF}$, $C_i = 1\text{pF}$
 - $F = 1/6$ (C_i hurts!)
- **Error specification: $<0.1\%$**
 - $FA_{vo} > 1000$
 - $A_{vo} > 6000$ over output range

Dynamic Errors

- **Many possible dynamic effects that impact settling error:**
 - Finite bandwidth
 - Feedforward zero
 - Non-dominant poles
 - Doublets
 - Slewing
- **Approximate analysis approach:**
 - Decompose each error source, isolate interactions
 - Add all errors together

Single Time Constant Linear Settling

- For dynamic settling (and for $T_0 \gg 1$), can generally ignore r_o



$$\frac{V_o}{V_i} = -c \frac{1 - s \frac{C_f}{G_m}}{1 + s \frac{C_L + (1-F)C_f}{FG_m}}$$

Time Domain Step Response

Frequency domain:

Time domain:

$$V_{o,step} = -c \frac{1 + s/z}{1 + s/p} \frac{V_{step}}{s}$$

Time Domain Step Response

Frequency domain:

input step:

$$V_{i,step} = \frac{V_{step}}{s}$$

output step:

$$\begin{aligned} V_{o,step} &= -c \frac{1+s/z}{1+s/p} V_{i,step} \\ &= -c \frac{1+s/z}{1+s/p} \frac{V_{step}}{s} \end{aligned}$$

Time domain:

(inverse Laplace transform)

$$v_{o,step}(t) = \underbrace{-V_{step}c}_{\text{ideal response}} \left[1 - \underbrace{\left(1 - \frac{p}{z} \right)}_{\substack{\text{initial error} \\ \text{(feedforward)}}} e^{-pt} \right]_{\text{exponentially decaying error}}$$

Case 1: $|p/z| \ll 1$

$$v_{o,step}(t) \cong \underbrace{-V_{step}c}_{\text{ideal response}} \left[1 - e^{-t/\tau} \right]$$

Relative settling error:

$$\varepsilon = \frac{v_o(t \rightarrow \infty) - v_o(t = t_s)}{v_o(t \rightarrow \infty)} = e^{-t_s/\tau}$$

$$\frac{t_s}{\tau} = -\ln \varepsilon$$

- **Easiest number to remember: 2.3τ per decade**
- **Example: 1% settling, 4.6ns clock cycle: $\tau = 1\text{ns}$**
- **$C_{L,eff}$ usually set by noise – use settling to determine required g_m**

Case 2: $|p/z|$ not negligible

$$v_{o,step}(t) \cong \underbrace{-V_{step}C}_{\text{ideal response}} \left[1 - \left(1 - \frac{p}{z} \right) e^{-t/\tau} \right]$$

Relative settling error:

$$\varepsilon = \frac{v_o(t \rightarrow \infty) - v_o(t = t_s)}{v_o(t \rightarrow \infty)} = \left(1 - \frac{p}{z} \right) e^{-t_s/\tau}$$

$$\frac{t_s}{\tau} = -\ln \left(\frac{\varepsilon}{1 + F C_f / C_{Leff}} \right)$$

• **Example:**

- $c = 0.25$, $C_f = 1\text{pF}$, $C_s = 250\text{fF}$, $C_i = 250\text{fF}$, $C_L = 1\text{pF}$
- $F = 0.67$, $C_{L,eff} = 1.33\text{pF}$

• $\varepsilon = 0.1\%$:

- t_s (no feedforward) = 6.9τ
- t_s (with feedforward) = $-\ln[1e-3/(1+0.67*0.75)]=7.3\tau$

Non-Dominant Pole

- **Ignore feed-forward zero for simplicity**
 - (Just increases final swing by $1 + FC_f/C_{L,eff}$)

$$H(s) = \frac{V_o}{V_{in}} = -c \frac{1}{1 + s \frac{C_{Leff}}{FG_m}}$$

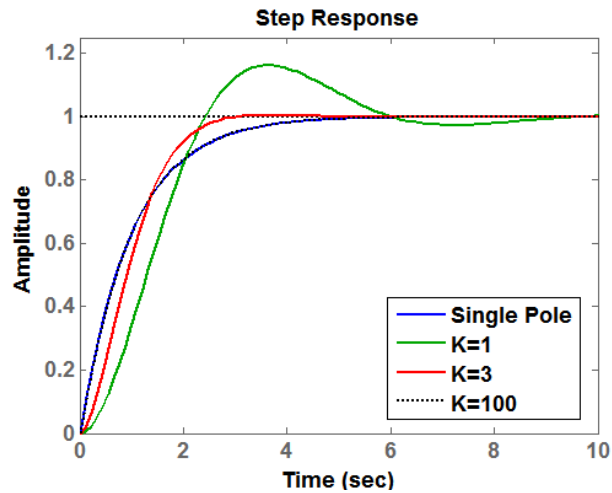
- **Model for non-dominant pole:**

$$G_m(s) = \frac{G_{m0}}{1 + s/p_2}$$

$$p_2 = K\omega_u$$

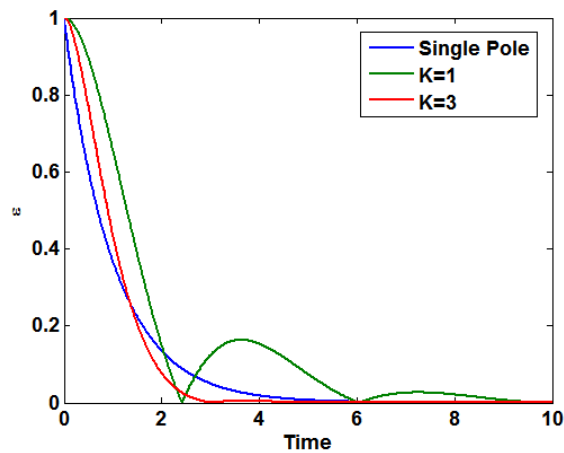
ω_u is unity gain bandwidth of $T(s)$

Step Response



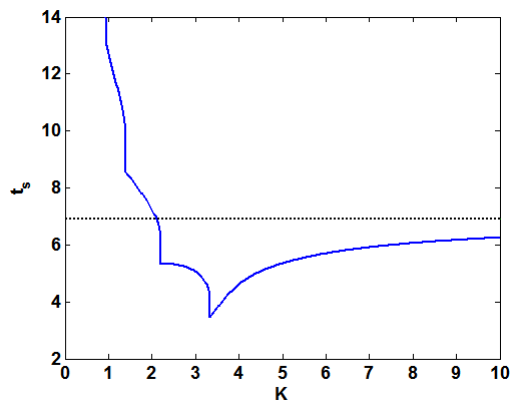
Non-Dominant Pole (cont.)

Relative error : $\varepsilon = |1 - v_s(t, K)|$



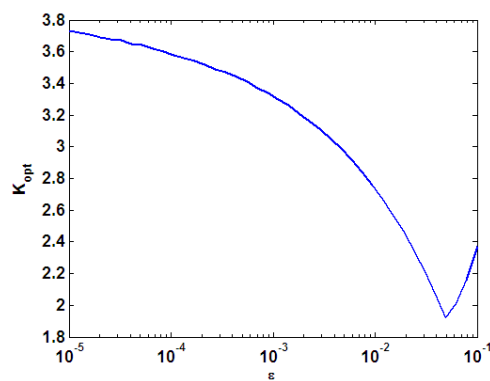
Settling Time

Settling time: $t_s(K)$ for $\varepsilon = 10^{-3}$, $\tau = 1$



- Optimum at $K=3.3$

Settling Time



- Optimum K actually depends on required accuracy
- Still, always want to avoid $K < \sim 2$

Doublets

- **Amplifier model:**

$$G_m(s) = G_{mo} \frac{1 + s/\omega_z}{1 + s/\omega_p} \quad \text{with} \quad \omega_z = \frac{\omega_p}{\alpha}$$

$\omega_p = \beta\omega_{-3dB}$, ω_{-3dB} is bandwidth of $T(s)$

$\alpha = 1 + \varepsilon \quad \text{with} \quad |\varepsilon| \ll 1$

- **Closed-loop gain (ignore feedforward zero):**

$$\frac{V_o}{V_{in}} = -c \frac{1}{1 + s \frac{C_{Leff}}{FG_m(s)}} \approx -\frac{c}{1 + s/\omega_{-3dB}} \left(\frac{1 + s/\omega_z}{1 + s/\omega_{pp}} \right)$$

$\omega_{-3dB} \cong \frac{FG_{mo}}{C_{Leff}}$

$\omega_{pp} \cong \omega_p$

Doublet Analysis

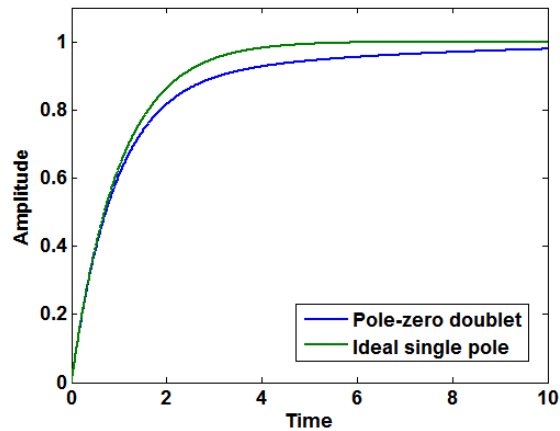
- **Step response**

$$v_{o,step}(t) = -cV_{step} \left(1 - Ae^{-t\omega_{-3dB}} - Be^{-t\omega_{pp}} \right)$$

$$B \approx \varepsilon \frac{\beta}{(1 - \beta)^2}$$

$$A \approx 1 - B \approx 1$$

Doublet Example



$$\alpha=1.5$$
$$\beta=0.3$$

Doublet Conclusions

- **Case A:** $\tau_2 \leq \tau_1$ i.e. $\beta \geq 1$
 - Doublet settles faster than amplifier
 - Has no impact on overall settling time
- **Case B:** $\tau_2 > \tau_1$
 - Doublet settles more slowly than amplifier
 - Determines overall settling time (unless ϵ within settling accuracy requirements)
- → Avoid “slow” doublets!

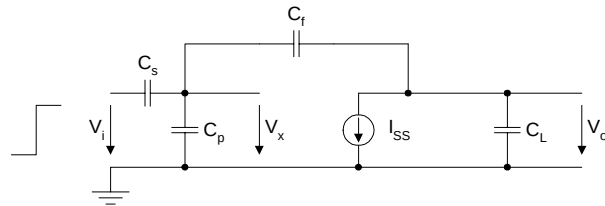
Final Note on Doublets

Slewing

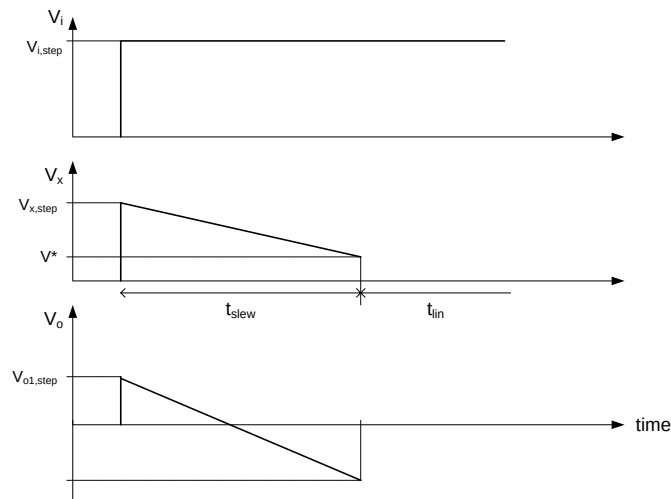
- Transconductor ΔI vs. ΔV :
- Model for (nonlinear) slewing amplifier
 - Piecewise linear approximation:
 - Slewing with constant current, followed by
 - Linear settling exponential
 - $t_s = t_{\text{slew}} + t_{s,\text{lin}}$

Slewing Analysis

- Circuit model during slewing:



Slewing Analysis (cont.)



Slewing Analysis (cont.)

- **Slewing period:**

$$V_{x,step} = V_{i,step} \frac{C_s}{C_s + C_2} \quad \text{with} \quad C_2 = C_i + \frac{C_f C_L}{C_f + C_L}$$

$$\Delta V_x = V_{x,step} - V^* \rightarrow \Delta V_o = \frac{\Delta V_x}{F}$$

$$t_{slew} = \frac{\Delta V_o}{SR} = \frac{\Delta V_x C_{Leff}}{F I_{SS}}$$

- **Linear settling during final V^* of swing at V_x :**

- **Step during linear settling:** V^*/F

- **Linear settling time:** $\frac{t_{s,lin}}{\tau} = -\ln \left(\varepsilon \frac{C V_{i,step} F}{V^*} \right)$

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Lecture 14: Design Example



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Design Example

- **What are the specs?**
 - And where do they come from?
- **Example: 50MS/s, ~11-bit ADC**
 - Accuracy:
 - Settling time:
 - Dynamic range:
 - Closed-loop gain:
 - Power:

Design Procedure

Design Procedure Cont'd

Finding C_L

Settling

Device Sizing

Settling?

Slewing

Single-ended vs. Differential

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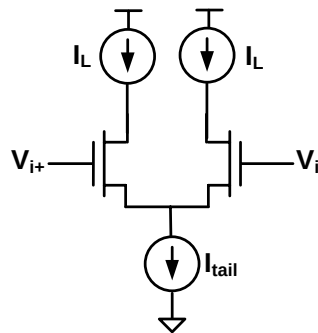
Lecture 15: Common-Mode Feedback



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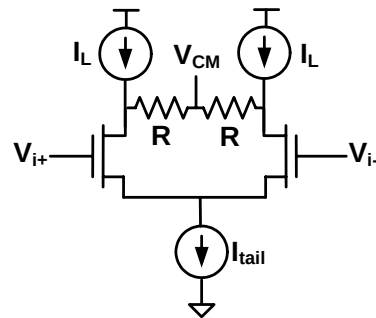
Problem with Common-Mode

- What if $I_L < I_{tail}/2$?
 - Will capacitive feedback solve this?
- Typical solution:
Common-mode feedback
 - Sense CM at output
 - Adjust some knob to alter CM

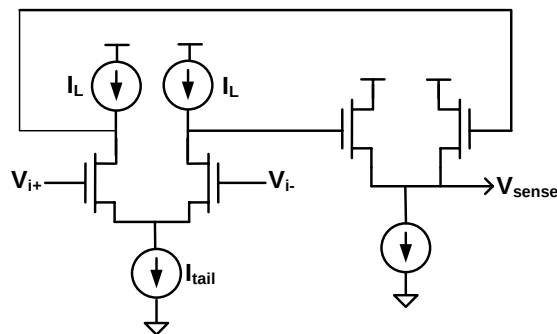


Common-Mode Sensing

- Simplest CM sensor: pair of resistors
- Resistors load the OTA (reduce gain)
 - If make R large, get slow V_{cm} tracking
 - Is this a problem?

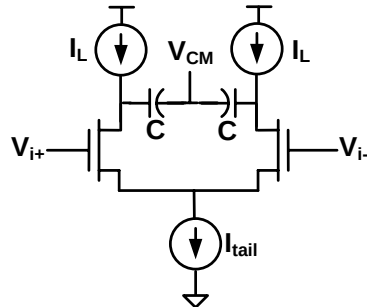


Sensing Scheme #2



- Isolated CM sensing
 - Works reasonably well
 - But hard to use with wide swing amplifier output

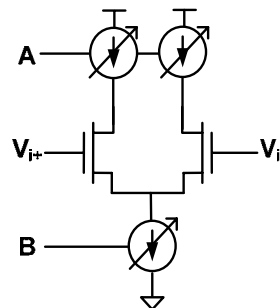
Capacitive Sensing



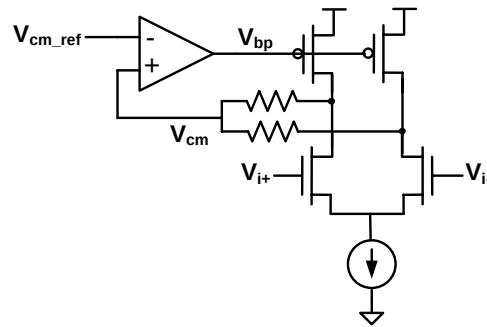
- **Capacitive sensing avoids DC loading**
 - (still creates AC load though)
- **Needs to be reset to remove initial offset**
 - Just like capacitive feedback

Adjusting Common-Mode

- **Really only two knobs:**
- **Knob A: adjust load current**
- **Knob B: adjust tail current**

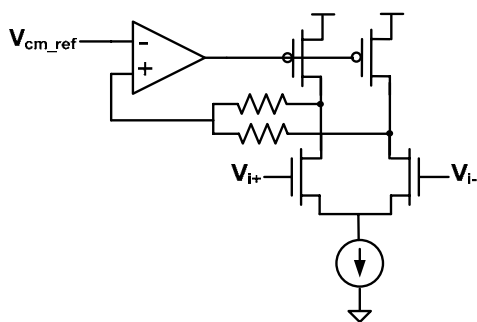


Example Common-Mode Feedback



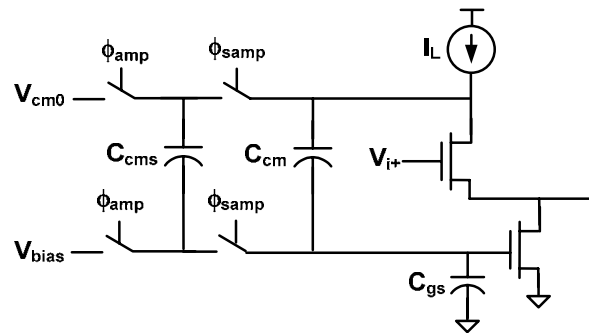
- Secondary amplifier enforces $V_{cm} = V_{cm_ref}$
- Place dominant pole at V_{bp} , or V_{cm} ?

CMRR Fix



- What if two PMOS transistors aren't perfectly matched?

“Continuous” CMFB



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Lecture 16: Multistage Amplifiers

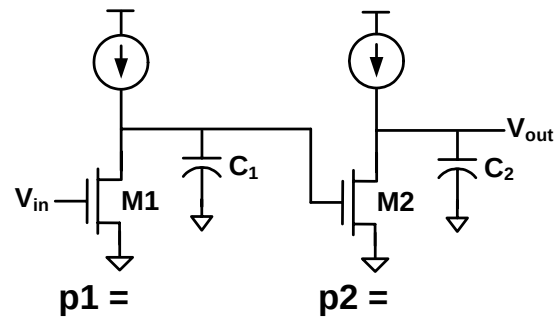


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Why use Multi-Stage Amplifiers?

- **Single-stage amplifier:**
 - Generally have to trade between swing and gain
 - (Need cascodes and/or large $V_{\min}$ for current sources)
- **Multi-stage amplifier:**
 - Higher gain without sacrificing swing
 - (Gain-boosted cascode is multi-stage amplifier in disguise)
- **Challenge: stability!**

Stability for Simple 2-Stage Amp



- Two closely spaced poles - is this circuit stable?

2-Stage Stability cont'd

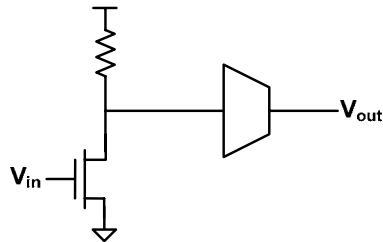
Compensation Techniques

- **Many options – best one depends on situation at hand**
- **Look at a few general categories:**
 - **Narrowbanding**
 - **Wideband input stage (pre-amp)**
 - **Miller**

Narrowbanding

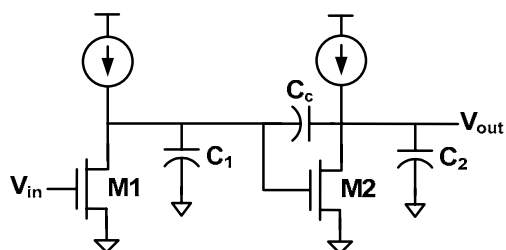
- **Narrowbanding**
 - **Lower one of the poles**
 - **Or introduce a new one**
- **Stability OK, but (feedback) bandwidth often low**
 - **Example: offset cancellation**

Pre-amp



- Build a pre-amp with bandwidth much higher than 2nd stage
 - Usually limits achievable pre-amp gain

Miller Compensation



- Very common form of compensation
 - Why is this “pole splitting” good?

Miller Compensated Poles/Zeros

Phase Margin Engineering

$$\omega_u \approx F \frac{g_{m1}}{C_c} \quad |p_2|, z \gg \omega_u \text{ of } T(s)$$

choose $|p_2| \geq K \omega_u$

$$C_c \geq K F C_2 \frac{g_{m1}}{g_{m2}}$$

$$\frac{z}{\omega_u} = \frac{1}{F} \frac{g_{m2}}{g_{m1}}$$

$$\frac{z}{|p_2|} \approx \frac{C_2}{C_c}$$

- Higher K $\rightarrow$ higher C_c

- For fixed C_c , larger $C_L = C_2$ *lowers* phase margin

- Zero can add significant phase lag
 - Unless $g_{m2} \gg F g_{m1}$

Nulling Resistor

$$z \rightarrow \frac{1}{\left(\frac{1}{g_{m2}} - R_z\right)C_c}$$

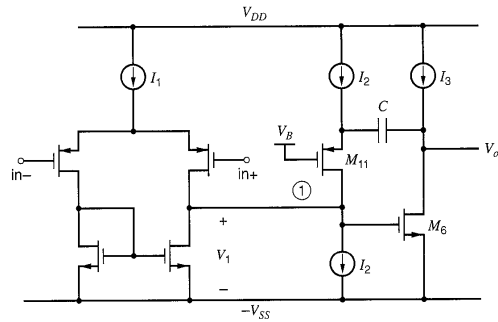
p_1, p_2 : no change

$$p_3 \approx -\frac{1}{R_z C_1}$$

- **R_z limits feedforward current at high frequency**
 - Pushes feedforward zero to higher frequency
- **Adds new pole p_3**

Nulling Resistor Implementation

Cascode Compensation (Ahuja)



- No RHP zero
- But cost in power can be high
 - (I_2 needs to slew C_c)

Cascode Compensation (Ribner)

Noise Analysis

- Need a simplified model:

Noise Analysis cont'd

$$v_o = \frac{1}{Fg_{m1}} \frac{1}{1 + \frac{s}{\omega_o Q} + \frac{s^2}{\omega_o^2}} \left(i_{n1} - i_{n2} \frac{sC_c}{g_{m2}} \right)$$

with

$$\omega_o^2 = \frac{Fg_{m1}g_{m2}}{C_c(C_c + C_L)}$$

$$\omega_o Q = \frac{Fg_{m1}}{C_c}$$

Total Noise at Output

$$\overline{v_{oT}^2} = \frac{k_B T}{C_c} \frac{\gamma}{F} + \frac{k_B T}{(C_c + C_L)} \gamma$$

$$\overline{v_{oT}^2} = \frac{k_B T}{C_c} \frac{\gamma}{F} \left(1 + \frac{F C_c}{C_c + C_L} \right)$$

- Noise from first stage dominates
- Noise capacitor: **C_c** (NOT C_L!)

2-Stage CMFB

2-Stage CMFB

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Lecture 17: Biasing and References



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Reference Circuits

- **External bias current not always available**
 - May need to generate reference internally
- **Tons of “constant X” reference circuits in the literature**
 - Important considerations include power, accuracy, PSRR, output impedance, etc.
- **Most important question: what do you really want to be constant?**

Constant Current Bias?

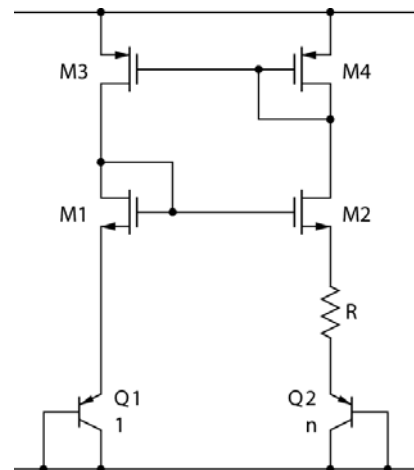
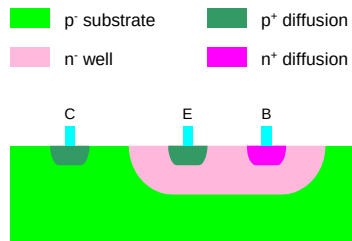
Constant Gain Example

Supply “Independent” Biasing

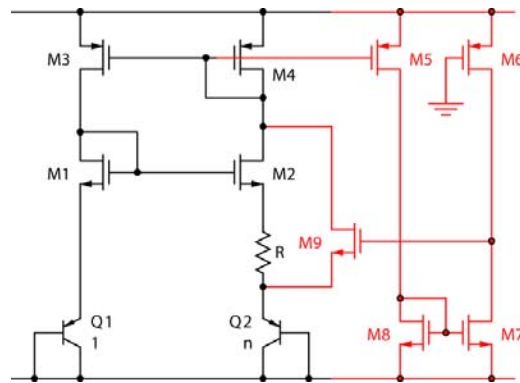
Improved V_{GS} Reference

PTAT Reference

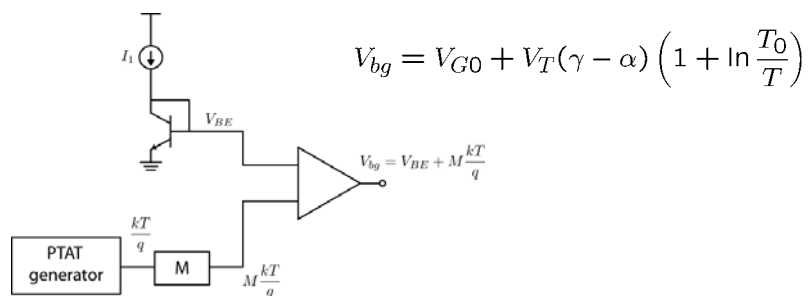
CMOS PTAT Reference



Startup Circuit



Conceptual Band-Gap



- V_{BE} has a tempco of roughly $-2 \text{ mV}/^\circ\text{C}$
- Add V_{BE} to PTAT voltage (with right M) $\rightarrow V_{bg}$ independent of T
- Reference derived from band-gap of Si (1.205V)

Constant g_m Reference

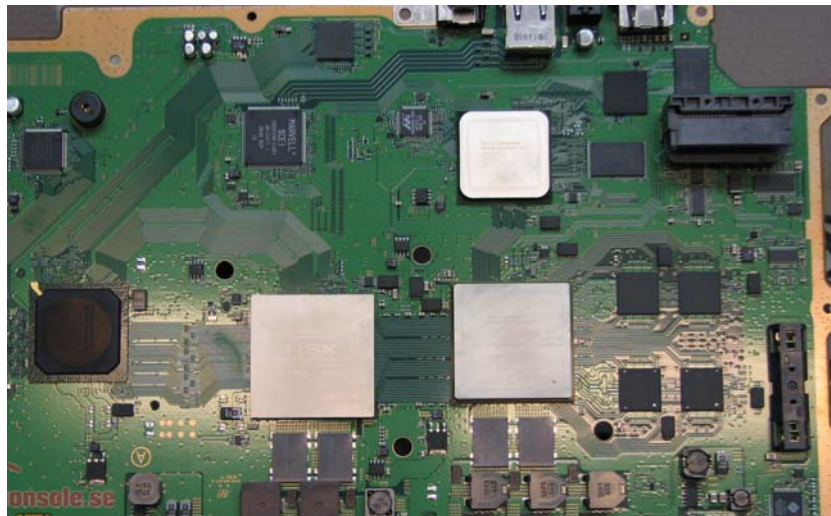
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Lecture 18: High-Speed Link Overview



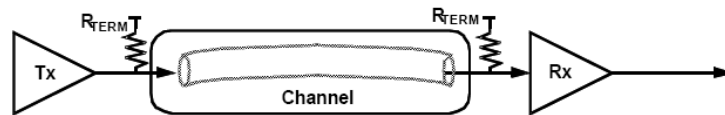
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Links are everywhere...

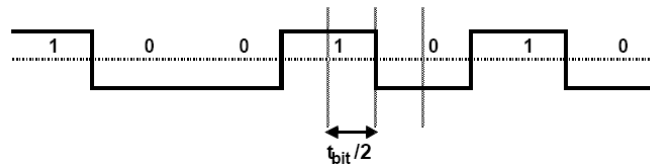


Basic Link Issues

- Signaling: getting bits from the TX to the RX



- Timing: determining which bit is which



Speed of Light

- Why is a link (i.e., off-chip I/O) different than on-chip wires?
 - Both send info back and forth
- Usually model on-chip wires with capacitor
 - Sometimes with resistance too
- On-chip model works because dimensions $\ll \lambda$
 - Not true for off-chip wires...

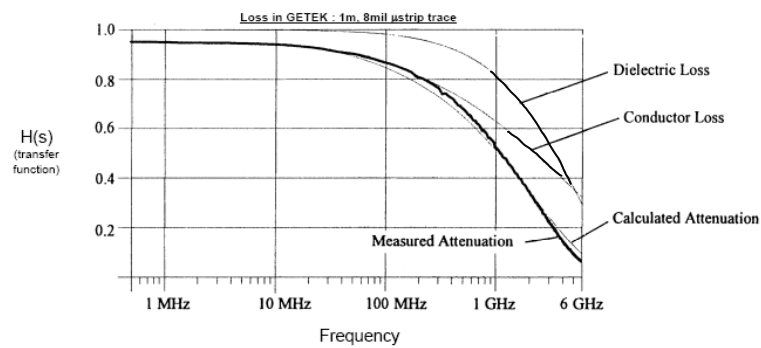
Transmission Lines

- Wire model when can't ignore c :
- Properties:
 - Delay
 - Characteristic impedance
 - Energy stored in E, B fields

Termination and Reflection

- Two constraints at any junction:
 - Voltage are equal
 - Power is conserved

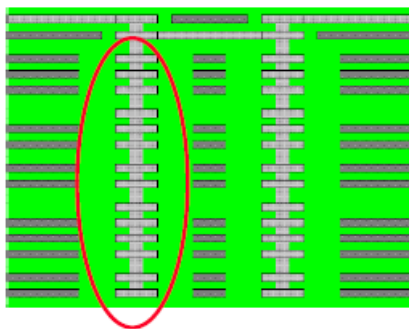
Loss



- **Real T-lines have loss too:**

- Skin loss $\propto \sqrt{f}$
- Dielectric loss $\propto f$

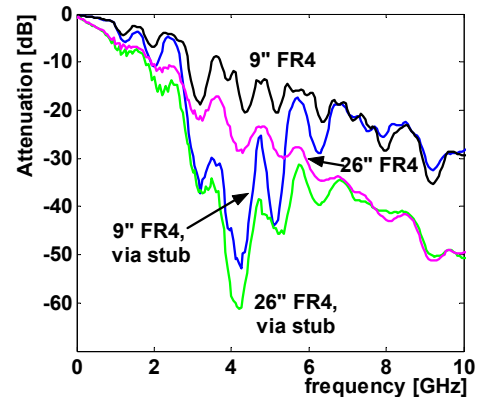
Not Just Material Issues...



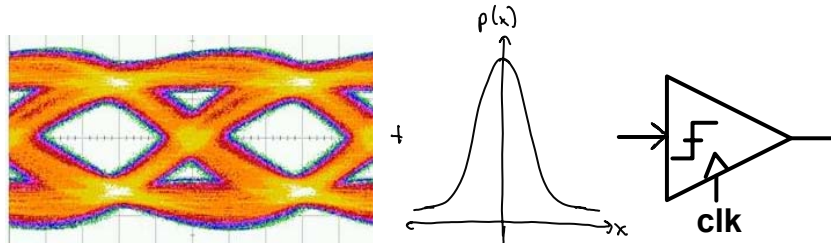
- **Energy splits at via**
- **Via stub looks like a capacitor – reflections**

Example Channels

- 20-30dB loss at 3GHz
- How bad is that?
- Two related issues:
 - (1) Noise and min. signal amplitude
 - (2) Intersymbol interference

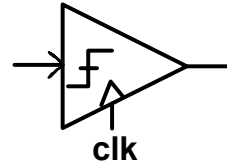
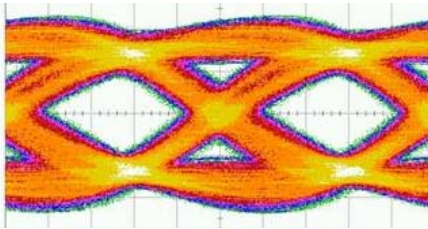


Noise and BER



- RX circuits always have noise
 - If noise is ever larger than the input signal (at sampling point), RX will decode the bit incorrectly
- BER = Bit Error Rate
 - I.e., average # of incorrectly received bits / total transmitted bits

Min. Signal Amplitude



- Min. signal set by noise σ and residual offset:

$$BER = \frac{1}{2} \operatorname{erfc} \left(\frac{V_{in, ampl} - V_{off}}{\sqrt{2} \sigma_{noise}} \right)$$

- $BER = 10^{-12}$: $(V_{in, ampl} - V_{off}) = 7\sigma_n$
- $BER = 10^{-20}$: $(V_{in, ampl} - V_{off}) = 9.25\sigma_n$

So What?

- Why not just hit the RX with a larger signal?
 - (Not a stupid question – this is often what people do)
- Simple (hand-wavy) answers:
 - Transmission line Z usually low ($\sim 50\Omega$)
 - 1V swing $\rightarrow$ 20mW
 - Larger swing doesn't help with ISI...
 - More next lecture
- Bottom line:
 - If can use lower swing, can get lower power
 - Good application of EE240 material!

Link Circuits: “Current-Mode” TX

- Often use differential signaling/circuits to reject supply/CM noise:

Receiver Termination Options

Basic Receiver

Front-end Amp Gain

Front-end Amp Bandwidth

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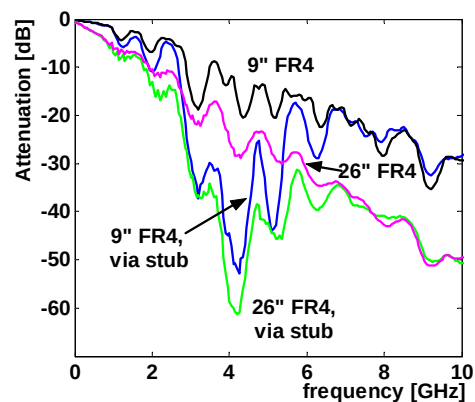
Lecture 19: High-Speed Filter Design



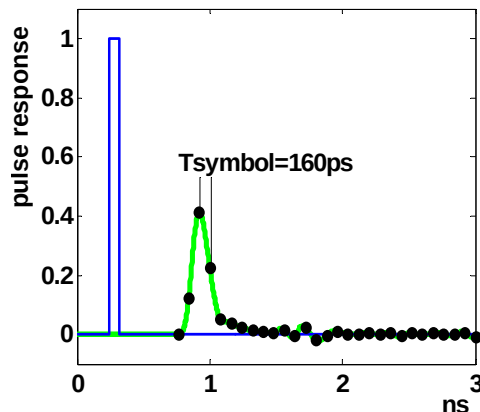
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Link Channels and ISI

- 20-30dB loss at 3GHz
- How bad is that?
- Two related issues:
 - (1) Noise and min. signal amplitude
 - (2) Intersymbol interference

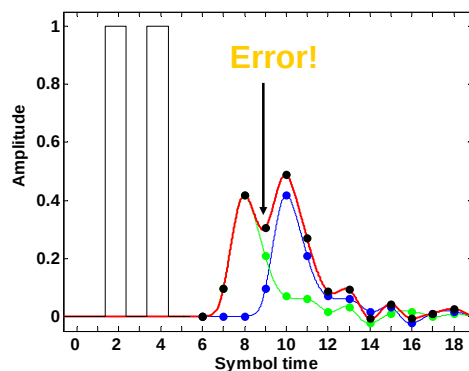


Inter-symbol interference (ISI)



- Channel is low pass
 - Short TX pulses get spread out

Impact of ISI



- Middle sample is corrupted by 0.2 trailing ISI (from previous symbol) and 0.1 leading ISI (from next symbol)
- Total ISI: 0.3 total ISI
 - Middle symbol incorrectly detected

Equalization

- **ISI is proportional to TX swing**
 - Generally can't just boost signal to overcome it
- **Solution: Equalization**
 - If channel applies filter $H(s)$
 - Pass the signal through another filter $H^{-1}(s)$

Equalization cont'd

- **Link channel basically low-pass**
 - Equalizer boosts high frequency, attenuates low frequency

Equalizer Requirements

Equalizer Requirements cont'd

Equalizer Options and Limitations

RX Equalizer

RX Equalizer Implementations

RX Equalizer Implementations

RX Equalizer Implementations

RX Equalizer Implementations

Programmability

Aside: Switched Cap. Resistor

Aside: Switched Cap Low-Pass Filter

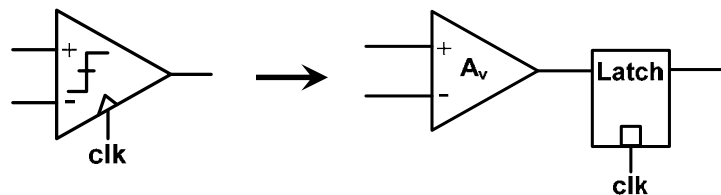
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Lecture 20: Comparators



Elad Alon
Dept. of EECS

Comparator

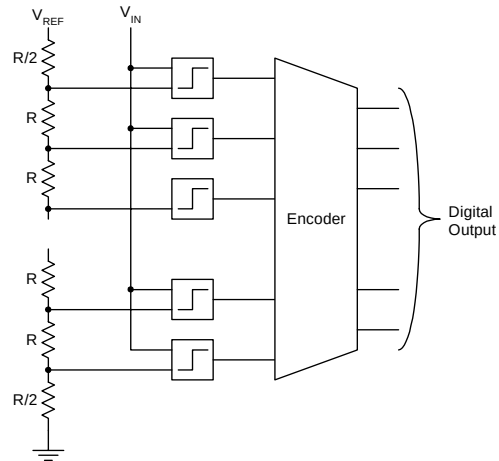


- **Specs and issues:**

- Clock rate f_s
- Offset
- Resolution
- Hysteresis
- Input cap
- Power dissipation
- CM rejection
- Kickback noise
- ...

Flash Converter

- **Fast: one clock cycle per conversion**
- **High complexity: $2^B - 1$ comparators**
- **High input capacitance**



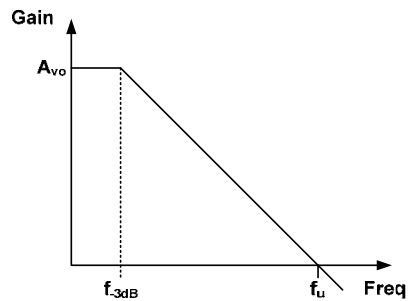
Comparator Gain-Bandwidth

Example:

- 4Gb/s link
- Minimum ΔV : 1mV
- $V_{dd} = 1V$

$$\rightarrow A_v > 1V / 1mV = 1000 \text{ in } < 250ps!$$

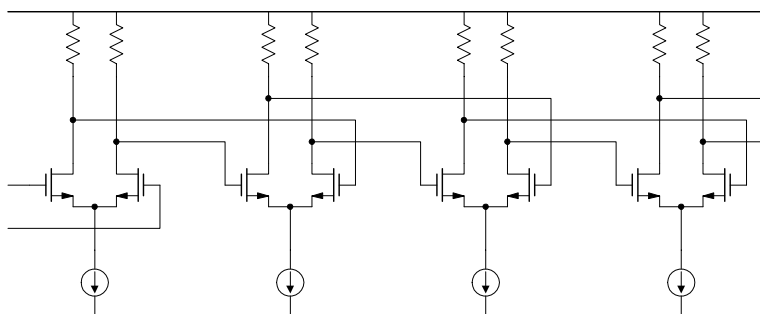
Operational Amplifier?



$$f_{-3dB} = \frac{f_u}{A_{vo}} = \frac{2}{3} \frac{1}{T_{bit}}$$

$$f_u = \frac{2A_{vo}}{3T_{bit}} = \frac{1000}{3 \times 250ps} = \underline{1.33THz}$$

Open-Loop Amplifier Cascade



Cascaded Amplifier

- **Simplified bandwidth analysis:**
 - **Open-circuit time constants**
 - **(Not most accurate, but leads to nearly the right answer for design optimization)**

Bandwidth/Gain Optimization

Bandwidth/Gain Optimization

Power Consumption

Regenerative Latch

CML Comparator (Latch)

StrongArm Latch

Another CMOS Comparator

Hysteresis

Overdrive Recovery

Kickback

Kickback cont'd

Kickback cont'd

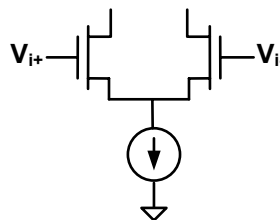
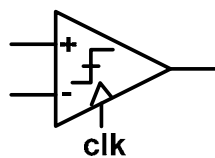
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Lecture 21: Matching



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Dept. of EECS

Offset



- To achieve zero offset, comparator devices must be perfectly matched to each other
- How well-matched can the devices be made?
 - Not arbitrary – direct function of design choices

Device Mismatch Categories

- **Die-to-die**
 - All devices on same chip (or wafer) have same characteristics
- **Within die (long-range)**
 - All devices within certain region have same characteristics
- **Local (short-range)**
 - Every device different, random
 - Usually most important source of mismatch

Sources of Local Variation

- **Deterministic sources:**
 - Local poly density
 - Sub-90nm: stress, litho interactions, ...
- **Random sources:**
 - Dopant fluctuations
 - Line-edge roughness
 - Oxide traps
- **Focus our modeling on random variations**
 - Deterministic handled with good layout practices

References

- M. J. M. Pelgrom, A. C. J. Duinmaijer, and A. P. G. Welbers, "Matching properties of MOS transistors," *IEEE JSSC*, vol. 24, pp. 1433 - 1439, Oct. 1989
 - Mismatch model
 - Statistical data for 2.5 μ m CMOS
- J. A. Croon, M. Rosmeulen, S. Decoutere, W. Sansen, and H. E. Maes, "An easy-to-use mismatch model for the MOS transistor," *IEEE JSSC*, vol. 37, pp. 1056 - 1064, Aug. 2002
 - 0.18 μ m CMOS data

Mismatch Statistics

- Total mismatch set by composite of many single, independent events
 - Correlation distance $\ll$ device dimensions
 - E.g., number of dopant atoms implanted into the channel
- Individual effects are small: linear superposition holds
- $\rightarrow$ Mismatch is zero mean, Gaussian distribution

Parameter Mismatch Model

$$\sigma^2(\Delta P) = \frac{A_p^2}{WL} + S_p^2 D_x^2$$

- $\sigma^2(\Delta P)$: standard deviation of P
 WL : active gate area
 D_x : distance between device centers
 A_p : measured area proportionality constant
 S_p : measured distance proportionality constant,
 $:$ $\cong 0$ for "good" layout

V_T Mismatch

- Mismatch in V_T between two identical devices:

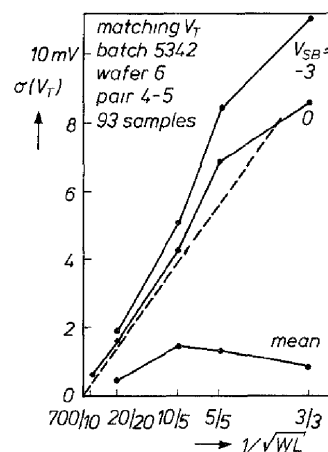
$$\sigma^2(\Delta V_T) = \frac{A_{V_T}^2}{WL} + S_{V_T}^2 D_x^2$$

2.5 μ m CMOS process:

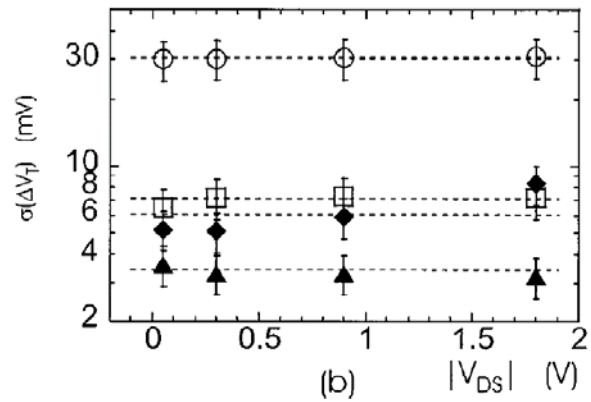
$$A_{V_T, NMOS} \cong 30 \text{ mV } \mu\text{m}$$

$$A_{V_T, PMOS} \cong 35 \text{ mV } \mu\text{m}$$

- Often largest source of offset

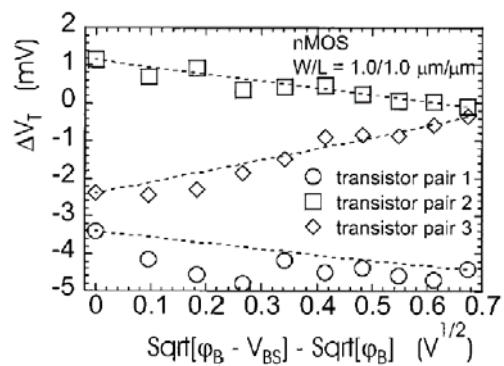


Drain Bias, V_{DS}



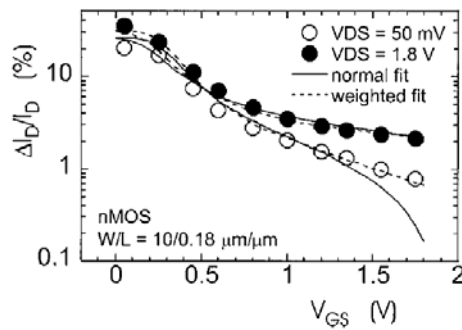
ΔV_T largely independent of V_{DS}

Back-Gate Bias, V_{SB}



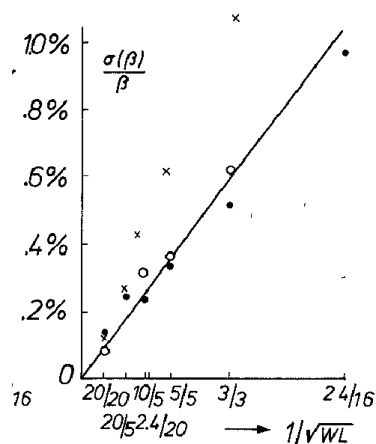
- Mismatch can depend on V_{SB}
- Why?

Current Matching, $\Delta I_D/I_D$



Strong bias dependence (we knew that already)

Current Factor



$$\beta = \mu C_{ox} \frac{W}{L}$$

Sources of β Mismatch

- **Mobility variations**
 - E.g., due to dopant variations, random defects
- **Oxide thickness variation**
 - Usually very well-controlled
- **Edge roughness**

Edge Model

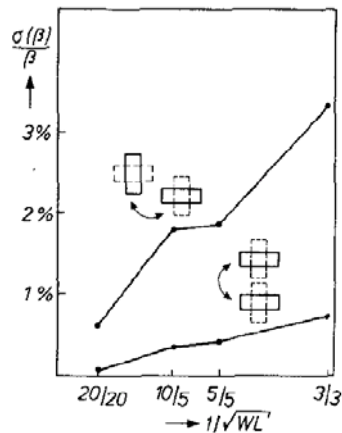
$$\frac{\sigma^2(\beta)}{\beta^2} = \frac{\sigma^2(W)}{W^2} + \frac{\sigma^2(L)}{L^2} + \frac{\sigma^2(C_{ox})}{C_{ox}^2} + \frac{\sigma^2(\mu_n)}{\mu_n^2}$$

For: $\sigma^2(W) \propto 1/L$ and $\sigma^2(L) \propto 1/W$

Simplifies to:

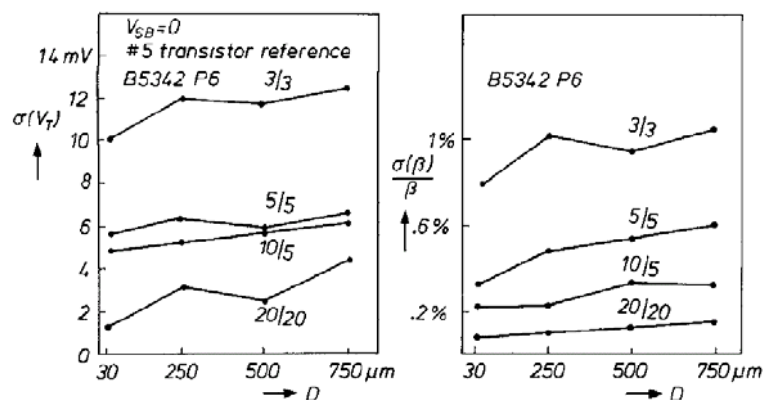
$$\frac{\sigma^2(\beta)}{\beta^2} = \frac{A_L^2}{WL^2} + \frac{A_W^2}{W^2L} + \frac{A_{C_{ox}}^2}{WL} + \frac{A_{\mu}^2}{WL} + S_{\beta}^2 D^2$$

Orientation Effects



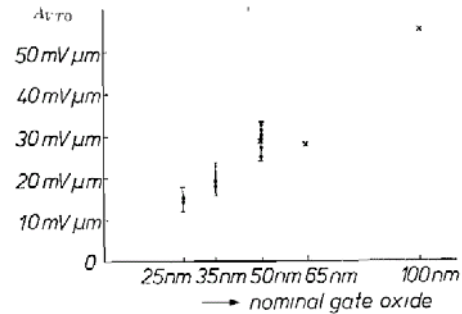
- Si and transistors are not (perfectly) isotropic
- → keep direction of current flow same!

Distance Effect

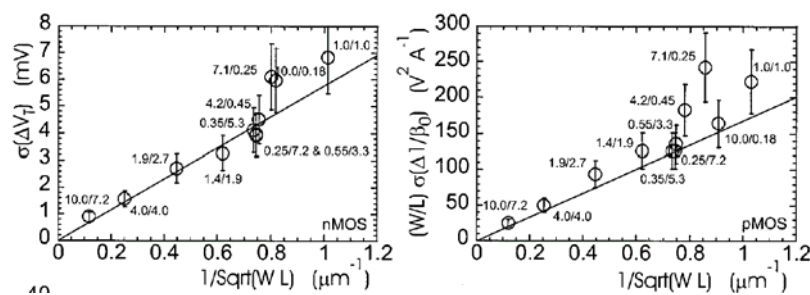


Process Dependence

- A_{vt} tends to scale with technology
- Proportional to t_{ox}
- Also depends on doping level



0.18 μ m CMOS



Current Matching

Voltage Matching

“Golden Rule” of Layout for Matching

- **Everything you can think of might matter**
 - **Even whether or not there is metal above the devices**
- **How to avoid systematic errors?**

Ref: A. Hastings, “The art of analog layout,” Prentice Hall, 2001

Common Centroid Layout

- **Cancels linear gradients**
- **Required for moderate matching**

Simulating Mismatch

- **Brute force: Monte Carlo**
 - HSPICE “throws the dice”...

Simulating Mismatch

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Lecture 22: Offset Cancellation



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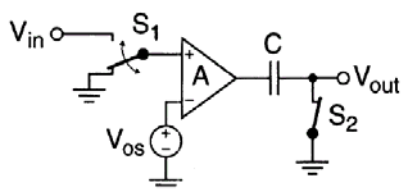
Offset Cancellation Overview

- Two main ideas/approaches
- Modulate and/or filter offset so that it is outside of signal band
 - CDS (auto-zeroing)
 - Chopping (synch. detection, DEM)
- Inject a DC signal that opposes the offset
 - Trimming
 - Often digitally controlled (especially for comparators)

Filtering/Modulating Offset

- **General idea:**
 - Put elements around the amplifier that treat offset differently than signal
- **CDS:**
 - Configure amplifier so that offset is (approx.) differentiated
- **Chopping:**
 - Modulate offset to frequencies beyond signal band, then filter it out

CDS #1: Output Offset Cancellation



Phase 1:

$$V_C = -AV_{os}$$

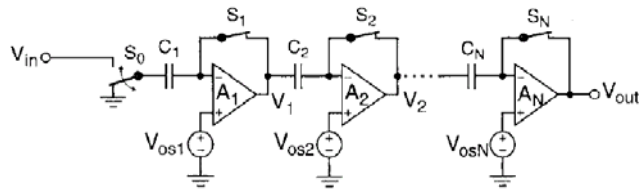
Phase 2:

$$\begin{aligned} V_{out} &= A(V_{in} - V_{os}) - V_C \\ &= AV_{in} \end{aligned}$$

- **Relatively insensitive to switch errors**
 - Storing amplified offset
- **But, what happens if gain is large?**

CDS #2: Input Offset Cancellation

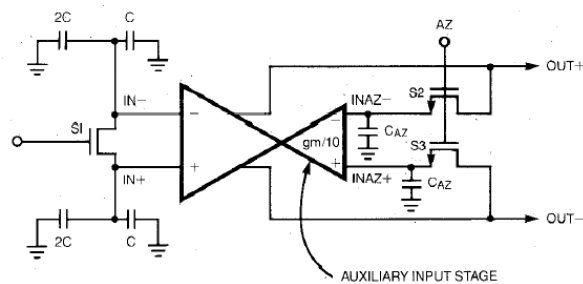
Multistage Cancellation



- Open switches left to right
 - Errors from $S_1 \dots S_{N-1}$ cancelled by final stage
- Application: continuous time comparators

Auxiliary Amplifier Offset Cancellation

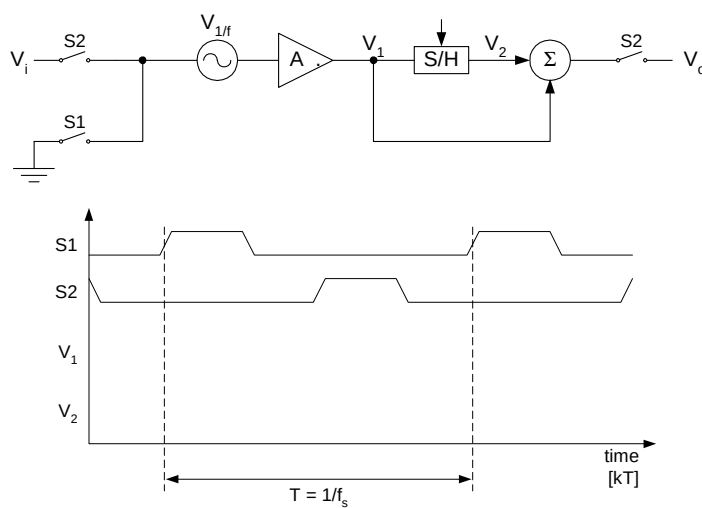
Aux. Amplifier Example



H. Ohara, H. X. Ngo, M. J. Armstrong, C. F. Rahim, and P. R. Gray, "A CMOS programmable self-calibrating 13-bit eight-channel data acquisition peripheral," *IEEE Journal of Solid-State Circuits*, vol. 22, pp. 930 - 938, December 1987.

Aux. Amplifier Implementation

CDS and Flicker Noise



Flicker Noise Analysis

$$V_o(kT) = A \left\{ \underbrace{V_i(kT)}_{\text{signal}} + \underbrace{V_{1/f}(kT) - V_{1/f}\left(kT - \frac{T}{2}\right)}_{\text{input referred error } V_{nieq}} \right\}$$

Laplace Transform

Delay by $t_d \rightarrow e^{-st_d}$

$$V_{nieq}(s) = V_{1/f}(s) \underbrace{\left\{ 1 - e^{-s\frac{T}{2}} \right\}}_{H_n(s)}$$

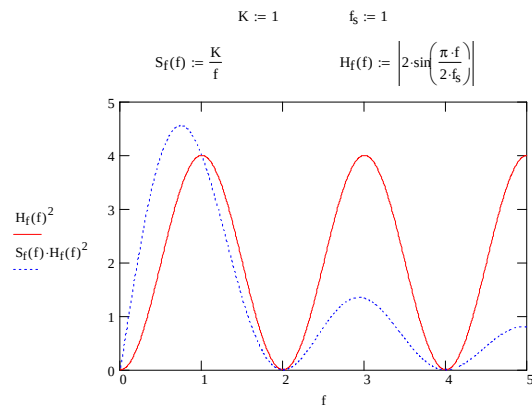
Flicker Noise Frequency Response

$$\begin{aligned} H_n(s) &= 1 - e^{-s\frac{T}{2}} \\ &= 1 - e^{-j\omega\frac{T}{2}} \\ &= 1 - \cos\frac{\omega T}{2} + j \sin\frac{\omega T}{2} \end{aligned}$$

$$\begin{aligned} |H_n(s)|_{s \rightarrow j\omega}^2 &= \left(1 - \cos\frac{\omega T}{2}\right)^2 + \left(\sin\frac{\omega T}{2}\right)^2 \\ &= 1 - 2\cos\frac{\omega T}{2} + \underbrace{\cos^2\frac{\omega T}{2} + \sin^2\frac{\omega T}{2}}_1 \\ &= 2\left(1 - \cos\frac{\omega T}{2}\right) \\ &= 4\sin^2\frac{\omega T}{4} \end{aligned}$$

$$|H_n(s)|_{s \rightarrow j\omega} = \left| 2 \sin\frac{\omega T}{4} \right| = \left| 2 \sin\frac{\pi f}{2 f_s} \right|$$

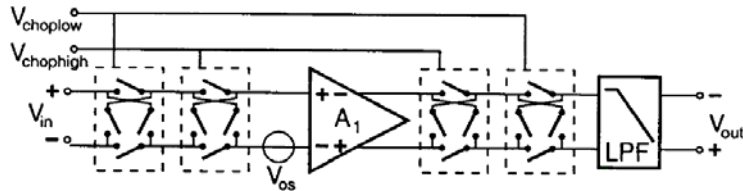
Flicker Noise Spectrum



- Flicker noise is differentiated
 - As is thermal noise
- Noise removed at low freq.
 - But amplified at “high” freq.
- Noise above $f_s/2$ folds to baseband

Chopping

Nested Chopper Amplifier



- Inner chopper at high freq. to remove $1/f$ noise
- Outer chopper at low frequency to minimize “spiking” and remove residual offset from inner chopper.

Offset Trimming

Digital Trimming

Comparator Trimming

Trim Implementation Issues

- **Infinite number of ways to introduce digitally controlled offset**
 - People have tried just about all of them
- **Key issues:**
 - Power overhead
 - Circuit Imbalance
 - Effective resolution
 - Area overhead

Comparator Trim Schemes

Pre-Amp Trim

Pre-Amp Trim

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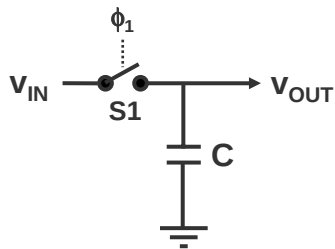
Lecture 23: MOS Sample and Hold



Elad Alon
Dept. of EECS

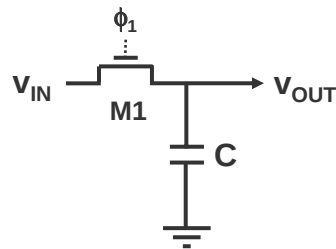
MOS Sample & Hold

Ideal Sampling



- Grab exact value of V_{in} when switch opens

Practical Sampling



- kT/C noise
- Limited bandwidth
- $R_{sw} = f(V_{in}) \rightarrow$ distortion
- Switch charge injection
- Clock jitter

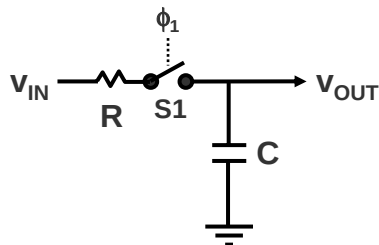
Switch Resistance

Acquisition Bandwidth

- Finite switch $R \rightarrow$ finite bandwidth

- Assuming constant V_{in} and C starts at 0V:

$$v_{out}(t) = v_{in} (1 - e^{-t/\tau})$$



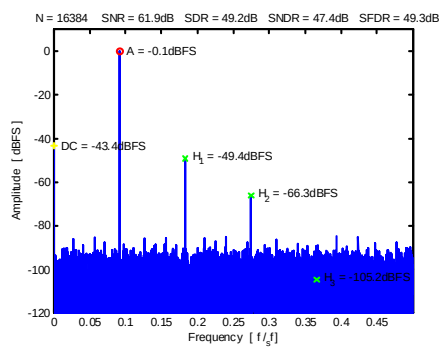
- Leads to min. switch size for given bandwidth, resolution
 - Linear settling calc. – remember may only get $T/2$
- (Will C always start at 0V?)

Switch R_{on} Non-Linearity

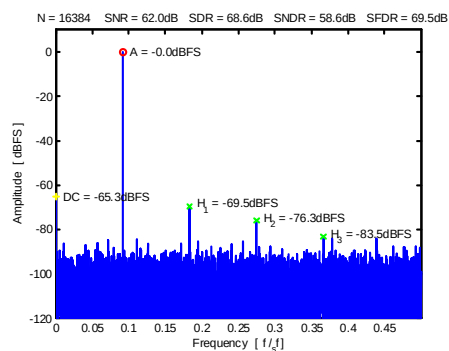
Sampling Distortion

$$v_{out} = v_{in} \left(1 - e^{-\frac{T}{2\tau} \left(1 - \frac{v_{in}}{V_{DD} - V_{TH}} \right)} \right)$$

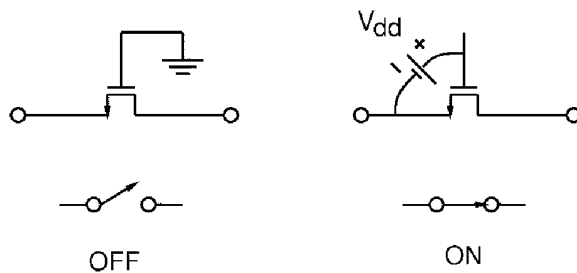
$T/\tau = 10$



$T/\tau = 20$



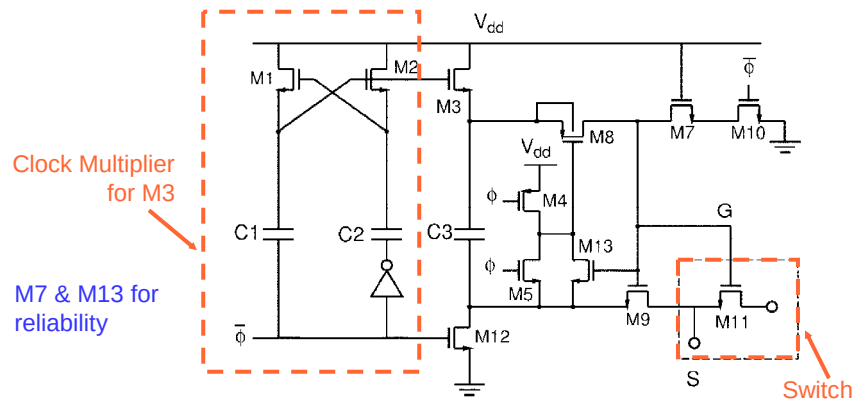
Constant V_{GS} Sampling



- Switch overdrive voltage is independent of signal
- Error from finite R_{ON} is linear (to first order)

Constant V_{GS} Sampling Circuit

Complete Circuit

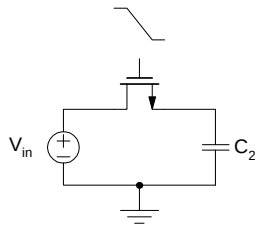


Ref: A. Abo et al, "A 1.5-V, 10-bit, 14.3-MS/s CMOS Pipeline Analog-to-Digital Converter," JSSC May 1999, pp. 599.

Charge Injection

- "Extra" charge dumped onto holding capacitor
 - Channel charge has to go somewhere
 - (Also get injection through C_{ov})
- Problems:
 - Offset
 - Distortion (error charge is function of V_{IN})

Worst-Case Error Example



channel charge :

$$Q_{CH} = WLC_{ox}(V_{DD} - V_{in} - V_{TH})$$

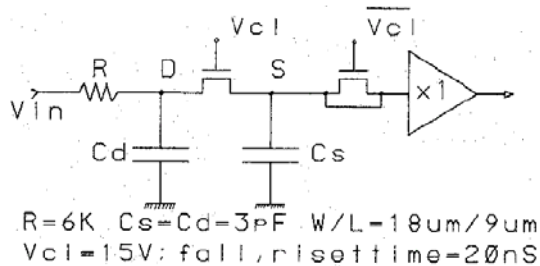
max pedestal error : $V_{in} = V_{SS}$

$$\Delta V = \frac{Q_{CH}}{C_2} = \frac{WLC_{ox}}{C_2}(V_{DD} - V_{SS} - V_{TH})$$

Example :

$$\Delta V = \frac{10 \times 0.35 \times 5}{1000}(3 - 0.6) = 42 \text{ mV}$$

Dummy Switch



- Dummy switch is half width
- Depends on equal split between source and drain
- Is split equal?

V_{in}	UNCOMPENSATED SWITCH	COMPENSATED WITH DUMMY	BALANCED SWITCH
0v	-160mV	-45mV	6mV
5v	-105mV	-30mV	1mV
10v	-40mV	-11mV	0.5mV

Ref: Bienstman et al, JSSC 12/1980, pp. 1051.

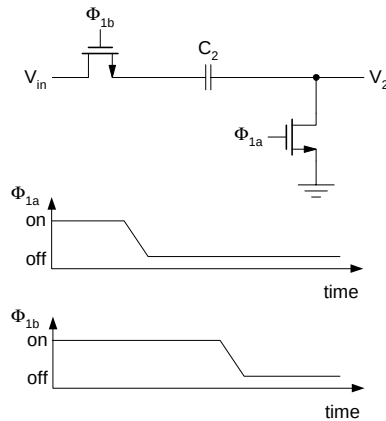
Eichenberger et al, JSSC 8/1989, pp. 1143.

Charge Injection Analysis

- Can perform more detailed, distributed analysis
 - See e.g. Wegmann et al, "Charge Injection in Analog MOS Switches," IEEE J. Solid-state Circuits, Dec. 1987.
 - Results depend on how fast switch is turned off
- Note that SPICE doesn't do this (lumped model) – uses "XPART" parameter instead:
 - XPART = 0: Source 60%, Drain 40%
 - XPART = 0.5: equal split
 - XPART = 1: 100% Drain

Rejecting Injection Error

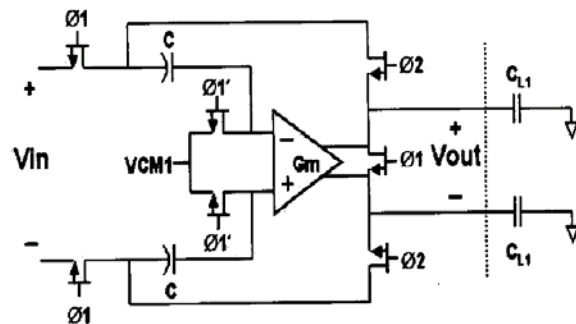
Bottom-Plate Sampling



- Turn off Φ_{1a} first
 - Injected charge is constant
 - Removed in differential output
- Switch Φ_{1b} opens later
 - C_2 disconnected
→ “zero” charge injected
- Is this useful?
 - $V_2 = 0V...$

Using Bottom-Plate Sampling

Using Bottom-Plate Sampling



Ref: W. Yang, D. Kelly, I. Mehr, M. T. Sayuk, and L. Singer, "A 3-V 340mW 14-b 75-Msample/s CMOS ADC with 85-dB SFDR at Nyquist input," *IEEE Journal of Solid-State Circuits*, vol. 36, pp. 1931 - 1936, December 2001.

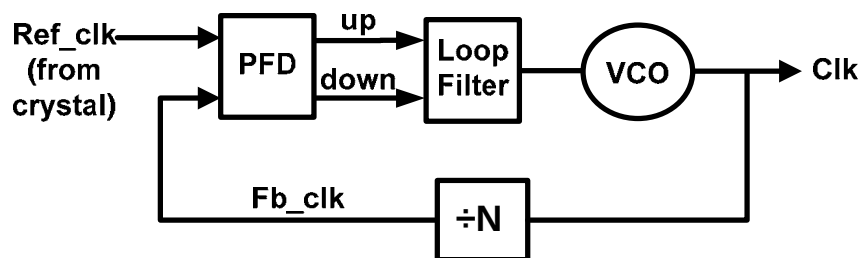
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Lecture 24: PLL and CDR Overview



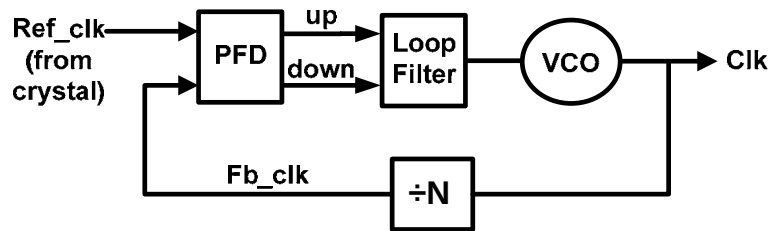
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Clock Generation



- Typical (low-cost) crystals give <500 MHz clock
 - 5 Gb/s link → where to get a 5 GHz clock?
- PLL: multiply frequency up, align phase
 - While maintaining low jitter, power

PLL: Linear Model



Linear Model cont'd

Stability and Loop Bandwidth

Loop Components: Phase Detectors

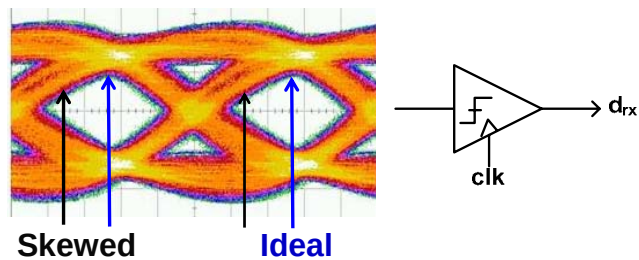
- **Basic idea: create pulses with width α to phase difference**

Loop Filter

VCOs

Noise and Jitter

Clock Recovery

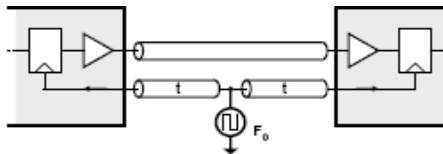


- Voltage margin strongly dependent on exact sampling position
- How do we set the clock at the “best” place?

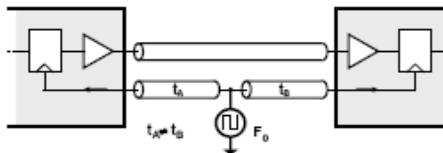
Conceptual CDR

System Types

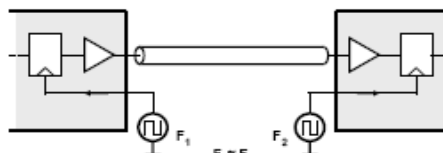
- **(Source) Synchronous**
 - Same frequency & phase



- **Mesochronous**
 - Same frequency, unknown phase



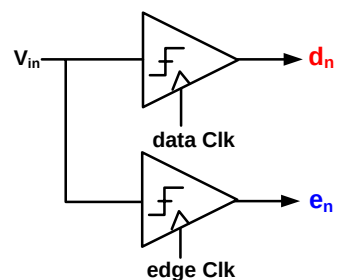
- **Plesiochronous**
 - Almost same frequency



* From EE371, Stanford University

Linear (Hogge) Phase Detector

Bang-Bang (Alexander) Phase Detector



- Edge clock $T_{\text{sym}}/2$ away from data
- Derive early/late from data and edge samples:
 - Dn: $(d_n \neq e_n) \ \& \ (d_{n-1} \neq d_n)$
 - Up: $(d_n == e_n) \ \& \ (d_{n-1} \neq d_n)$

